Human review packet: JGS23SupplySideRecommenderSystems

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: JGS23SupplySideRecommenderSystems
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How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) AppliedModelingLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-repository-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper JGS23SupplySideRecommenderSystems --serve
```

Open http://127.0.0.1:8765/ in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's local reviewer-owned review record.

Contents

Governing Lean declarations

These exact graph-bound declaration bodies are retained by the displayed specifications or reviewed prerequisites. They are supporting context and do not add source-result rows or semantic judgments.

AppliedModelingLib.FiniteDimensionalNorms.l2Sq

Declaration location: reusable library

Lean declaration body

```
type:
{ι : Type u_1} → [Fintype ι] → (ι → ℝ) → ℝ

value:
fun {ι : Type u_1} [Fintype ι] (x : ι → ℝ) => ∑ i : ι, x i ^ 2
```

AppliedModelingLib.FiniteDimensionalNorms.l2

Declaration location: reusable library

Lean declaration body

```
type:
{ι : Type u_1} → [Fintype ι] → (ι → ℝ) → ℝ

value:
fun {ι : Type u_1} [Fintype ι] (x : ι → ℝ) => √(AppliedModelingLib.FiniteDimensionalNorms.l2Sq x)
```

Direct retained dependencies

- AppliedModelingLib.FiniteDimensionalNorms.l2Sq

AppliedModelingLib.FiniteDimensionalNorms.lpPower

Declaration location: reusable library

Lean declaration body

```
type:
{ι : Type u_1} → [Fintype ι] → ℝ → (ι → ℝ) → ℝ

value:
fun {ι : Type u_1} [Fintype ι] (p : ℝ) (x : ι → ℝ) => ∑ i : ι, |x i| ^ p
```

AppliedModelingLib.FiniteDimensionalNorms.lp

Declaration location: reusable library

Lean declaration body

```
type:
{ι : Type u_1} → [Fintype ι] → ℝ → (ι → ℝ) → ℝ

value:
fun {ι : Type u_1} [Fintype ι] (p : ℝ) (x : ι → ℝ) => AppliedModelingLib.FiniteDimensionalNorms.lpPower p x ^ (1 / p)
```

Direct retained dependencies

- AppliedModelingLib.FiniteDimensionalNorms.lpPower

AppliedModelingLib.Optimization.IsMaximizerOn

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → (α → Prop) → (α → ℝ) → α → Prop

value:
fun {α : Type u_1} (feasible : α → Prop) (objective : α → ℝ) (x : α) =>
  feasible x ∧ ∀ (y : α), feasible y → objective y ≤ objective x
```

AppliedModelingLib.Optimization.IsMinimizerOn

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → (α → Prop) → (α → ℝ) → α → Prop

value:
fun {α : Type u_1} (feasible : α → Prop) (objective : α → ℝ) (x : α) =>
  feasible x ∧ ∀ (y : α), feasible y → objective x ≤ objective y
```

AppliedModelingLib.Probability.iidMaximumCdf

Declaration location: reusable library

Lean declaration body

```
type:
{n : ℕ} → MeasureTheory.Measure ℝ → ℝ → ℝ

value:
fun {n : ℕ} (μ : MeasureTheory.Measure ℝ) (x : ℝ) =>
  (MeasureTheory.Measure.pi fun (x : Fin n) => μ).real (Set.univ.pi fun (x_1 : Fin n) => Set.Iic x)
```

AppliedModelingLib.Probability.lowerCDFMass

Declaration location: reusable library

Lean declaration body

```
type:
MeasureTheory.Measure ℝ → ℝ → ℝ

value:
fun (μ : MeasureTheory.Measure ℝ) (x : ℝ) => μ.real (Set.Iic x)
```

AppliedModelingLib.Probability.unitIntervalPowerSample

Declaration location: reusable library

Lean declaration body

```
type:
ℝ → ↑unitInterval → ℝ

value:
fun (a : ℝ) (u : ↑unitInterval) => ↑u ^ a
```

JGS23SupplySideRecommenderSystems.Content

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → Type

value:
fun (D : ℕ) => Fin D → ℝ
```

JGS23SupplySideRecommenderSystems.MixedContentStrategy

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → Type

value:
fun (D : ℕ) => MeasureTheory.Measure (JGS23SupplySideRecommenderSystems.Content D)
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content

JGS23SupplySideRecommenderSystems.HasContentPointMass

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → JGS23SupplySideRecommenderSystems.MixedContentStrategy D → JGS23SupplySideRecommenderSystems.Content D → Prop

value:
fun {D : ℕ} (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D)
    (p : JGS23SupplySideRecommenderSystems.Content D) =>
  μ {p} ≠ 0
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy

JGS23SupplySideRecommenderSystems.MixedContentStrategyAtomless

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → JGS23SupplySideRecommenderSystems.MixedContentStrategy D → Prop

value:
fun {D : ℕ} (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) =>
  ∀ (p : JGS23SupplySideRecommenderSystems.Content D), ¬JGS23SupplySideRecommenderSystems.HasContentPointMass μ p
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.HasContentPointMass
- JGS23SupplySideRecommenderSystems.MixedContentStrategy

JGS23SupplySideRecommenderSystems.NonnegativeContent

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → JGS23SupplySideRecommenderSystems.Content D → Prop

value:
fun {D : ℕ} (p : JGS23SupplySideRecommenderSystems.Content D) => ∀ (d : Fin D), 0 ≤ p d
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content

JGS23SupplySideRecommenderSystems.NonzeroContent

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → JGS23SupplySideRecommenderSystems.Content D → Prop

value:
fun {D : ℕ} (p : JGS23SupplySideRecommenderSystems.Content D) => ∃ (d : Fin D), p d ≠ 0
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content

JGS23SupplySideRecommenderSystems.PerturbationCostContinuous

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → (JGS23SupplySideRecommenderSystems.Content D → ℝ) → Prop

value:
fun {D : ℕ} (cost : JGS23SupplySideRecommenderSystems.Content D → ℝ) =>
  ∀ (p u : JGS23SupplySideRecommenderSystems.Content D),
    Filter.Tendsto (fun (ε : ℝ) => cost fun (d : Fin D) => p d + ε * u d) (nhdsWithin 0 (Set.Ioi 0)) (nhds (cost p))
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content

JGS23SupplySideRecommenderSystems.SourceMixedPurePayoff

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} →
  (JGS23SupplySideRecommenderSystems.Content D → JGS23SupplySideRecommenderSystems.MixedContentStrategy D → ℝ) →
    (JGS23SupplySideRecommenderSystems.Content D → ℝ) →
      JGS23SupplySideRecommenderSystems.Content D → JGS23SupplySideRecommenderSystems.MixedContentStrategy D → ℝ

value:
fun {D : ℕ}
    (expectedUsersWon :
      JGS23SupplySideRecommenderSystems.Content D → JGS23SupplySideRecommenderSystems.MixedContentStrategy D → ℝ)
    (cost : JGS23SupplySideRecommenderSystems.Content D → ℝ) (p : JGS23SupplySideRecommenderSystems.Content D)
    (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) =>
  expectedUsersWon p μ - cost p
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy

JGS23SupplySideRecommenderSystems.content2

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → Fin 2 → ℝ

value:
fun (x y : ℝ) (a : Fin 2) => if a = 0 then x else y
```

JGS23SupplySideRecommenderSystems.canonicalTwoUserContentOfValues

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℝ → Fin 2 → ℝ

value:
fun (θ z1 z2 : ℝ) (a : Fin 2) => JGS23SupplySideRecommenderSystems.content2 z1 ((z2 - z1 * Real.cos θ) / Real.sin θ) a
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.content2

JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst

Declaration location: paper-local

Lean declaration body

```
type:
Fin 2 → ℝ

value:
fun (a : Fin 2) => JGS23SupplySideRecommenderSystems.content2 1 0 a
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.content2

JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → Fin 2 → ℝ

value:
fun (θ : ℝ) (a : Fin 2) => JGS23SupplySideRecommenderSystems.content2 (Real.cos θ) (Real.sin θ) a
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.content2

JGS23SupplySideRecommenderSystems.coordinateProduct

Declaration location: paper-local

Lean declaration body

```
type:
{N : ℕ} → (Fin N → ℝ) → ℝ

value:
fun {N : ℕ} (y : Fin N → ℝ) => ∏ i : Fin N, y i
```

JGS23SupplySideRecommenderSystems.IsCoordinateProductMaximizer

Declaration location: paper-local

Lean declaration body

```
type:
{N : ℕ} → Set (Fin N → ℝ) → (Fin N → ℝ) → Prop

value:
fun {N : ℕ} (R : Set (Fin N → ℝ)) (yStar : Fin N → ℝ) =>
  AppliedModelingLib.Optimization.IsMaximizerOn (fun (y : Fin N → ℝ) => y ∈ R)
    JGS23SupplySideRecommenderSystems.coordinateProduct yStar
```

Direct retained dependencies

- AppliedModelingLib.Optimization.IsMaximizerOn
- JGS23SupplySideRecommenderSystems.coordinateProduct

JGS23SupplySideRecommenderSystems.coordinateProductSup

Declaration location: paper-local

Lean declaration body

```
type:
{N : ℕ} → Set (Fin N → ℝ) → ℝ

value:
fun {N : ℕ} (S : Set (Fin N → ℝ)) => sSup (JGS23SupplySideRecommenderSystems.coordinateProduct '' S)
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.coordinateProduct

JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition

Declaration location: paper-local

Lean declaration body

```
type:
{N : ℕ} → Set (Fin N → ℝ) → Prop

value:
fun {N : ℕ} (S : Set (Fin N → ℝ)) =>
  JGS23SupplySideRecommenderSystems.coordinateProductSup S =
    JGS23SupplySideRecommenderSystems.coordinateProductSup ((convexHull ℝ) S)
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.coordinateProductSup

JGS23SupplySideRecommenderSystems.deviateProfile

Declaration location: paper-local

Lean declaration body

```
type:
{D P : ℕ} →
  (Fin P → JGS23SupplySideRecommenderSystems.Content D) →
    Fin P → JGS23SupplySideRecommenderSystems.Content D → Fin P → Fin D → ℝ

value:
fun {D P : ℕ} (profile : Fin P → JGS23SupplySideRecommenderSystems.Content D) (j : Fin P)
    (p : JGS23SupplySideRecommenderSystems.Content D) (a : Fin P) (a_1 : Fin D) =>
  Function.update profile j p a a_1
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content

JGS23SupplySideRecommenderSystems.finitePSourceCurveSample

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ↑unitInterval → ℝ × ℝ

value:
fun (P : ℕ) (u : ↑unitInterval) =>
  have exponent : ℝ := (↑P - 1) / 2;
  (↑u ^ exponent, (1 - ↑u) ^ exponent)
```

JGS23SupplySideRecommenderSystems.finitePSourceCurveMeasure

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → MeasureTheory.Measure (ℝ × ℝ)

value:
fun (P : ℕ) =>
  MeasureTheory.Measure.map (JGS23SupplySideRecommenderSystems.finitePSourceCurveSample P) MeasureTheory.volume
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.finitePSourceCurveSample

JGS23SupplySideRecommenderSystems.infiniteGenreFirstGenre

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → Fin 2 → ℝ

value:
fun (phi : ℝ) (a : Fin 2) => JGS23SupplySideRecommenderSystems.content2 (Real.cos phi) (Real.sin phi) a
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.content2

JGS23SupplySideRecommenderSystems.infiniteGenreSecondGenre

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → Fin 2 → ℝ

value:
fun (theta phi : ℝ) (a : Fin 2) =>
  JGS23SupplySideRecommenderSystems.content2 (Real.cos (theta - phi)) (Real.sin (theta - phi)) a
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.content2

JGS23SupplySideRecommenderSystems.oneDimensionalUnitContent

Declaration location: paper-local

Lean declaration body

```
type:
Fin 1 → ℝ

value:
fun (a : Fin 1) => 1
```

JGS23SupplySideRecommenderSystems.pTwoContentOfCoordinates

Declaration location: paper-local

Lean declaration body

```
type:
ℝ × ℝ → Fin 2 → ℝ

value:
fun (z : ℝ × ℝ) (a : Fin 2) => JGS23SupplySideRecommenderSystems.content2 z.1 z.2 a
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.content2

JGS23SupplySideRecommenderSystems.finitePSourceContentMeasure

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → MeasureTheory.Measure (JGS23SupplySideRecommenderSystems.Content 2)

value:
fun (P : ℕ) =>
  MeasureTheory.Measure.map JGS23SupplySideRecommenderSystems.pTwoContentOfCoordinates
    (JGS23SupplySideRecommenderSystems.finitePSourceCurveMeasure P)
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.finitePSourceCurveMeasure
- JGS23SupplySideRecommenderSystems.pTwoContentOfCoordinates

JGS23SupplySideRecommenderSystems.pTwoSourceCircleSample

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ↑unitInterval → ℝ × ℝ

value:
fun (beta : ℝ) (u : ↑unitInterval) =>
  have radius : ℝ := (2 / beta) ^ (1 / beta);
  (radius * √(1 - ↑u), radius * √↑u)
```

JGS23SupplySideRecommenderSystems.pTwoSourceCircleMeasure

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → MeasureTheory.Measure (ℝ × ℝ)

value:
fun (beta : ℝ) =>
  MeasureTheory.Measure.map (JGS23SupplySideRecommenderSystems.pTwoSourceCircleSample beta) MeasureTheory.volume
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.pTwoSourceCircleSample

JGS23SupplySideRecommenderSystems.pTwoSourceContentMeasure

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → MeasureTheory.Measure (JGS23SupplySideRecommenderSystems.Content 2)

value:
fun (beta : ℝ) =>
  MeasureTheory.Measure.map JGS23SupplySideRecommenderSystems.pTwoContentOfCoordinates
    (JGS23SupplySideRecommenderSystems.pTwoSourceCircleMeasure beta)
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.pTwoContentOfCoordinates
- JGS23SupplySideRecommenderSystems.pTwoSourceCircleMeasure

JGS23SupplySideRecommenderSystems.pTwoStandardBasisUsers

Declaration location: paper-local

Lean declaration body

```
type:
Fin 2 → Fin 2 → ℝ

value:
fun (i a : Fin 2) => JGS23SupplySideRecommenderSystems.content2 (if i = 0 then 1 else 0) (if i = 1 then 1 else 0) a
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.content2

JGS23SupplySideRecommenderSystems.paper_mixed_content_strategy

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → Type

value:
fun (D : ℕ) => JGS23SupplySideRecommenderSystems.MixedContentStrategy D
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.MixedContentStrategy

JGS23SupplySideRecommenderSystems.paper_mixed_content_strategy_atomless

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → JGS23SupplySideRecommenderSystems.paper_mixed_content_strategy D → Prop

value:
fun {D : ℕ} (μ : JGS23SupplySideRecommenderSystems.paper_mixed_content_strategy D) =>
  JGS23SupplySideRecommenderSystems.MixedContentStrategyAtomless μ
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.MixedContentStrategyAtomless
- JGS23SupplySideRecommenderSystems.paper_mixed_content_strategy

JGS23SupplySideRecommenderSystems.ratioObjective

Declaration location: paper-local

Lean declaration body

```
type:
{N : ℕ} → (Fin N → ℝ) → (Fin N → ℝ) → ℝ

value:
fun {N : ℕ} (y y' : Fin N → ℝ) => ∑ i : Fin N, y' i / y i
```

JGS23SupplySideRecommenderSystems.paper_ratio_objective

Declaration location: paper-local

Lean declaration body

```
type:
{N : ℕ} → (Fin N → ℝ) → (Fin N → ℝ) → ℝ

value:
fun {N : ℕ} (y y' : Fin N → ℝ) => JGS23SupplySideRecommenderSystems.ratioObjective y y'
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.ratioObjective

JGS23SupplySideRecommenderSystems.scaleContent

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → ℝ → JGS23SupplySideRecommenderSystems.Content D → Fin D → ℝ

value:
fun {D : ℕ} (a : ℝ) (p : JGS23SupplySideRecommenderSystems.Content D) (a_1 : Fin D) => a * p a_1
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content

JGS23SupplySideRecommenderSystems.FiniteGenreConditionalNormLawAnyDim

Declaration location: paper-local

Lean declaration body

```
field weight:
{D G : ℕ} →
  {mu : JGS23SupplySideRecommenderSystems.MixedContentStrategy D} →
    {direction : Fin G → JGS23SupplySideRecommenderSystems.Content D} →
      JGS23SupplySideRecommenderSystems.FiniteGenreConditionalNormLawAnyDim mu direction → Fin G → ℝ

field radial:
{D G : ℕ} →
  {mu : JGS23SupplySideRecommenderSystems.MixedContentStrategy D} →
    {direction : Fin G → JGS23SupplySideRecommenderSystems.Content D} →
      JGS23SupplySideRecommenderSystems.FiniteGenreConditionalNormLawAnyDim mu direction →
        Fin G → MeasureTheory.Measure ℝ

field direction_nonnegative:
∀ {D G : ℕ} {mu : JGS23SupplySideRecommenderSystems.MixedContentStrategy D}
  {direction : Fin G → JGS23SupplySideRecommenderSystems.Content D}
  (self : JGS23SupplySideRecommenderSystems.FiniteGenreConditionalNormLawAnyDim mu direction) (i : Fin G),
  JGS23SupplySideRecommenderSystems.NonnegativeContent (direction i)

field direction_l2_norm:
∀ {D G : ℕ} {mu : JGS23SupplySideRecommenderSystems.MixedContentStrategy D}
  {direction : Fin G → JGS23SupplySideRecommenderSystems.Content D}
  (self : JGS23SupplySideRecommenderSystems.FiniteGenreConditionalNormLawAnyDim mu direction) (i : Fin G),
  AppliedModelingLib.FiniteDimensionalNorms.l2 (direction i) = 1

field weight_pos:
∀ {D G : ℕ} {mu : JGS23SupplySideRecommenderSystems.MixedContentStrategy D}
  {direction : Fin G → JGS23SupplySideRecommenderSystems.Content D}
  (self : JGS23SupplySideRecommenderSystems.FiniteGenreConditionalNormLawAnyDim mu direction) (i : Fin G),
  0 < self.weight i

field weights_sum:
∀ {D G : ℕ} {mu : JGS23SupplySideRecommenderSystems.MixedContentStrategy D}
  {direction : Fin G → JGS23SupplySideRecommenderSystems.Content D}
  (self : JGS23SupplySideRecommenderSystems.FiniteGenreConditionalNormLawAnyDim mu direction),
  ∑ i : Fin G, self.weight i = 1

field radial_probability:
∀ {D G : ℕ} {mu : JGS23SupplySideRecommenderSystems.MixedContentStrategy D}
  {direction : Fin G → JGS23SupplySideRecommenderSystems.Content D}
  (self : JGS23SupplySideRecommenderSystems.FiniteGenreConditionalNormLawAnyDim mu direction) (i : Fin G),
  MeasureTheory.IsProbabilityMeasure (self.radial i)

field radial_support_nonnegative:
∀ {D G : ℕ} {mu : JGS23SupplySideRecommenderSystems.MixedContentStrategy D}
  {direction : Fin G → JGS23SupplySideRecommenderSystems.Content D}
  (self : JGS23SupplySideRecommenderSystems.FiniteGenreConditionalNormLawAnyDim mu direction) (i : Fin G),
  (self.radial i).support ⊆ Set.Ici 0

field radial_has_positive_support:
∀ {D G : ℕ} {mu : JGS23SupplySideRecommenderSystems.MixedContentStrategy D}
  {direction : Fin G → JGS23SupplySideRecommenderSystems.Content D}
  (self : JGS23SupplySideRecommenderSystems.FiniteGenreConditionalNormLawAnyDim mu direction) (i : Fin G),
  ∃ r ∈ (self.radial i).support, 0 < r

field radial_cdf_c1:
∀ {D G : ℕ} {mu : JGS23SupplySideRecommenderSystems.MixedContentStrategy D}
  {direction : Fin G → JGS23SupplySideRecommenderSystems.Content D}
  (self : JGS23SupplySideRecommenderSystems.FiniteGenreConditionalNormLawAnyDim mu direction) (i : Fin G),
  ContDiff ℝ 1 (AppliedModelingLib.Probability.lowerCDFMass (self.radial i))

field decomposition:
∀ {D G : ℕ} {mu : JGS23SupplySideRecommenderSystems.MixedContentStrategy D}
  {direction : Fin G → JGS23SupplySideRecommenderSystems.Content D}
  (self : JGS23SupplySideRecommenderSystems.FiniteGenreConditionalNormLawAnyDim mu direction),
  mu =
    ∑ i : Fin G,
      ENNReal.ofReal (self.weight i) •
        MeasureTheory.Measure.map (fun (r : ℝ) => JGS23SupplySideRecommenderSystems.scaleContent r (direction i))
          (self.radial i)
```

Direct retained dependencies

- AppliedModelingLib.FiniteDimensionalNorms.l2
- AppliedModelingLib.Probability.lowerCDFMass
- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.scaleContent

JGS23SupplySideRecommenderSystems.SourceNorm

Declaration location: paper-local

Lean declaration body

```
field norm:
{D : ℕ} → JGS23SupplySideRecommenderSystems.SourceNorm D → JGS23SupplySideRecommenderSystems.Content D → ℝ

field nonneg:
∀ {D : ℕ} (self : JGS23SupplySideRecommenderSystems.SourceNorm D) (p : JGS23SupplySideRecommenderSystems.Content D),
  0 ≤ self.norm p

field pos_of_nonzero:
∀ {D : ℕ} (self : JGS23SupplySideRecommenderSystems.SourceNorm D) {p : JGS23SupplySideRecommenderSystems.Content D},
  JGS23SupplySideRecommenderSystems.NonzeroContent p → 0 < self.norm p

field scale_nonneg:
∀ {D : ℕ} (self : JGS23SupplySideRecommenderSystems.SourceNorm D) {a : ℝ}
  {p : JGS23SupplySideRecommenderSystems.Content D},
  0 ≤ a → self.norm (JGS23SupplySideRecommenderSystems.scaleContent a p) = a * self.norm p
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.scaleContent

JGS23SupplySideRecommenderSystems.SourceNorm.l2

Declaration location: paper-local

Lean declaration body

```
type:
(D : ℕ) → JGS23SupplySideRecommenderSystems.SourceNorm D

value:
fun (D : ℕ) =>
  { norm := fun (p : JGS23SupplySideRecommenderSystems.Content D) => AppliedModelingLib.FiniteDimensionalNorms.l2 p,
    nonneg := JGS23SupplySideRecommenderSystems.SourceNorm.l2._proof_1 D,
    pos_of_nonzero := @JGS23SupplySideRecommenderSystems.SourceNorm.l2._proof_2 D,
    scale_nonneg := @JGS23SupplySideRecommenderSystems.SourceNorm.l2._proof_3 D }
```

Direct retained dependencies

- AppliedModelingLib.FiniteDimensionalNorms.l2
- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SourceNorm

JGS23SupplySideRecommenderSystems.SourceNorm.lp

Declaration location: paper-local

Lean declaration body

```
type:
(D : ℕ) → {q : ℝ} → 0 < q → JGS23SupplySideRecommenderSystems.SourceNorm D

value:
fun (D : ℕ) {q : ℝ} (hq : 0 < q) =>
  { norm := fun (p : JGS23SupplySideRecommenderSystems.Content D) => AppliedModelingLib.FiniteDimensionalNorms.lp q p,
    nonneg := JGS23SupplySideRecommenderSystems.SourceNorm.lp._proof_1 D,
    pos_of_nonzero := @JGS23SupplySideRecommenderSystems.SourceNorm.lp._proof_2 D q hq,
    scale_nonneg := @JGS23SupplySideRecommenderSystems.SourceNorm.lp._proof_3 D q hq }
```

Direct retained dependencies

- AppliedModelingLib.FiniteDimensionalNorms.lp
- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SourceNorm

JGS23SupplySideRecommenderSystems.SourceNorm.normalizedContent

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → JGS23SupplySideRecommenderSystems.SourceNorm D → JGS23SupplySideRecommenderSystems.Content D → Fin D → ℝ

value:
fun {D : ℕ} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D) (p : JGS23SupplySideRecommenderSystems.Content D)
    (a : Fin D) =>
  JGS23SupplySideRecommenderSystems.scaleContent (ν.norm p)⁻¹ p a
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.scaleContent

JGS23SupplySideRecommenderSystems.SourceNonzeroSupportGenres

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} →
  JGS23SupplySideRecommenderSystems.SourceNorm D →
    JGS23SupplySideRecommenderSystems.MixedContentStrategy D → JGS23SupplySideRecommenderSystems.Content D → Prop

value:
fun {D : ℕ} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D)
    (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) (a : JGS23SupplySideRecommenderSystems.Content D) =>
  {genre : JGS23SupplySideRecommenderSystems.Content D |
      ∃ p ∈ MeasureTheory.Measure.support μ,
        JGS23SupplySideRecommenderSystems.NonzeroContent p ∧ genre = ν.normalizedContent p}
    a
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.SourceNorm.normalizedContent

JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → JGS23SupplySideRecommenderSystems.SourceNorm D → Prop

value:
fun {D : ℕ} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D) =>
  ∀ (R : ℝ), IsCompact {p : JGS23SupplySideRecommenderSystems.Content D | ν.norm p ≤ R}
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.SourceNorm

JGS23SupplySideRecommenderSystems.SourceNormConvexSublevels

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → JGS23SupplySideRecommenderSystems.SourceNorm D → Prop

value:
fun {D : ℕ} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D) =>
  ∀ (R : ℝ), Convex ℝ {p : JGS23SupplySideRecommenderSystems.Content D | ν.norm p ≤ R}
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.SourceNorm

JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → JGS23SupplySideRecommenderSystems.SourceNorm D → Prop

value:
fun {D : ℕ} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D) =>
  ∀ (p u : JGS23SupplySideRecommenderSystems.Content D),
    Filter.Tendsto (fun (ε : ℝ) => ν.norm fun (d : Fin D) => p d + ε * u d) (nhdsWithin 0 (Set.Ioi 0)) (nhds (ν.norm p))
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.SourceNorm

JGS23SupplySideRecommenderSystems.finiteGenreConditionalNormLawSpec

Declaration location: paper-local

Lean declaration body

```
field weight:
{G : ℕ} →
  {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy 2} →
    {angle : Fin G → ℝ} → JGS23SupplySideRecommenderSystems.finiteGenreConditionalNormLawSpec μ angle → Fin G → ℝ

field radial:
{G : ℕ} →
  {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy 2} →
    {angle : Fin G → ℝ} →
      JGS23SupplySideRecommenderSystems.finiteGenreConditionalNormLawSpec μ angle → Fin G → MeasureTheory.Measure ℝ

field weight_pos:
∀ {G : ℕ} {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy 2} {angle : Fin G → ℝ}
  (self : JGS23SupplySideRecommenderSystems.finiteGenreConditionalNormLawSpec μ angle) (i : Fin G), 0 < self.weight i

field weights_sum:
∀ {G : ℕ} {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy 2} {angle : Fin G → ℝ}
  (self : JGS23SupplySideRecommenderSystems.finiteGenreConditionalNormLawSpec μ angle), ∑ i : Fin G, self.weight i = 1

field radial_probability:
∀ {G : ℕ} {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy 2} {angle : Fin G → ℝ}
  (self : JGS23SupplySideRecommenderSystems.finiteGenreConditionalNormLawSpec μ angle) (i : Fin G),
  MeasureTheory.IsProbabilityMeasure (self.radial i)

field radial_support_nonnegative:
∀ {G : ℕ} {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy 2} {angle : Fin G → ℝ}
  (self : JGS23SupplySideRecommenderSystems.finiteGenreConditionalNormLawSpec μ angle) (i : Fin G),
  (self.radial i).support ⊆ Set.Ici 0

field radial_has_positive_support:
∀ {G : ℕ} {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy 2} {angle : Fin G → ℝ}
  (self : JGS23SupplySideRecommenderSystems.finiteGenreConditionalNormLawSpec μ angle) (i : Fin G),
  ∃ r ∈ (self.radial i).support, 0 < r

field radial_cdf_c1:
∀ {G : ℕ} {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy 2} {angle : Fin G → ℝ}
  (self : JGS23SupplySideRecommenderSystems.finiteGenreConditionalNormLawSpec μ angle) (i : Fin G),
  ContDiff ℝ 1 (AppliedModelingLib.Probability.lowerCDFMass (self.radial i))

field decomposition:
∀ {G : ℕ} {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy 2} {angle : Fin G → ℝ}
  (self : JGS23SupplySideRecommenderSystems.finiteGenreConditionalNormLawSpec μ angle),
  μ =
    ∑ i : Fin G,
      ENNReal.ofReal (self.weight i) •
```

```
  MeasureTheory.Measure.map
    (fun (r : ℝ) =>
      JGS23SupplySideRecommenderSystems.scaleContent r
        (JGS23SupplySideRecommenderSystems.content2 (Real.cos (angle i)) (Real.sin (angle i))))
    (self.radial i)
```

Direct retained dependencies

- AppliedModelingLib.Probability.lowerCDFMass
- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.content2
- JGS23SupplySideRecommenderSystems.scaleContent

JGS23SupplySideRecommenderSystems.l2NormalizedUsers

Declaration location: paper-local

Lean declaration body

```
type:
{D N : ℕ} → (Fin N → JGS23SupplySideRecommenderSystems.Content D) → Fin N → Fin D → ℝ

value:
fun {D N : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D) (a : Fin N) (a_1 : Fin D) =>
  (JGS23SupplySideRecommenderSystems.SourceNorm.l2 D).normalizedContent (users a) a_1
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.SourceNorm.l2
- JGS23SupplySideRecommenderSystems.SourceNorm.normalizedContent

JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} →
  JGS23SupplySideRecommenderSystems.SourceNorm D →
    JGS23SupplySideRecommenderSystems.MixedContentStrategy D → JGS23SupplySideRecommenderSystems.Content D → Prop

value:
fun {D : ℕ} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D)
    (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) (a : JGS23SupplySideRecommenderSystems.Content D) =>
  {genre : JGS23SupplySideRecommenderSystems.Content D |
      ∃ p ∈ MeasureTheory.Measure.support μ,
        JGS23SupplySideRecommenderSystems.NonzeroContent p ∧ genre = ν.normalizedContent p}
    a
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.SourceNorm.normalizedContent

JGS23SupplySideRecommenderSystems.normRpowCost

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → JGS23SupplySideRecommenderSystems.SourceNorm D → ℝ → JGS23SupplySideRecommenderSystems.Content D → ℝ

value:
fun {D : ℕ} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D) (β : ℝ)
    (p : JGS23SupplySideRecommenderSystems.Content D) =>
  ν.norm p ^ β
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.SourceNorm

JGS23SupplySideRecommenderSystems.normPowerCostSpec

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → JGS23SupplySideRecommenderSystems.SourceNorm D → ℝ → JGS23SupplySideRecommenderSystems.Content D → ℝ

value:
fun {D : ℕ} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D) (β : ℝ)
    (a : JGS23SupplySideRecommenderSystems.Content D) =>
  JGS23SupplySideRecommenderSystems.normRpowCost ν β a
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.normRpowCost

JGS23SupplySideRecommenderSystems.score

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → JGS23SupplySideRecommenderSystems.Content D → JGS23SupplySideRecommenderSystems.Content D → ℝ

value:
fun {D : ℕ} (u p : JGS23SupplySideRecommenderSystems.Content D) => ∑ d : Fin D, u d * p d
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content

JGS23SupplySideRecommenderSystems.InUnitImage

Declaration location: paper-local

Lean declaration body

```
type:
{D N : ℕ} →
  (Fin N → JGS23SupplySideRecommenderSystems.Content D) →
    (JGS23SupplySideRecommenderSystems.Content D → Prop) → (Fin N → ℝ) → Prop

value:
fun {D N : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (unitFeasible : JGS23SupplySideRecommenderSystems.Content D → Prop) (y : Fin N → ℝ) =>
  ∃ (p : JGS23SupplySideRecommenderSystems.Content D),
    unitFeasible p ∧ y = fun (i : Fin N) => JGS23SupplySideRecommenderSystems.score (users i) p
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.score

JGS23SupplySideRecommenderSystems.InPoweredUnitImage

Declaration location: paper-local

Lean declaration body

```
type:
{D N : ℕ} →
  (Fin N → JGS23SupplySideRecommenderSystems.Content D) →
    (JGS23SupplySideRecommenderSystems.Content D → Prop) → ℝ → (Fin N → ℝ) → Prop

value:
fun {D N : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (unitFeasible : JGS23SupplySideRecommenderSystems.Content D → Prop) (β : ℝ) (y : Fin N → ℝ) =>
  ∃ (base : Fin N → ℝ),
    JGS23SupplySideRecommenderSystems.InUnitImage users unitFeasible base ∧ y = fun (i : Fin N) => base i ^ β
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.InUnitImage

JGS23SupplySideRecommenderSystems.canonicalTwoUserValueMap

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → JGS23SupplySideRecommenderSystems.Content 2 → ℝ × ℝ

value:
fun (θ : ℝ) (p : JGS23SupplySideRecommenderSystems.Content 2) =>
  (JGS23SupplySideRecommenderSystems.score JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst p,
    JGS23SupplySideRecommenderSystems.score (JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ) p)
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
- JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond
- JGS23SupplySideRecommenderSystems.score

JGS23SupplySideRecommenderSystems.infiniteGenreContentObjectiveSpec

Declaration location: paper-local

Lean declaration body

```
type:
{D N G : ℕ} →
  (Fin N → JGS23SupplySideRecommenderSystems.Content D) →
    (JGS23SupplySideRecommenderSystems.Content D → ℝ) →
      (Fin G → JGS23SupplySideRecommenderSystems.Content D) →
        (Fin G → ℝ → ℝ) → (Fin G → ℝ) → JGS23SupplySideRecommenderSystems.Content D → ℝ

value:
fun {D N G : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (cost : JGS23SupplySideRecommenderSystems.Content D → ℝ)
    (genres : Fin G → JGS23SupplySideRecommenderSystems.Content D) (maximumQualityCdf : Fin G → ℝ → ℝ)
    (weights : Fin G → ℝ) (p : JGS23SupplySideRecommenderSystems.Content D) =>
  ∑ i : Fin N,
      ∏ g : Fin G,
        maximumQualityCdf g
            (JGS23SupplySideRecommenderSystems.score (users i) p /
              JGS23SupplySideRecommenderSystems.score (users i) (genres g)) ^
          weights g -
    cost p
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.score

JGS23SupplySideRecommenderSystems.infiniteTwoGenreContentObjectiveSpec

Declaration location: paper-local

Lean declaration body

```
type:
(ℝ → ℝ) →
  ℝ →
    ℝ →
      JGS23SupplySideRecommenderSystems.Content 2 →
        JGS23SupplySideRecommenderSystems.Content 2 → JGS23SupplySideRecommenderSystems.Content 2 → ℝ

value:
fun (cdf : ℝ → ℝ) (β θ : ℝ) (firstGenre secondGenre p : JGS23SupplySideRecommenderSystems.Content 2) =>
  √(cdf
          (JGS23SupplySideRecommenderSystems.score JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst p /
            JGS23SupplySideRecommenderSystems.score JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
              firstGenre) *
        cdf
          (JGS23SupplySideRecommenderSystems.score JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst p /
            JGS23SupplySideRecommenderSystems.score JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
              secondGenre)) +
      √(cdf
          (JGS23SupplySideRecommenderSystems.score (JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ) p /
            JGS23SupplySideRecommenderSystems.score (JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
              firstGenre) *
        cdf
          (JGS23SupplySideRecommenderSystems.score (JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ) p /
            JGS23SupplySideRecommenderSystems.score (JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
              secondGenre)) -
    JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2) β p
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.SourceNorm.l2
- JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
- JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.score

JGS23SupplySideRecommenderSystems.infiniteProducerCandidateSpec

Declaration location: paper-local

Lean declaration body

```
type:
(ℝ → ℝ) → ℝ → ℝ → ℝ → Prop

value:
fun (cdf : ℝ → ℝ) (β θ phi : ℝ) =>
  ∃ (firstGenre : JGS23SupplySideRecommenderSystems.Content 2) (secondGenre :
    JGS23SupplySideRecommenderSystems.Content 2) (conditionalQuality : MeasureTheory.Measure ℝ) (firstWeight : ℝ)
    (secondWeight : ℝ),
    firstGenre = JGS23SupplySideRecommenderSystems.infiniteGenreFirstGenre phi ∧
      secondGenre = JGS23SupplySideRecommenderSystems.infiniteGenreSecondGenre θ phi ∧
        firstGenre ≠ secondGenre ∧
          JGS23SupplySideRecommenderSystems.NonnegativeContent firstGenre ∧
            JGS23SupplySideRecommenderSystems.NonnegativeContent secondGenre ∧
              (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2).norm firstGenre = 1 ∧
                (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2).norm secondGenre = 1 ∧
                  MeasureTheory.IsProbabilityMeasure conditionalQuality ∧
                    (∀ (q : ℝ), conditionalQuality (Set.Iic q) = ENNReal.ofReal (cdf q)) ∧
                      conditionalQuality.support ⊆ Set.Ici 0 ∧
                        0 ≤ firstWeight ∧
                          0 ≤ secondWeight ∧
                            firstWeight = 1 / 2 ∧
                              secondWeight = 1 / 2 ∧
                                firstWeight + secondWeight = 1 ∧
                                  (∀ {{r : ℝ}},
                                      r ∈ conditionalQuality.support →
                                        JGS23SupplySideRecommenderSystems.scaleContent r firstGenre ∈
                                            {p : JGS23SupplySideRecommenderSystems.Content 2 |
                                              JGS23SupplySideRecommenderSystems.NonnegativeContent p} ∧
                                          IsMaxOn
                                            (JGS23SupplySideRecommenderSystems.infiniteTwoGenreContentObjectiveSpec cdf
                                              β θ firstGenre secondGenre)
                                            {p : JGS23SupplySideRecommenderSystems.Content 2 |
                                              JGS23SupplySideRecommenderSystems.NonnegativeContent p}
                                            (JGS23SupplySideRecommenderSystems.scaleContent r firstGenre)) ∧
                                    ∀ {{r : ℝ}},
                                      r ∈ conditionalQuality.support →
```

```
JGS23SupplySideRecommenderSystems.scaleContent r secondGenre ∈
    {p : JGS23SupplySideRecommenderSystems.Content 2 |
      JGS23SupplySideRecommenderSystems.NonnegativeContent p} ∧
  IsMaxOn
    (JGS23SupplySideRecommenderSystems.infiniteTwoGenreContentObjectiveSpec cdf
      β θ firstGenre secondGenre)
    {p : JGS23SupplySideRecommenderSystems.Content 2 |
      JGS23SupplySideRecommenderSystems.NonnegativeContent p}
    (JGS23SupplySideRecommenderSystems.scaleContent r secondGenre)
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.SourceNorm.l2
- JGS23SupplySideRecommenderSystems.infiniteGenreFirstGenre
- JGS23SupplySideRecommenderSystems.infiniteGenreSecondGenre
- JGS23SupplySideRecommenderSystems.infiniteTwoGenreContentObjectiveSpec
- JGS23SupplySideRecommenderSystems.scaleContent

JGS23SupplySideRecommenderSystems.pTwoStandardBasisValueMap

Declaration location: paper-local

Lean declaration body

```
type:
JGS23SupplySideRecommenderSystems.Content 2 → ℝ × ℝ

value:
fun (p : JGS23SupplySideRecommenderSystems.Content 2) =>
  (JGS23SupplySideRecommenderSystems.score (JGS23SupplySideRecommenderSystems.pTwoStandardBasisUsers 0) p,
    JGS23SupplySideRecommenderSystems.score (JGS23SupplySideRecommenderSystems.pTwoStandardBasisUsers 1) p)
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.pTwoStandardBasisUsers
- JGS23SupplySideRecommenderSystems.score

JGS23SupplySideRecommenderSystems.paper_inferred_user_value

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → JGS23SupplySideRecommenderSystems.Content D → JGS23SupplySideRecommenderSystems.Content D → ℝ

value:
fun {D : ℕ} (u p : JGS23SupplySideRecommenderSystems.Content D) => JGS23SupplySideRecommenderSystems.score u p
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.score

JGS23SupplySideRecommenderSystems.paper_powered_unit_image

Declaration location: paper-local

Lean declaration body

```
type:
{D N : ℕ} →
  (Fin N → JGS23SupplySideRecommenderSystems.Content D) →
    (JGS23SupplySideRecommenderSystems.Content D → Prop) → ℝ → (Fin N → ℝ) → Prop

value:
fun {D N : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (unitFeasible : JGS23SupplySideRecommenderSystems.Content D → Prop) (β : ℝ) (y : Fin N → ℝ) =>
  JGS23SupplySideRecommenderSystems.InPoweredUnitImage users unitFeasible β y
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.InPoweredUnitImage

JGS23SupplySideRecommenderSystems.singleGenreQualitySample

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℕ → ℝ → ↑unitInterval → ℝ

value:
fun (N P : ℕ) (β : ℝ) (u : ↑unitInterval) =>
  ↑N ^ β⁻¹ * AppliedModelingLib.Probability.unitIntervalPowerSample ((↑P - 1) / β) u
```

Direct retained dependencies

- AppliedModelingLib.Probability.unitIntervalPowerSample

JGS23SupplySideRecommenderSystems.singleGenreContentSample

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → ℕ → ℕ → ℝ → JGS23SupplySideRecommenderSystems.Content D → ↑unitInterval → Fin D → ℝ

value:
fun {D : ℕ} (N P : ℕ) (β : ℝ) (genre : JGS23SupplySideRecommenderSystems.Content D) (u : ↑unitInterval) (a : Fin D) =>
  JGS23SupplySideRecommenderSystems.scaleContent (JGS23SupplySideRecommenderSystems.singleGenreQualitySample N P β u)
    genre a
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.scaleContent
- JGS23SupplySideRecommenderSystems.singleGenreQualitySample

JGS23SupplySideRecommenderSystems.singleGenreContentLaw

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} →
  ℕ →
    ℕ →
      ℝ →
        JGS23SupplySideRecommenderSystems.Content D →
          MeasureTheory.Measure (JGS23SupplySideRecommenderSystems.Content D)

value:
fun {D : ℕ} (N P : ℕ) (β : ℝ) (genre : JGS23SupplySideRecommenderSystems.Content D) =>
  MeasureTheory.Measure.map (JGS23SupplySideRecommenderSystems.singleGenreContentSample N P β genre)
    MeasureTheory.volume
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.singleGenreContentSample

JGS23SupplySideRecommenderSystems.standardBasisContent

Declaration location: paper-local

Lean declaration body

```
type:
(D : ℕ) → Fin D → Fin D → ℝ

value:
fun (D : ℕ) (i a : Fin D) => if a = i then 1 else 0
```

JGS23SupplySideRecommenderSystems.standardBasisUsers

Declaration location: paper-local

Lean declaration body

```
type:
(D : ℕ) → Fin D → Fin D → ℝ

value:
fun (D : ℕ) (i a : Fin D) => JGS23SupplySideRecommenderSystems.standardBasisContent D i a
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.standardBasisContent

JGS23SupplySideRecommenderSystems.twoPopulationUsers

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → (K : ℕ) → (Fin 2 → JGS23SupplySideRecommenderSystems.Content D) → Fin (2 * K) → Fin D → ℝ

value:
fun {D : ℕ} (K : ℕ) (users : Fin 2 → JGS23SupplySideRecommenderSystems.Content D) (a : Fin (2 * K)) (a_1 : Fin D) =>
  users (finProdFinEquiv.symm a).1 a_1
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content

JGS23SupplySideRecommenderSystems.twoUserInducedCostNumerator

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℝ → ℝ

value:
fun (θ z1 z2 : ℝ) => z1 ^ 2 + z2 ^ 2 - 2 * z1 * z2 * Real.cos θ
```

JGS23SupplySideRecommenderSystems.twoUserInducedCost

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℝ → ℝ → ℝ → ℝ

value:
fun (α β θ z1 z2 : ℝ) =>
  α * Real.sin θ ^ (-β) * JGS23SupplySideRecommenderSystems.twoUserInducedCostNumerator θ z1 z2 ^ (β / 2)
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.twoUserInducedCostNumerator

JGS23SupplySideRecommenderSystems.paper_two_user_induced_cost

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℝ → ℝ → ℝ → ℝ

value:
fun (α β θ z1 z2 : ℝ) => JGS23SupplySideRecommenderSystems.twoUserInducedCost α β θ z1 z2
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.twoUserInducedCost

JGS23SupplySideRecommenderSystems.twoUserInducedCostCrossPartial

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℝ → ℝ → ℝ → ℝ

value:
fun (α β θ z1 z2 : ℝ) =>
  β * α * Real.sin θ ^ (-β) *
    ((β - 2) * JGS23SupplySideRecommenderSystems.twoUserInducedCostNumerator θ z1 z2 ^ (β / 2 - 2) *
          (z2 - z1 * Real.cos θ) *
        (z1 - z2 * Real.cos θ) -
      Real.cos θ * JGS23SupplySideRecommenderSystems.twoUserInducedCostNumerator θ z1 z2 ^ (β / 2 - 1))
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.twoUserInducedCostNumerator

JGS23SupplySideRecommenderSystems.paper_two_user_induced_cost_cross_partial

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℝ → ℝ → ℝ → ℝ

value:
fun (α β θ z1 z2 : ℝ) => JGS23SupplySideRecommenderSystems.twoUserInducedCostCrossPartial α β θ z1 z2
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.twoUserInducedCostCrossPartial

JGS23SupplySideRecommenderSystems.twoUserNormalizedContent

Declaration location: paper-local

Lean declaration body

```
type:
{D : ℕ} → JGS23SupplySideRecommenderSystems.Content D → Fin D → ℝ

value:
fun {D : ℕ} (u : JGS23SupplySideRecommenderSystems.Content D) (a : Fin D) =>
  JGS23SupplySideRecommenderSystems.scaleContent (AppliedModelingLib.FiniteDimensionalNorms.l2 u)⁻¹ u a
```

Direct retained dependencies

- AppliedModelingLib.FiniteDimensionalNorms.l2
- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.scaleContent

JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ

value:
fun (θ : ℝ) => 2 / (1 - Real.cos θ)
```

JGS23SupplySideRecommenderSystems.twoUserScoreMap

Declaration location: paper-local

Lean declaration body

```
type:
JGS23SupplySideRecommenderSystems.Content 2 →
  JGS23SupplySideRecommenderSystems.Content 2 → JGS23SupplySideRecommenderSystems.Content 2 → ℝ × ℝ

value:
fun (u v p : JGS23SupplySideRecommenderSystems.Content 2) =>
  (JGS23SupplySideRecommenderSystems.score u p, JGS23SupplySideRecommenderSystems.score v p)
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.score

JGS23SupplySideRecommenderSystems.twoUserScoreFeasibleSet

Declaration location: paper-local

Lean declaration body

```
type:
JGS23SupplySideRecommenderSystems.Content 2 → JGS23SupplySideRecommenderSystems.Content 2 → ℝ × ℝ → Prop

value:
fun (u v : JGS23SupplySideRecommenderSystems.Content 2) (a : ℝ × ℝ) =>
  (JGS23SupplySideRecommenderSystems.twoUserScoreMap u v ''
      {p : JGS23SupplySideRecommenderSystems.Content 2 | JGS23SupplySideRecommenderSystems.NonnegativeContent p})
    a
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.twoUserScoreMap

JGS23SupplySideRecommenderSystems.twoUserSecondDerivSignBracket

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℝ → ℝ

value:
fun (β θ φ : ℝ) => (β - 2) / β * Real.cos (θ - 2 * φ) - Real.cos θ
```

JGS23SupplySideRecommenderSystems.paper_two_user_second_deriv_sign_bracket

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℝ → ℝ

value:
fun (β θ φ : ℝ) => JGS23SupplySideRecommenderSystems.twoUserSecondDerivSignBracket β θ φ
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.twoUserSecondDerivSignBracket

JGS23SupplySideRecommenderSystems.winningProducers

Declaration location: paper-local

Lean declaration body

```
type:
{D P : ℕ} →
  JGS23SupplySideRecommenderSystems.Content D → (Fin P → JGS23SupplySideRecommenderSystems.Content D) → Finset (Fin P)

value:
fun {D P : ℕ} (u : JGS23SupplySideRecommenderSystems.Content D)
    (profile : Fin P → JGS23SupplySideRecommenderSystems.Content D) =>
  {j : Fin P |
    ∀ (k : Fin P),
      JGS23SupplySideRecommenderSystems.score u (profile k) ≤ JGS23SupplySideRecommenderSystems.score u (profile j)}
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.score

JGS23SupplySideRecommenderSystems.tieShare

Declaration location: paper-local

Lean declaration body

```
type:
{D P : ℕ} →
  JGS23SupplySideRecommenderSystems.Content D → (Fin P → JGS23SupplySideRecommenderSystems.Content D) → Fin P → ℝ

value:
fun {D P : ℕ} (u : JGS23SupplySideRecommenderSystems.Content D)
    (profile : Fin P → JGS23SupplySideRecommenderSystems.Content D) (j : Fin P) =>
  if j ∈ JGS23SupplySideRecommenderSystems.winningProducers u profile then
    (↑(JGS23SupplySideRecommenderSystems.winningProducers u profile).card)⁻¹
  else 0
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.winningProducers

JGS23SupplySideRecommenderSystems.usersWon

Declaration location: paper-local

Lean declaration body

```
type:
{D N P : ℕ} →
  (Fin N → JGS23SupplySideRecommenderSystems.Content D) →
    (Fin P → JGS23SupplySideRecommenderSystems.Content D) → Fin P → ℝ

value:
fun {D N P : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (profile : Fin P → JGS23SupplySideRecommenderSystems.Content D) (j : Fin P) =>
  ∑ i : Fin N, JGS23SupplySideRecommenderSystems.tieShare (users i) profile j
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.tieShare

JGS23SupplySideRecommenderSystems.ExpectedUsersWonAgainstSymmetricMixed

Declaration location: paper-local

Lean declaration body

```
type:
{D N P : ℕ} →
  (Fin N → JGS23SupplySideRecommenderSystems.Content D) →
    Fin P →
      JGS23SupplySideRecommenderSystems.Content D →
        MeasureTheory.Measure (JGS23SupplySideRecommenderSystems.Content D) → ℝ

value:
fun {D N P : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D) (j : Fin P)
    (p : JGS23SupplySideRecommenderSystems.Content D)
    (μ : MeasureTheory.Measure (JGS23SupplySideRecommenderSystems.Content D)) =>
  ∫ (profile : Fin P → JGS23SupplySideRecommenderSystems.Content D),
    JGS23SupplySideRecommenderSystems.usersWon users (JGS23SupplySideRecommenderSystems.deviateProfile profile j p)
      j ∂MeasureTheory.Measure.pi fun (x : Fin P) => μ
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.deviateProfile
- JGS23SupplySideRecommenderSystems.usersWon

JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash

Declaration location: paper-local

Lean declaration body

```
type:
{D N : ℕ} →
  {P : ℕ} →
    (Fin N → JGS23SupplySideRecommenderSystems.Content D) →
      (JGS23SupplySideRecommenderSystems.Content D → ℝ) →
        JGS23SupplySideRecommenderSystems.MixedContentStrategy D → Prop

value:
fun {D N P : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (cost : JGS23SupplySideRecommenderSystems.Content D → ℝ)
    (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) =>
  MeasureTheory.IsProbabilityMeasure μ ∧
    MeasureTheory.Measure.support μ ⊆
        {p : JGS23SupplySideRecommenderSystems.Content D | JGS23SupplySideRecommenderSystems.NonnegativeContent p} ∧
      ∀ (j : Fin P),
        ∀ p ∈ MeasureTheory.Measure.support μ,
          ∀ (q : JGS23SupplySideRecommenderSystems.Content D),
            JGS23SupplySideRecommenderSystems.NonnegativeContent q →
              JGS23SupplySideRecommenderSystems.SourceMixedPurePayoff
                  (JGS23SupplySideRecommenderSystems.ExpectedUsersWonAgainstSymmetricMixed users j) cost q μ ≤
                JGS23SupplySideRecommenderSystems.SourceMixedPurePayoff
                  (JGS23SupplySideRecommenderSystems.ExpectedUsersWonAgainstSymmetricMixed users j) cost p μ
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.ExpectedUsersWonAgainstSymmetricMixed
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.SourceMixedPurePayoff

JGS23SupplySideRecommenderSystems.pureProfit

Declaration location: paper-local

Lean declaration body

```
type:
{D N P : ℕ} →
  (Fin N → JGS23SupplySideRecommenderSystems.Content D) →
    (JGS23SupplySideRecommenderSystems.Content D → ℝ) →
      (Fin P → JGS23SupplySideRecommenderSystems.Content D) → Fin P → ℝ

value:
fun {D N P : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (cost : JGS23SupplySideRecommenderSystems.Content D → ℝ)
    (profile : Fin P → JGS23SupplySideRecommenderSystems.Content D) (j : Fin P) =>
  JGS23SupplySideRecommenderSystems.usersWon users profile j - cost (profile j)
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.usersWon

JGS23SupplySideRecommenderSystems.PureNashEquilibrium

Declaration location: paper-local

Lean declaration body

```
type:
{D N P : ℕ} →
  (Fin N → JGS23SupplySideRecommenderSystems.Content D) →
    (JGS23SupplySideRecommenderSystems.Content D → ℝ) → (Fin P → JGS23SupplySideRecommenderSystems.Content D) → Prop

value:
fun {D N P : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (cost : JGS23SupplySideRecommenderSystems.Content D → ℝ)
    (profile : Fin P → JGS23SupplySideRecommenderSystems.Content D) =>
  (∀ (j : Fin P), JGS23SupplySideRecommenderSystems.NonnegativeContent (profile j)) ∧
    ∀ (j : Fin P) (p : JGS23SupplySideRecommenderSystems.Content D),
      JGS23SupplySideRecommenderSystems.NonnegativeContent p →
        JGS23SupplySideRecommenderSystems.pureProfit users cost
            (JGS23SupplySideRecommenderSystems.deviateProfile profile j p) j ≤
          JGS23SupplySideRecommenderSystems.pureProfit users cost profile j
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.deviateProfile
- JGS23SupplySideRecommenderSystems.pureProfit

JGS23SupplySideRecommenderSystems.paper_pure_nash_equilibrium

Declaration location: paper-local

Lean declaration body

```
type:
{D N P : ℕ} →
  (Fin N → JGS23SupplySideRecommenderSystems.Content D) →
    (JGS23SupplySideRecommenderSystems.Content D → ℝ) → (Fin P → JGS23SupplySideRecommenderSystems.Content D) → Prop

value:
fun {D N P : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (cost : JGS23SupplySideRecommenderSystems.Content D → ℝ)
    (profile : Fin P → JGS23SupplySideRecommenderSystems.Content D) =>
  JGS23SupplySideRecommenderSystems.PureNashEquilibrium users cost profile
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.PureNashEquilibrium

Paper-specific semantic prerequisites

JGS23SupplySideRecommenderSystems.poweredScoreGeometrySpec

Paper-local declaration:
JGS23SupplySideRecommenderSystems.poweredScoreGeometrySpec
Source locator: cited publication, lines 409-414

Verbatim paper-source connection

```
We first provide a tight geometric characterization of when a marketplace is in the single-genre regime versus in the multi-genre regime. More formally, let
↪   $\UMatrix = [\u_1; \cdots; \u_N]$ be the $N \times D$ matrix of user vectors, and let $\Set$ denote the image of the unit ball under $\UMatrix$:
\begin{equation}
  \Set := \left\{\UMatrix\p \mid \|\p\| \le 1, \p \in \bR_{\ge 0}^D  \right\}
\end{equation}
Each element of $\Set$ is an $N$-dimensional vector, which represents the inferred user values for some unit-norm producer $\p$. Additionally, let
↪   $\Set^{\beta}$ be the image of $\Set$ under coordinate-wise powers, i.e.~if $(z_1, \ldots, z_N) \in \Set$ then $(z_1^{\beta}, \ldots, z_N^{\beta}) \in
↪   \Set^{\beta}$.
We show that genres emerge when $\Set^{\beta}$ is sufficiently different from its convex hull $\bar{\Set}^{\beta}$:
```

Lean-expanded paper semantic target

```
type:
{D N : ℕ} →
  (Fin N → JGS23SupplySideRecommenderSystems.Content D) →
    JGS23SupplySideRecommenderSystems.SourceNorm D → ℝ → (Fin N → ℝ) → Prop

value:
fun {D N : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (ν : JGS23SupplySideRecommenderSystems.SourceNorm D) (β : ℝ) (a : Fin N → ℝ) =>
  {y : Fin N → ℝ |
      JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
        (fun (p : JGS23SupplySideRecommenderSystems.Content D) =>
          JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧ ν.norm p ≤ 1)
        β y}
    a
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.paper_powered_unit_image

Recorded source-to-declaration screening **Verdict:** matches

Reason: It is exactly the coordinatewise beta-powered image of scores from nonnegative content in the source norm unit ball; the set-comprehension form preserves the source user-matrix image.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec

Paper-local declaration:
JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec
Source locator: cited publication, lines 436-438

Verbatim paper-source connection

```
Theorem \ref{thm:singlegenre} further shows that the boundary between the single-genre and multi-genre regimes can be represented by a
↪   \textit{threshold} defined as follows
\[\beta^* := \sup \left\{\beta \ge 1 \mid \max_{y \in \Set^{\beta}} \prod_{i=1}^{N} y_i  = \max_{y \in \bar{\Set}^{\beta}} \prod_{i=1}^{N} y_i \right\}.
↪   \]
where single-genre equilibria exist exactly when $\beta \leq \beta^*$. Conceptually, larger $\beta$ make producer costs more superlinear, which eventually
↪   discentivizes producers from attempting to perform well on all dimensions at once.
```

Lean-expanded paper semantic target

```
type:
{D N : ℕ} →
  (Fin N → JGS23SupplySideRecommenderSystems.Content D) → JGS23SupplySideRecommenderSystems.SourceNorm D → WithTop ℝ

value:
fun {D N : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (ν : JGS23SupplySideRecommenderSystems.SourceNorm D) =>
  sSup
    ((fun (β : ℝ) => ↑β) ''
      {β : ℝ |
        1 ≤ β ∧
          JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition
            (JGS23SupplySideRecommenderSystems.poweredScoreGeometrySpec users ν β)})
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.poweredScoreGeometrySpec

Recorded source-to-declaration screening **Verdict:** matches

Reason: The source defines beta-star as the supremum of real exponents beta >= 1 for which the maximum coordinate product over S^beta equals that over its convex hull. The expanded target takes sSup of exactly the embedded real beta satisfying 1 <= beta and SingleGenreProductSupCondition on poweredScoreGeometrySpec. The displayed prerequisites make that condition equality of the coordinate-product suprema on the powered image and its real convex hull; define the product as the finite product of coordinates; and define the powered image from score rows U p with p nonnegative, norm at most one, followed by coordinatewise beta powers. Thus it specializes to the complete source formula on the source norm and user model. The WithTop codomain preserves every finite source threshold and represents an unbounded exponent set by top, without excluding or changing any source-admissible finite case.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Paper-local declaration:
JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec
Source locator: cited publication, lines 291-318

Verbatim paper-source connection

```
\subsection{Equilibrium concept and existence of equilibrium}
\label{sec:equilibrium-def}


We study the Nash equilibria of the game between producers. In particular, each producer $j$ chooses a (random) strategy over content, given by a probability measure $\mu_j$ over the content embedding space $\bR_{\geq 0}^D$. The strategies $(\mu_1, \ldots, \mu_P)$ form a \textit{Nash equilibrium} if no producer---given the strategies of other producers---can chose a different strategy where they achieve higher expected profit: that is, for every $j \in [P]$ and every $\p_j \in \text{supp}(\mu_j)$, it holds that $\p_j \in \argmax_{\p \in \mathbb{R}_{\ge 0}^D} \mathbb{E}_{\p_{-j} \sim \mu_{-j}}[\Profit(\p; \p_{-j})]$. A salient feature of our model is that there are  discontinuities in the producer utility function in equation \eqref{eq:utility}, since the function $\argmax_{1 \le i \le P} \langle u_i, \p_j \rangle$ changes discontinuously with the producer vectors $\p_j$. Due to these discontinuities, pure strategy equilibria do not exist.\footnote{A Nash equilibrium $(\mu_{1}, \mu_2, \ldots, \mu_P)$ is a \textit{pure strategy equilibrium} if each $\mu_j$ contains only one vector in its support; otherwise, it is a \textit{mixed strategy equilibrium}.}
\begin{restatable}{proposition}{pure}
\label{prop:pure}
For any set of users and any $\beta \ge 1$, a pure strategy equilibrium does not exist.
\end{restatable}
\noindent The intuition is that if two producers are tied, then a producer can increase their utility by infinitesimally increasing the magnitude of their content.


Since pure strategy equilibria do not exist, we must turn to mixed strategy equilibria. Using the technology of equilibria in discontinuous games \citep{reny99}, we show that a mixed strategy equilibrium exists. In fact, because of the symmetries in the producer utility functions, we can actually show that a
\textit{symmetric} mixed strategy equilibrium  (i.e. an equilibrium where $\mu_1 = \ldots = \mu_P$) exists.
\begin{restatable}{proposition}{existence}
\label{prop:existence}
For any set of users and any $\beta \ge 1$, a symmetric mixed equilibrium exists.
\end{restatable}

Interestingly, symmetric mixed equilibria must exhibit significant randomness across different content embeddings. (Note that every symmetric equilibrium must exhibits some randomization, since pure strategy equilibria do not exist.) In particular, we show that a symmetric mixed equilibrium cannot contain point masses.
\begin{proposition}
\label{prop:atom}
For any set of users and any $\beta \ge 1$, every symmetric mixed equilibrium is atomless.
\end{proposition}
\noindent Proposition \ref{prop:atom} implies that a symmetric mixed equilibrium has \textit{infinite support}. The randomness can come from randomness over \textit{quality} $\|\p\|$ as well as randomness over \textit{genres} $p/\|p\|$.


We take the symmetric mixed equilibria of this game as the main object of our study, since they are both tractable to analyze and rich enough to capture asymmetric solution concepts such as specialization. In terms of tractability, a symmetric mixed equilibrium (unlike an asymmetric equilibrium) can be represented as a \textit{single} distribution $\mu$. Despite this simplicity, we can still study specialization---which is an asymmetric concept---within the family of symmetric equilibria as we formalize in Section \ref{sec:specializationdef}.
```

Lean-expanded paper semantic target

```
type:
{D N : ℕ} →
  {P : ℕ} →
    (Fin N → JGS23SupplySideRecommenderSystems.Content D) →
      (JGS23SupplySideRecommenderSystems.Content D → ℝ) →
        JGS23SupplySideRecommenderSystems.MixedContentStrategy D → Prop

value:
fun {D N P : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (cost : JGS23SupplySideRecommenderSystems.Content D → ℝ)
    (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) =>
  MeasureTheory.IsProbabilityMeasure μ ∧
    MeasureTheory.Measure.support μ ⊆
        {p : JGS23SupplySideRecommenderSystems.Content D | JGS23SupplySideRecommenderSystems.NonnegativeContent p} ∧
      ∀ (j : Fin P),
        ∀ p ∈ MeasureTheory.Measure.support μ,
          ∀ (q : JGS23SupplySideRecommenderSystems.Content D),
            JGS23SupplySideRecommenderSystems.NonnegativeContent q →
              JGS23SupplySideRecommenderSystems.SourceMixedPurePayoff
                  (JGS23SupplySideRecommenderSystems.ExpectedUsersWonAgainstSymmetricMixed users j) cost q μ ≤
                JGS23SupplySideRecommenderSystems.SourceMixedPurePayoff
                  (JGS23SupplySideRecommenderSystems.ExpectedUsersWonAgainstSymmetricMixed users j) cost p μ
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.ExpectedUsersWonAgainstSymmetricMixed
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.SourceMixedPurePayoff

Recorded source-to-declaration screening **Verdict:** matches

Reason: One common probability law is supported on nonnegative content, and every support action weakly beats every nonnegative deviation against independent symmetric opponents under the tie-aware expected payoff, matching source symmetric mixed Nash.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

JGS23SupplySideRecommenderSystems.equilibriumProfitModelSpec

Paper-local declaration:
JGS23SupplySideRecommenderSystems.equilibriumProfitModelSpec
Source locator: cited publication, lines 713-716

Verbatim paper-source connection

```
More formally, we can quantify the producer profit at equilibrium as follows. At a symmetric equilibria $\mu$, all producers receive the same expected
↪   profit given by:
\begin{equation}
  \ProfitEq(\mu) := \mathbb{E}_{\p_1, \ldots, \p_P \sim \mu} [\Profit(\p_1; \p_{-1})] =  \mathbb{E} \Big[ \Big(\sum_{i=1}^N \Indicator[j^*(\u_i;\;\p_{1:P}) =
  ↪   1]\Big) - \|\p_1\|^{\beta}  \Big],
\end{equation}

[Next verbatim source excerpt]

Proposition \ref{thm:utility} provides insight into how the geometry of the users and structure of producer costs impact whether producers can achieve
↪   positive profit. To interpret Proposition \ref{thm:utility}, let us examine the quantity $Q := \max_{\|\p\| \le 1} \min_{i=1}^N  \langle \p,
↪   \frac{u_i}{||u_i||} \rangle$ that appears on the left-hand side of \eqref{eq:positiveprofit}. Intuitively, $Q$ captures how easy it is to produce content
↪   that appeals simultaneously to all users. It is larger when the users are close together and smaller when they are spread out. For any set of vectors we
↪   see that $Q \le 1$, with strict inequality if the set of vectors is non-degenerate. The right-hand side of \eqref{eq:positiveprofit}, on the other hand,
↪   goes to $1$ as $\beta \rightarrow \infty$. Thus, for any non-degenerate set of users, if $\beta$ is sufficiently large, the condition in
↪   Proposition~\ref{thm:utility} is met and producer profit is strictly positive. The value of $\beta$ at which positive profit is guaranteed by Proposition
↪   \ref{thm:utility} decreases as the user vectors become more spread out.
```

Lean-expanded paper semantic target

```
type:
{D N : ℕ} →
  {P : ℕ} →
    (Fin N → JGS23SupplySideRecommenderSystems.Content D) →
      JGS23SupplySideRecommenderSystems.SourceNorm D →
        (JGS23SupplySideRecommenderSystems.Content D → ℝ) →
          JGS23SupplySideRecommenderSystems.MixedContentStrategy D → ℝ → ℝ → Prop

value:
fun {D N P : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (ν : JGS23SupplySideRecommenderSystems.SourceNorm D) (cost : JGS23SupplySideRecommenderSystems.Content D → ℝ)
    (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) (profit Q : ℝ) =>
  JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users cost μ ∧
    (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)) ∧
      (∀ (j : Fin P),
          ∫ (p : JGS23SupplySideRecommenderSystems.Content D),
              JGS23SupplySideRecommenderSystems.SourceMixedPurePayoff
                (JGS23SupplySideRecommenderSystems.ExpectedUsersWonAgainstSymmetricMixed users j) cost p μ ∂μ =
            profit) ∧
        Q =
          sSup
            {q : ℝ |
              ∃ (p : JGS23SupplySideRecommenderSystems.Content D),
                JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧
                  ν.norm p ≤ 1 ∧
                    ∀ (i : Fin N), q ≤ JGS23SupplySideRecommenderSystems.score p (ν.normalizedContent (users i))}
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.ExpectedUsersWonAgainstSymmetricMixed
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SourceMixedPurePayoff
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.SourceNorm.normalizedContent
- JGS23SupplySideRecommenderSystems.score
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Recorded source-to-declaration screening **Verdict:** matches

Reason: It retains common symmetric-equilibrium expected payoff and the nonnegative unit-ball max-min alignment quantity Q; sSup records the same source value without silently selecting a maximizer.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

JGS23SupplySideRecommenderSystems.globalMarketModelSpec

Paper-local declaration:
JGS23SupplySideRecommenderSystems.globalMarketModelSpec
Source locator: cited publication, lines 265-285

Verbatim paper-source connection

```
We introduce a game-theoretic model for supply-side competition in recommender systems. Consider a platform with $N \ge 1$ heterogeneous users who are offered personalized recommendations and $P \ge 2$ producers who strategically decide what digital good to create.


\paragraph{Embeddings of users and digital goods.}
Each user $i$ is associated with a $D$-dimensional embedding $\u_i$ that captures their preferences. We assume that $u_i \in \bR_{\geq 0}^D \setminus \left\{\vec{0} \right\}$---i.e., the coordinates of each embedding are nonnegative and each embedding is nonzero. While user vectors are fixed, producers \textit{choose} what content to create. Each producer $j$ creates a single digital good, which is associated with a content vector $\p_j \in \bR_{\geq 0}^D$. The inferred user value of good $\p$ for user $u$ is $\langle u, p \rangle$.

\paragraph{Personalized recommendations.}
After the producers decide what content to create, the platform offers personalized recommendations to each user. We consider a stylized model where the platform has complete knowledge of the user and content vectors. The platform recommends to each user the content with the maximal inferred user value for them, assigning them to the producer who created this content. Mathematically, the platform assigns a user $\u$ to the producer $j^*$, where $j^*(\u;\;\p_{1:P}) = \argmax_{1 \le j \le P} \langle \u, \p_j\rangle$. If there are ties, the platform sets $j^*(\u;\;\p_{1:P})$ to be a producer chosen uniformly at random from the argmax.

\paragraph{Producer cost function.} Each producer faces a \textit{fixed (one-time) cost} for producing content $\p$, which depends on the magnitude of $\p$. Since the good is digital and thus cheap to replicate, the production cost does not scale with the number of users.
We assume that the cost function $c(\p)$ takes the form $\|\p\|^{\beta}$, where $\|\cdot\|$ is any norm and the exponent $\beta$ is at least $1$. The magnitude $\|\p\|$ captures the \textit{quality} of the content: in particular, if a producer chooses content $\lambda \p$, they win at least as many users as if they choose $\lambda' \p$ for $\lambda' < \lambda$. (This relies on the fact that all vectors are in the positive orthant.) The norm and $\beta$ together encode the cost of producing a content vector $v$, and reflect cost tradeoffs for excelling in different dimensions (for example, producing a movie that is both a drama and a comedy). Large $\beta$, for instance, means that this cost grows superlinearly. In Section \ref{sec:specialization}, we will see that these tradeoffs capture the extent to which producers are incentivized to specialize.


\paragraph{Producer profit.} A producer receives profit equal to the number of users who are recommended their content minus the cost of producing the content. The profit of producer $j$ is equal to:
\begin{equation}
\label{eq:utility}
 \Profit(\p_j; \p_{-j}) =  \mathbb{E}\Big[ \Big(\sum_{i=1}^n \Indicator[j^*(u_i;\; \p_{1:P}) = j]\Big) - \|\p_j\|^{\beta}  \Big],
\end{equation}
where
$\p_{-j} = [\p_1, \ldots, \p_{j-1}, \p_{j+1}, \ldots, \p_P]$ denotes the content produced by all of the other producers and
where the expectation comes from the randomness over platform recommendations in the case of ties.
```

Lean-expanded paper semantic target

```
type:
{D N : ℕ} →
  {P : ℕ} →
    [Nonempty (Fin N)] →
      (Fin N → JGS23SupplySideRecommenderSystems.Content D) →
        JGS23SupplySideRecommenderSystems.SourceNorm D →
          ℝ → JGS23SupplySideRecommenderSystems.MixedContentStrategy D → Prop

value:
fun {D N P : ℕ} [Nonempty (Fin N)] (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (ν : JGS23SupplySideRecommenderSystems.SourceNorm D) (β : ℝ)
    (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) =>
  2 ≤ P ∧
    1 ≤ β ∧
      (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) ∧
        (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)) ∧
          JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
            (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Recorded source-to-declaration screening **Verdict:** matches

Reason: It preserves the finite market's nonzero nonnegative users, at least two producers, norm-power cost, uniform-tie payoff, nonnegative content, and symmetric mixed Nash condition. The source-norm regularity convention is explicitly supplied.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

JGS23SupplySideRecommenderSystems.infiniteGenreDefinitionSpec

Paper-local declaration:
JGS23SupplySideRecommenderSystems.infiniteGenreDefinitionSpec
Source locator: cited publication, lines 1747-1754

Verbatim paper-source connection

```
\begin{definition}[Finite-genre equilibria for $P = \infty$]
\label{def:infinitegenre}
Let $u_1, \ldots, u_N \in \mathbb{R}_{\ge 0}^D$ be a set of users and let $c(\p) = \|p\|_2^{\beta}$ be the cost function. A set of \textit{genres} $d_1,
↪    \ldots, d_G \in \mathbb{R}_{\ge 0}^D$ such that $\|d_i\|_2 = 1$ for all $1 \le g \le G$, a set of \textit{conditional quality distributions} $F_1, F_2,
↪    \ldots, F_G$ over $\mathbb{R}_{\ge 0}$, and a set of weights $\alpha_1, \ldots, \alpha_G \ge 0$ such that $\sum_{g=1}^G \alpha_g = 1$ forms a
↪    finite-genre equilibrium if the following condition holds for
\begin{equation}
\label{eq:infinitecondition}
\p^* \in \argmax_{\p \in \mathbb{R}_{\ge 0}^D} \left(\sum_{i=1}^N \left(\prod_{g=1}^G \left(\FMax_g \left(\frac{\langle u_i, \p \rangle}{\langle u_i, d_g
↪    \rangle} \right)\right)^{\alpha_i} \right) - c(\p) \right)
\end{equation}
for any $\p^* = q_i d_i$ such that $1 \le i \le G$ and $q_i \in \text{supp}(F_i)$.
```

Formalized review target The archival source above is not asserted equivalent to this formalized target.

```
The corrected finite-genre infinite-producer definition uses normalized nonnegative genre weights, conditional quality CDFs, and support-wise global
↪    best-response optimality in one coherent predicate.
```

Archival source anchor: cited publication, lines 1747-1754

Lean-expanded paper semantic target

```
type:
{D N G : ℕ} →
  (Fin N → JGS23SupplySideRecommenderSystems.Content D) →
    (JGS23SupplySideRecommenderSystems.Content D → ℝ) →
      (Fin G → JGS23SupplySideRecommenderSystems.Content D) →
        (Fin G → MeasureTheory.Measure ℝ) → (Fin G → ℝ → ℝ) → (Fin G → ℝ) → Prop

value:
fun {D N G : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (cost : JGS23SupplySideRecommenderSystems.Content D → ℝ)
    (genres : Fin G → JGS23SupplySideRecommenderSystems.Content D)
    (conditionalQuality : Fin G → MeasureTheory.Measure ℝ) (maximumQualityCdf : Fin G → ℝ → ℝ) (weights : Fin G → ℝ) =>
  (∀ (g : Fin G),
      JGS23SupplySideRecommenderSystems.NonnegativeContent (genres g) ∧
        (JGS23SupplySideRecommenderSystems.SourceNorm.l2 D).norm (genres g) = 1) ∧
    (∀ (g : Fin G),
        MeasureTheory.IsProbabilityMeasure (conditionalQuality g) ∧
          (∀ (q : ℝ), (conditionalQuality g) (Set.Iic q) = ENNReal.ofReal (maximumQualityCdf g q)) ∧
            (conditionalQuality g).support ⊆ Set.Ici 0) ∧
      (∀ (g : Fin G), 0 ≤ weights g) ∧
        ∑ g : Fin G, weights g = 1 ∧
          ∀ (g : Fin G) {{r : ℝ}},
            r ∈ (conditionalQuality g).support →
              JGS23SupplySideRecommenderSystems.scaleContent r (genres g) ∈
                  {p : JGS23SupplySideRecommenderSystems.Content D |
                    JGS23SupplySideRecommenderSystems.NonnegativeContent p} ∧
                IsMaxOn
                  (JGS23SupplySideRecommenderSystems.infiniteGenreContentObjectiveSpec users cost genres
                    maximumQualityCdf weights)
                  {p : JGS23SupplySideRecommenderSystems.Content D |
                    JGS23SupplySideRecommenderSystems.NonnegativeContent p}
                  (JGS23SupplySideRecommenderSystems.scaleContent r (genres g))
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.SourceNorm.l2
- JGS23SupplySideRecommenderSystems.infiniteGenreContentObjectiveSpec
- JGS23SupplySideRecommenderSystems.scaleContent

Recorded source-to-declaration screening **Verdict:** matches_approved_corrected_target

Reason: The archival display has inconsistent CDF names and indexes the product exponent by the user rather than the genre, so it is not a raw-source match. The approval-pinned corrected target is realized by nonnegative L2-unit genres, nonnegative-support probability quality laws with their CDFs, nonnegative weights summing to one, and support-wise global best responses for the product objective weighted by the genre weights.
Reviewer check: ☐ Matches formalized target
If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

JGS23SupplySideRecommenderSystems.infiniteProducerModelSpec

Paper-local declaration:
JGS23SupplySideRecommenderSystems.infiniteProducerModelSpec
Source locator: cited publication, lines 1725-1745

Verbatim paper-source connection

```
Since our characterization result (Theorem \ref{thm:infinitegenre}) focuses on finite-genre equilibria,  we restrict our formal definition of the
↪    infinite-producer limit to case of finite genres for technical convenience.

We arrive at a formalism by taking a limit of the conditions  in Lemma \ref{claim:necessarysuff} as $P \rightarrow \infty$. Let $\mu$ be a finite-genre
↪    distribution over $\mathbb{R}_{\ge 0}^D$. We can specify $\mu$ by the three attributes: the genres $d_1, \ldots, d_G$, the distributions $F_g$ over
↪    $\mathbb{R}_{\ge 0}$ corresponding to the distribution of $\|p\|$ for $p$ drawn from $\mu$ conditioned on $\p$ pointing in the direction of $d_g$,
↪    and the weights $\alpha_g$ corresponding to the probability that $\p \sim \mu$ points in the direction of $d_g$. In particular, $\mu$ can be described
↪    as follows: with probability $\alpha_g$, choose the vector $q_g d_g$ where $q_g$ is drawn from a distribution with cdf $F_g$. We see that the
↪    corresponding function $H_i$ from Lemma \ref{claim:necessarysuff} will be equal to:
\[H_i(z_i) = \left(\sum_{g=1}^G \alpha_g F_g\left(\frac{\langle u_i, \p \rangle}{\langle u_i, d_g \rangle} \right) \right)^{P-1}. \]
Note that the conditions (C2) and (C3) are essentially satisfied by construction; condition (C1) requires that
\[\max_{\p \in \mathbb{R}_{\ge 0}^D} \left(\sum_{i=1}^N H_i(\langle u_i, \p\rangle) - c(\p) \right)\] is maximized for any $\p \in \text{supp}(\mu)$. This
↪    can be rewritten as requiring that any $\p^* \in \text{supp}(\mu)$ satisfies:
\[
  \p^* \in \argmax_{\p \in \mathbb{R}_{\ge 0}^D} \left(\sum_{i=1}^N \left(\sum_{g=1}^G \alpha_g F_g\left(\frac{\langle u_i, \p \rangle}{\langle u_i, d_g
  ↪    \rangle} \right) \right)^{P-1} - c(\p) \right).\]
Let's rewrite this in terms of winning producers: more formally, let $\FMax_g(\cdot) = (F_g(\cdot))^{P-1}$ denote the cumulative distribution function of the
↪    \textit{maximum} quality in a genre, conditioned on all producers choosing that genre. We call the distributions $\FMax_1, \ldots, \FMax_G$ the
↪    \textit{conditional quality distributions}. Then we obtain the following:
\begin{equation}
\label{eq:finitegenrep}
  \p^* \in \argmax_{\p \in \mathbb{R}_{\ge 0}^D} \left(\sum_{i=1}^N \left(\sum_{g=1}^G \alpha_g \left(\FMax_g\left(\frac{\langle u_i, \p \rangle}{\langle
  ↪    u_i, d_g \rangle} \right)\right)^{1/(P-1)} \right)^{P-1} - c(\p) \right).
\end{equation}
Taking a limit as $P \rightarrow \infty$, we see that
\[\left(\sum_{g=1}^G \alpha_g F_g\left(\frac{\langle u_i, \p \rangle}{\langle u_i, d_g \rangle} \right)^{1/(P-1)}\right)^{P-1} \rightarrow \prod_{g=1}^G
↪    \left(F_g\left(\frac{\langle u_i, \p \rangle}{\langle u_i, d_g \rangle} \right)\right)^{\alpha_i}.\]
Thus, equation \ref{eq:finitegenrep} (informally speaking) approaches the following condition in the limit:
\begin{equation}
\label{eq:infinitegenrecondition}
   \p^* \in \argmax_{\p \in \mathbb{R}_{\ge 0}^D} \left(\sum_{i=1}^N \left(\sum_{g=1}^G \alpha_g F_g\left(\frac{\langle u_i, \p \rangle}{\langle u_i, d_g
   ↪    \rangle} \right)^{1/(P-1)} \right)^{P-1} - c(\p) \right).
\end{equation}
```

Formalized review target The archival source above is not asserted equivalent to this formalized target.

```
The corrected finite-genre infinite-producer model uses normalized nonnegative genre weights, conditional quality CDFs, and support-wise global
↪    best-response optimality in one coherent predicate.
```

Archival source anchor: cited publication, lines 1725-1745

Lean-expanded paper semantic target

```
type:
{D N G : ℕ} →
  (Fin N → JGS23SupplySideRecommenderSystems.Content D) →
    (JGS23SupplySideRecommenderSystems.Content D → ℝ) →
      (Fin G → JGS23SupplySideRecommenderSystems.Content D) →
        (Fin G → MeasureTheory.Measure ℝ) → (Fin G → ℝ → ℝ) → (Fin G → ℝ) → Prop

value:
fun {D N G : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (cost : JGS23SupplySideRecommenderSystems.Content D → ℝ)
    (genres : Fin G → JGS23SupplySideRecommenderSystems.Content D)
    (conditionalQuality : Fin G → MeasureTheory.Measure ℝ) (maximumQualityCdf : Fin G → ℝ → ℝ) (weights : Fin G → ℝ) =>
  JGS23SupplySideRecommenderSystems.infiniteGenreDefinitionSpec users cost genres conditionalQuality maximumQualityCdf
    weights
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.infiniteGenreDefinitionSpec

Recorded source-to-declaration screening **Verdict:** matches_approved_corrected_target

Reason: The source describes a finite-genre distribution by nonnegative genre directions, conditional radial/maximum-quality laws, probability weights, and support-wise maximization, but its limiting display has index/weight inconsistencies. The displayed approved correction calls for the coherent normalized model; the Lean target supplies exactly those probability laws, normalized nonnegative genres and weights, genre-indexed product objective, and support-wise global best responses.
Reviewer check: ☐ Matches formalized target
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

JGS23SupplySideRecommenderSystems.qualityAndGenreModelSpec

Paper-local declaration:
JGS23SupplySideRecommenderSystems.qualityAndGenreModelSpec
Source locator: cited publication, lines 274-275

Verbatim paper-source connection

```
\paragraph{Producer cost function.} Each producer faces a \textit{fixed (one-time) cost} for producing content $\p$, which depends on the magnitude of
↪    $\p$. Since the good is digital and thus cheap to replicate, the production cost does not scale with the number of users.
We assume that the cost function $c(\p)$ takes the form $\|\p\|^{\beta}$, where $\|\cdot\|$ is any norm and the exponent $\beta$ is at least $1$. The
↪    magnitude $\|\p\|$ captures the \textit{quality} of the content: in particular, if a producer chooses content $\lambda \p$, they win at least as many
↪    users as if they choose $\lambda' \p$ for $\lambda' < \lambda$. (This relies on the fact that all vectors are in the positive orthant.) The norm and
↪    $\beta$ together encode the cost of producing a content vector $v$, and reflect cost tradeoffs for excelling in different dimensions (for example,
↪    producing a movie that is both a drama and a comedy). Large $\beta$, for instance, means that this cost grows superlinearly. In Section
↪    \ref{sec:specialization}, we will see that these tradeoffs capture the extent to which producers are incentivized to specialize.

[Next verbatim source excerpt]

To formalize specialization, we need to disentangle two forms of differentiation: (1) differentiation along direction (genre), and (2) differentiation along
↪    magnitude (quality). We define specialization as differentiation along genres, and not as differentiation along quality. To focus on the former, we define
↪    \textit{genres} as the set of \textit{directions} that arise at a symmetric mixed Nash equilibrium $\mu$:
\begin{equation}
\label{eq:genredef}
 \Genre(\mu) := \Big\{\frac{\p}{\|\p\|} \mid \p \in \text{supp}(\mu) \Big\},
\end{equation}
where we normalize by $\|\p\|$ to separate out the quality (norm) from the genre (direction). The set of genres $ \Genre(\mu)$ captures the set of content
↪    that may arise on the platform in some realization of randomness of the producers' strategies. When an equilibrium has a single genre, all producers
↪    cater to an average user, and only a single type of content appears  on the platform. On the other hand, when an equilibrium has multiple genres,
↪    many types of digital content are likely to appear on the platform.



We thus say that \textit{specialization} occurs at an equilibrium $\mu$ if and only if $\mu$ has more than one \textit{genre} (i.e., if and only if
↪    $|\Genre(\mu)| > 1$). When specialization does occur at $\mu$, the form of specialization is further captured by the number of genres $|\Genre(\mu)|$
↪    and other properties of the set of genres $\Genre(\mu)$. Note that we define specialization in terms of the \textit{support} of a symmetric mixed
↪    equilibrium distribution. In this definition, we implicitly interpret the randomness in the producer strategies as differentiation between producers; this
↪    formalization of specialization obviates the need to reason about asymmetric equilibria, thus making the model much more tractable to analyze.
```

Formalized review target   The archival source above is not asserted equivalent to this formalized target.

Quality is the source norm of nonzero content, and its genre is the corresponding nonzero-support normalized direction; specialization is defined only from
↪   those nonzero support genres.

Archival source anchor: cited publication, lines 369-378

Lean-expanded paper semantic target

```
type:
{D : ℕ} →
  JGS23SupplySideRecommenderSystems.SourceNorm D →
    JGS23SupplySideRecommenderSystems.MixedContentStrategy D → JGS23SupplySideRecommenderSystems.Content D → Prop

value:
fun {D : ℕ} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D)
    (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) (a : JGS23SupplySideRecommenderSystems.Content D) =>
  {genre : JGS23SupplySideRecommenderSystems.Content D |
      ∃ p ∈ MeasureTheory.Measure.support μ,
        JGS23SupplySideRecommenderSystems.NonzeroContent p ∧ genre = ν.normalizedContent p}
    a
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.SourceNorm.normalizedContent

Recorded source-to-declaration screening   **Verdict:** matches_approved_corrected_target
Reason: It implements the approval-pinned zero-safe convention: quality is the source norm and genres are normalized directions only of nonzero support content. It therefore does not claim the archival zero-division display is literal.
Reviewer check: ☐ Matches formalized target
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

JGS23SupplySideRecommenderSystems.reparameterizedEquilibriumModelSpec

Paper-local declaration:
JGS23SupplySideRecommenderSystems.reparameterizedEquilibriumModelSpec
Source locator: cited publication, lines 677-693

Verbatim paper-source connection

```
To prove our results in this section, our first step is establish a useful characterization of equilibria that enables us to separately account for the geometry of the users and the number of producers. This takes the form of necessary and sufficient conditions that decouple in terms of two quantities: a set of \textit{marginal distributions} $H_i$,
and the \textit{support} $S \subseteq \mathbb{R}_{\ge 0}^N$.
\begin{restatable}{lemma}{necessarysuff}
\label{claim:necessarysuff}
Let $\UMatrix = [u_1; u_2; \ldots; u_N]$ be the $N \times D$ matrix of users vectors. Given a set $S \subseteq \mathbb{R}_{\ge 0}^N$ and distributions $H_1, \ldots, H_N$ over $\mathbb{R}_{\ge 0}$, suppose that the following conditions hold:
\begin{description}
  \item{(C1)} Every $z^* \in S$ is a maximizer of the equation:
  \begin{equation}
\label{eq:reparam}
 \max_{z \in \mathbb{R}_{\ge 0}^D} \sum_{i=1}^N H_i(z_i) - c_{\UMatrix}(z),
\end{equation}
where $c_{\UMatrix}(z) := \min \left\{c(\p) \mid \p \in \mathbb{R}_{\ge 0}^D, \UMatrix p=z  \right\}$.
  \item{(C2)} There exists a random variable $Z$ with support $S$, such that the marginal distribution $Z_i$ has cdf equal to $H_i(z)^{1/(P-1)}$.
  \item{(C3)} $Z$ is distributed as $\UMatrix Y$ with $Y \sim \mu$, for some distribution $\mu$ over $\mathbb{R}_{\ge 0}^D$.
\end{description}
Then, the distribution $\mu$ from (C3) is a symmetric mixed Nash equilibrium. Moreover, every symmetric mixed Nash equilibrium $\mu$ is associated with some $(H_1, \ldots, H_N, S)$ that satisfy (C1)-(C3).
\end{restatable}

[Next verbatim source excerpt]

\begin{proof}[Proof of Lemma \ref{claim:necessarysuff}]
The intuition is that the the set $S$ captures the support of the realized inferred user values $[\langle u_1, \p \rangle, \ldots, \langle u_N, \p \rangle]$ for $\p \sim \mu$ and the distribution $H_i$ captures the distribution of the maximum inferred user value$\max_{1 \le j \le P -1} \langle u_i, \p_j\rangle$ for user $u_i$.

To formalize this, we reparameterize from content vectors in $\mathbb{R}_{\ge 0}^D$ to realized inferred user values in $\mathbb{R}_{\ge 0}^N$. That is, we transform the content vector $\p \in \mathbb{R}_{\ge 0}^D$ into the vector of realized inferred user values given by $z = \UMatrix p$. This reparameterization allows us to cleanly reason about the number of users that a producer wins: a producer wins a user $u_i$ if and only if they have the highest value in the $i$th coordinate of $z$. In this parametrization, the cost of production can be computed through an induced function $c_{\UMatrix}$ given by $c_{\UMatrix}(z) := \min \left\{c(\p) \mid \p \in \mathbb{R}_{\ge 0}^D, z = \UMatrix p  \right\}$ if $z \in \left\{\UMatrix p \mid \p \in \mathbb{R}_{\ge 0}^D\right\}$.

In this reparameterization, the producer profit takes a clean form. If producer $1$ chooses $z \in \mathbb{R}^N$, and other producers follow a distribution $\mu_Z$ over $\mathbb{R}^N$, then the expected profit of producer $1$ is:
\[
 \sum_{i=1}^N H_i(z_i) - c_{\UMatrix}(z),
\]
where $H_i(\cdot)$ is the cumulative distribution function of the {maximum realized inferred user value} over the other $P-1$ producers, i.e.~of the random variable $\max_{2 \le j \le P} (z_j)_i$ where $z_2, \ldots, z_{P} \sim \mu_Z$.


Recall that a distribution $\mu$ corresponds to a symmetric mixed Nash equilibrium if and only if every $z$ in the support $S:= \text{supp}(\mu_Z)$ is a maximizer of equation \eqref{eq:reparam} (where $\mu_Z$ is the distribution over $\UMatrix p$ for $p \sim \mu$).
```

Formalized review target
The archival source above is not asserted equivalent to this formalized target.

The reparameterized equilibrium model uses the explicit uniform-tie expected-winner objective, with score-level nullness wherever a CDF shortcut replaces tie-aware payoff probabilities.

Archival source anchor: cited publication, lines 677-693

Lean-expanded paper semantic target

```
type:
{D N : ℕ} →
  {P : ℕ} →
    (Fin N → JGS23SupplySideRecommenderSystems.Content D) →
      JGS23SupplySideRecommenderSystems.MixedContentStrategy D →
        (JGS23SupplySideRecommenderSystems.Content D → ℝ) → Prop

value:
fun {D N P : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D)
    (cost : JGS23SupplySideRecommenderSystems.Content D → ℝ) =>
  (∀ (i : Fin N) (z : ℝ),
      μ {q : JGS23SupplySideRecommenderSystems.Content D | JGS23SupplySideRecommenderSystems.score (users i) q = z} =
        0) ∧
    ∀ (j : Fin P) (p : JGS23SupplySideRecommenderSystems.Content D),
      JGS23SupplySideRecommenderSystems.SourceMixedPurePayoff
          (JGS23SupplySideRecommenderSystems.ExpectedUsersWonAgainstSymmetricMixed users j) cost p μ =
        ∑ i : Fin N,
            MeasureTheory.Measure.real μ
                {q : JGS23SupplySideRecommenderSystems.Content D |
                  JGS23SupplySideRecommenderSystems.score (users i) q <
                    JGS23SupplySideRecommenderSystems.score (users i) p} ^
              (P - 1) -
          cost p
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.ExpectedUsersWonAgainstSymmetricMixed
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.SourceMixedPurePayoff
- JGS23SupplySideRecommenderSystems.score

Recorded source-to-declaration screening **Verdict:** matches_approved_corrected_target

Reason: It uses the uniform-tie expected payoff and imposes score-fiber nullness before replacing it by the strict-CDF expression, exactly the authorized tie-aware repair rather than the archival C1--C3 statement.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

JGS23SupplySideRecommenderSystems.singleAndMultiGenreRegimesSpec

Paper-local declaration:
JGS23SupplySideRecommenderSystems.singleAndMultiGenreRegimesSpec
Source locator: cited publication, lines 395-402

Verbatim paper-source connection

```
\section{When does specialization occur?}\label{sec:specialization}

In order to investigate whether specialization occurs in a given marketplace, we investigate when the set of genres $\Genre(\mu)$ of an equilibrium $\mu$
↪    contains more than one direction. We distinguish between two regimes of marketplaces based on whether or not a single-genre equilibrium exists:
\begin{enumerate}
    \item A marketplace is in the \textit{single-genre regime} if there exists an equilibrium $\mu$ such that $|\Genre(\mu)| = 1$. All producers thus create
    ↪    content of the same genre.
    \item A marketplace is in the \textit{multi-genre regime} if all equilibria $\mu$ satisfy $|\Genre(\mu)| > 1$. Producers thus necessarily differentiate in
    ↪    the genre of content that they produce.
\end{enumerate}
To understand these regimes, we ask: \textit{what conditions on the user vectors $u_1, \ldots, u_N$ and the cost function parameters $\|\cdot \|$ and
↪    $\beta$ determine which regime the marketplace is in?}
```

Formalized review target The archival source above is not asserted equivalent to this formalized target.

```
A single-genre regime has an equilibrium with exactly one nonzero support genre, while a multi-genre regime requires every equilibrium to have a
↪    non-singleton nonzero-support genre set.
```

Archival source anchor: cited publication, lines 395-402

Lean-expanded paper semantic target

```
type:
{D N : ℕ} →
  {P : ℕ} →
    (Fin N → JGS23SupplySideRecommenderSystems.Content D) → JGS23SupplySideRecommenderSystems.SourceNorm D → ℝ → Prop

value:
fun {D N P : ℕ} (users : Fin N → JGS23SupplySideRecommenderSystems.Content D)
    (ν : JGS23SupplySideRecommenderSystems.SourceNorm D) (β : ℝ) =>
  (∃ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) (genre :
      JGS23SupplySideRecommenderSystems.Content D),
      JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
          (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ ∧
        JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ = {genre}) ∨
    ∀ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D)
      (genre : JGS23SupplySideRecommenderSystems.Content D),
      JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
          (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ →
        JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ ≠ {genre}
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Recorded source-to-declaration screening **Verdict:** matches_approved_corrected_target

Reason: It preserves the existential single-genre and universal multi-genre quantifier directions using the approved nonzero-support genre set, rather than claiming the archival normalization is defined at zero.
Reviewer check: ☐ Matches formalized target
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

JGS23SupplySideRecommenderSystems.twoPopulationModelSpec

Paper-local declaration:
JGS23SupplySideRecommenderSystems.twoPopulationModelSpec
Source locator: cited publication, lines 571-573

Verbatim paper-source connection

```
\section{Equilibrium structure for two equally sized populations of users}\label{sec:specialcases}

We next investigate the form of specialization exhibited by multi-genre equilibria, focusing on the case of {two equally sized populations} and producer
↪    cost functions given by powers of the $\ell_2$ norm. More formally, there are $N$ users split equally between two linearly independently vectors $u_1,
↪    u_2 \in \mathbb{R}_{\ge 0}^D$, and the the cost function is $c(\p) = \|p\|_2^{\beta}$. We establish \textit{structural properties} of the equilibria
↪    (see Section \ref{sec:structural}). We next concretely compute the equilibria $\mu$ in several special instances that permit \textit{closed-form
↪    solutions} (see Section \ref{sec:closedformsb}-\ref{sec:closedforminfinite}). We then provide an overview of proof techniques, which involves
↪    developing machinery to characterize these equilibria (see Section \ref{sec:proofoverview}).

[Next verbatim source excerpt]

\subsection{Closed-form equilibria for the standard basis vectors}\label{sec:closedformsb}
We next compute the equilibria in the special case of user vectors located at the standard basis vectors, and we analyze the form of specialization that the
↪    equilibria exhibit. For ease of notation, for the remainder of the section, we assume these populations each consist of a \textit{single} user (these
↪    results can be easily adapted to the case of $N/2$ users in each population).

Interestingly, all of these multi-genre equilibria exhibit the following \textit{relaxation of pure horizontal differentiation}: producers can differentiate along
↪    genre, but the genre of content fully specifies the content's quality. More specifically, for any genre $\p^* \in \Genre(\mu)$, the set $\Genre(\mu) \cap
↪    \left\{ q \cdot \p^* \mid q \in \mathbb{R}^{\ge 0}\right\}$ contains exactly one single element.\footnote{Pure horizontal differentiation is not
↪    satisfied, since content in different genres may not have the same quality (see Figure \ref{fig:equilibriaP}).} This stands in contrast to single-genre
↪    equilibria, which by definition exhibit pure vertical differentiation.\footnote{Pure vertical differentiation is when producers only differentiate along
↪    quality, not along direction.}
```

Lean-expanded paper semantic target

```
type:
{D : ℕ} →
  {K P : ℕ} →
    (Fin 2 → JGS23SupplySideRecommenderSystems.Content D) →
      ℝ → JGS23SupplySideRecommenderSystems.MixedContentStrategy D → Prop

value:
fun {D K P : ℕ} (users : Fin 2 → JGS23SupplySideRecommenderSystems.Content D) (β : ℝ)
    (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) =>
  0 < K ∧
    2 ≤ P ∧
      1 ≤ β ∧
        LinearIndependent ℝ users ∧
          (∀ (i : Fin 2), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) ∧
            JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec
              (JGS23SupplySideRecommenderSystems.twoPopulationUsers K users)
              (JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 D)
                β)
              μ
```

Direct retained dependencies

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.SourceNorm.l2
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec
- JGS23SupplySideRecommenderSystems.twoPopulationUsers

Recorded source-to-declaration screening **Verdict:** matches

Reason: It represents two equal positive populations by K copies of each linearly independent nonnegative user type, with L2 power cost and a symmetric mixed Nash equilibrium. Linear independence supplies the source nonzero condition.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 296-299

Verbatim source input

```
\begin{restatable}{proposition}{pure}
\label{prop:pure}
For any set of users and any $\beta \ge 1$, a pure strategy equilibrium does not exist.
\end{restatable}
```

Expanded PaperInterface specification

```
∀ {D N P : ℕ} [Nontrivial (Fin P)] {users : Fin N → JGS23SupplySideRecommenderSystems.Content D}
  {profile : Fin P → JGS23SupplySideRecommenderSystems.Content D} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D)
  {β : ℝ},
  0 < β →
    (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
      (∃ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)) →
        JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous ν →
          ¬JGS23SupplySideRecommenderSystems.paper_pure_nash_equilibrium users
            (JGS23SupplySideRecommenderSystems.normRpowCost ν β) profile
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous
- JGS23SupplySideRecommenderSystems.normRpowCost
- JGS23SupplySideRecommenderSystems.paper_pure_nash_equilibrium

Lean-elaborated claim roles

```
1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nontrivial (Fin P)
5. parameter: Fin N → JGS23SupplySideRecommenderSystems.Content D
6. parameter: Fin P → JGS23SupplySideRecommenderSystems.Content D
7. parameter: JGS23SupplySideRecommenderSystems.SourceNorm D
8. parameter: ℝ
9. assumption: 0 < β
10. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)
11. assumption: ∃ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)
12. assumption: JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous ν
13. conclusion: ¬JGS23SupplySideRecommenderSystems.paper_pure_nash_equilibrium users
    (JGS23SupplySideRecommenderSystems.normRpowCost ν β) profile
```

Lean proof endpoint (built separately)

```
JGS23SupplySideRecommenderSystems.propositionPure
```

Recorded source-to-Spec screening **Verdict:** matches

Reason: Anchor 296-299 says no pure equilibrium exists in the paper's nonzero-user powered-norm model. The target preserves that result (indeed for beta > 0) with the source perturbation-continuity convention explicit.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 305-308

Verbatim source input

```
\begin{restatable}{proposition}{existence}
\label{prop:existence}
For any set of users and any $\beta \ge 1$, a symmetric mixed equilibrium exists.
\end{restatable}
```

Expanded PaperInterface specification

```
∀ {D N P : ℕ} [Nonempty (Fin N)] [Nontrivial (Fin P)] {users : Fin N → JGS23SupplySideRecommenderSystems.Content D}
  {ν : JGS23SupplySideRecommenderSystems.SourceNorm D} {β : ℝ},
  (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
    (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)) →
      JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν →
        Continuous ν.norm →
          JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous ν →
            1 ≤ β →
              ∃ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D),
                JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
                  (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels
- JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

```
1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nonempty (Fin N)
5. assumption: Nontrivial (Fin P)
6. parameter: Fin N → JGS23SupplySideRecommenderSystems.Content D
7. parameter: JGS23SupplySideRecommenderSystems.SourceNorm D
8. parameter: ℝ
9. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)
10. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)
11. assumption: JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν
12. assumption: Continuous ν.norm
13. assumption: JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous ν
14. assumption: 1 ≤ β
15. conclusion: ∃ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D),
  JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
    (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ
```

Lean proof endpoint (built separately)

```
JGS23SupplySideRecommenderSystems.propositionExistence
```

Recorded source-to-Spec screening **Verdict:** matches

Reason: Anchor 305-308 asserts a symmetric mixed equilibrium for the paper's finite-dimensional norm model and beta >= 1. The target preserves this conclusion and makes the source-norm compactness and continuity conventions explicit.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 3

Source locator: cited publication, lines 311-314

Verbatim source input

```
\begin{proposition}
\label{prop:atom}
For any set of users and any $\beta \ge 1$, every symmetric mixed equilibrium is atomless.
\end{proposition}
```

Expanded PaperInterface specification

```
∀ {D N P : ℕ} [Nonempty (Fin P)] [Nontrivial (Fin P)] {users : Fin N → JGS23SupplySideRecommenderSystems.Content D}
  {cost : JGS23SupplySideRecommenderSystems.Content D → ℝ}
  {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D} {i0 : Fin N},
  JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users cost μ →
    JGS23SupplySideRecommenderSystems.PerturbationCostContinuous cost →
      (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
        JGS23SupplySideRecommenderSystems.NonzeroContent (users i0) →
          JGS23SupplySideRecommenderSystems.paper_mixed_content_strategy_atomless μ
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.PerturbationCostContinuous
- JGS23SupplySideRecommenderSystems.paper_mixed_content_strategy_atomless
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

```
1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nonempty (Fin P)
5. assumption: Nontrivial (Fin P)
6. parameter: Fin N → JGS23SupplySideRecommenderSystems.Content D
7. parameter: JGS23SupplySideRecommenderSystems.Content D → ℝ
8. parameter: JGS23SupplySideRecommenderSystems.MixedContentStrategy D
9. parameter: Fin N
10. assumption: JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users cost μ
11. assumption: JGS23SupplySideRecommenderSystems.PerturbationCostContinuous cost
12. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)
13. assumption: JGS23SupplySideRecommenderSystems.NonzeroContent (users i0)
14. conclusion: JGS23SupplySideRecommenderSystems.paper_mixed_content_strategy_atomless μ
```

Lean proof endpoint (built separately)

```
JGS23SupplySideRecommenderSystems.propositionAtom
```

Recorded source-to-Spec screening **Verdict:** matches

Reason: Anchor 311-314 says every symmetric mixed equilibrium is atomless. The target states that conclusion under the source model's nonzero-user and continuous powered-cost conditions, expressed as an explicit perturbation-continuity premise.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 358-361

Verbatim source input

```
\begin{corollary}
\label{cor:onepopulation}
Suppose that there is a single population of $N$ users, all of whose embeddings are at the same vector $u$. Then, there is a symmetric mixed Nash
↪    equilibrium $\mu$ supported on $\left\{ q p^* \mid q \in [0, N^{\frac{1}{\beta}}] \right\}$ where $p^* \in \argmax_{\|p \| = 1 } \langle p, u \rangle$.
↪    The cumulative distribution function of $q = \|p\| \sim \mu$ is $F(q) = \left(q/N\right)^{\beta / (P-1)}$.
\end{corollary}
```

Formalized review target The archival source above is not asserted equivalent to this formalized target.

```
For homogeneous nonzero users and a norm-maximizing nonnegative unit genre, the specified single-ray law is a symmetric mixed equilibrium; its norm
↪    CDF is the clipped corrected radial law F(q) = ((q^beta)/N)^(1/(P-1)).
```

Archival source anchor: cited publication, lines 358-361

Expanded PaperInterface specification

```
∀ {D N P : ℕ} [Nonempty (Fin N)] [Nonempty (Fin P)] [Nontrivial (Fin P)] {β : ℝ}
  {u genre : JGS23SupplySideRecommenderSystems.Content D} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D),
  0 < β →
    JGS23SupplySideRecommenderSystems.NonnegativeContent u →
      JGS23SupplySideRecommenderSystems.NonnegativeContent genre →
        ν.norm genre = 1 →
          0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value u genre →
            (∀ (d : JGS23SupplySideRecommenderSystems.Content D),
                JGS23SupplySideRecommenderSystems.NonnegativeContent d →
                  ν.norm d = 1 →
                    JGS23SupplySideRecommenderSystems.paper_inferred_user_value u d ≤
                      JGS23SupplySideRecommenderSystems.paper_inferred_user_value u genre) →
              Measurable ν.norm →
                JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec (fun (x : Fin N) => u)
                    (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β)
                    (JGS23SupplySideRecommenderSystems.singleGenreContentLaw N P β genre) ∧
                  (JGS23SupplySideRecommenderSystems.singleGenreContentLaw N P β genre).support ⊆
                      {p : JGS23SupplySideRecommenderSystems.Content D |
                        ∃ q ∈ Set.Icc 0 (↑N ^ β⁻¹), p = JGS23SupplySideRecommenderSystems.scaleContent q genre} ∧
                    ∀ (z : ℝ),
                      0 ≤ z →
                        (MeasureTheory.Measure.map ν.norm
                              (JGS23SupplySideRecommenderSystems.singleGenreContentLaw N P β genre))
                            (Set.Iic z) =
                          ENNReal.ofReal (min 1 ((z ^ β / ↑N) ^ (↑P - 1)⁻¹))
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.paper_inferred_user_value
- JGS23SupplySideRecommenderSystems.scaleContent
- JGS23SupplySideRecommenderSystems.singleGenreContentLaw
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

```
1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nonempty (Fin N)
5. assumption: Nonempty (Fin P)
6. assumption: Nontrivial (Fin P)
7. parameter: ℝ
8. parameter: JGS23SupplySideRecommenderSystems.Content D
9. parameter: JGS23SupplySideRecommenderSystems.Content D
10. parameter: JGS23SupplySideRecommenderSystems.SourceNorm D
11. assumption: 0 < β
12. assumption: JGS23SupplySideRecommenderSystems.NonnegativeContent u
13. assumption: JGS23SupplySideRecommenderSystems.NonnegativeContent genre
14. assumption: ν.norm genre = 1
15. assumption: 0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value u genre
16. assumption: ∀ (d : JGS23SupplySideRecommenderSystems.Content D),
  JGS23SupplySideRecommenderSystems.NonnegativeContent d →
```

```
  ν.norm d = 1 →
    JGS23SupplySideRecommenderSystems.paper_inferred_user_value u d ≤
      JGS23SupplySideRecommenderSystems.paper_inferred_user_value u genre
17. assumption: Measurable ν.norm
18. conclusion: JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec (fun (x : Fin N) => u)
    (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β)
    (JGS23SupplySideRecommenderSystems.singleGenreContentLaw N P β genre) ∧
  (JGS23SupplySideRecommenderSystems.singleGenreContentLaw N P β genre).support ⊆
    {p : JGS23SupplySideRecommenderSystems.Content D |
      ∃ q ∈ Set.Icc 0 (↑N ^ β⁻¹), p = JGS23SupplySideRecommenderSystems.scaleContent q genre} ∧
  ∀ (z : ℝ),
    0 ≤ z →
      (MeasureTheory.Measure.map ν.norm (JGS23SupplySideRecommenderSystems.singleGenreContentLaw N P β genre))
          (Set.Iic z) =
        ENNReal.ofReal (min 1 ((z ^ β / ↑N) ^ (↑P - 1)⁻¹))
```

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.corollaryOnePopulation

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: Anchor 358-361 has the inconsistent printed CDF; the approved replacement retains the homogeneous ray equilibrium and gives the corrected clipped radial law ((z^beta)/N)^(1/(P-1)).

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 415-423

Verbatim source input

```
\begin{restatable}{theorem}{singlegenre}
\label{thm:singlegenre}
Let $\UMatrix := [\u_1; \cdots; \u_N]$, let $\Set$ be $\left\{\UMatrix\p \mid \|\p\| \le 1, \p \in \bR_{\ge 0}^D  \right\}$, and let $\Set^{\beta}$ be the image
↪   of $\Set$ under coordinate-wise powers. Then, there is a symmetric equilibrium $\mu$ with $|\Genre(\mu)| = 1$ if and only if
\begin{equation}
\label{eq:maxcondition}
  \max_{y \in \Set^{\beta}} \prod_{i=1}^{N} y_i  = \max_{y \in \bar{\Set}^{\beta}} \prod_{i=1}^{N} y_i.
\end{equation}
Otherwise, all symmetric equilibria have multiple genres. Moreover, if \eqref{eq:maxcondition} holds for some $\beta$, it also holds for every $\beta' \leq
↪   \beta$.
\end{restatable}
```

Formalized review target   The archival source above is not asserted equivalent to this formalized target.

```
Under the stated finite-dimensional norm regularity, positive nonzero users, and the zero-safe nonzero-support genre convention, a single-genre symmetric
↪   equilibrium exists exactly when the powered-score product-supremum condition holds.
```

Archival source anchor: cited publication, lines 415-423

Expanded PaperInterface specification

```
∀ {D N P : ℕ} [Nonempty (Fin N)] [Nonempty (Fin P)] [Nontrivial (Fin P)]
  {users : Fin N → JGS23SupplySideRecommenderSystems.Content D} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D)
  {β : ℝ},
  1 ≤ β →
    (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
      (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)) →
        JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous ν →
          JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν →
            Continuous ν.norm →
              ((∃ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) (genre :
                    JGS23SupplySideRecommenderSystems.Content D),
                    JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
                        (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ ∧
                      JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ = {genre}) ↔
                  JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition
                    {y : Fin N → ℝ |
                      JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
                        (fun (p : JGS23SupplySideRecommenderSystems.Content D) =>
                          JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧ ν.norm p ≤ 1)
                        β y}) ∧
                (¬JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition
                        {y : Fin N → ℝ |
                          JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
                            (fun (p : JGS23SupplySideRecommenderSystems.Content D) =>
                              JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧ ν.norm p ≤ 1)
                            β y} →
                    ∀ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D)
                      (genre : JGS23SupplySideRecommenderSystems.Content D),
                      JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
                          (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ →
                        JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ ≠ {genre}) ∧
                  ∀ {β' : ℝ},
                    0 < β' →
                      β' ≤ β →
                        JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition
                            {y : Fin N → ℝ |
                              JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
                                (fun (p : JGS23SupplySideRecommenderSystems.Content D) =>
                                  JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧ ν.norm p ≤ 1)
                                β y} →
                          JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition
                            {y : Fin N → ℝ |
                              JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
                                (fun (p : JGS23SupplySideRecommenderSystems.Content D) =>
                                  JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧ ν.norm p ≤ 1)
                                β' y}
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition
- JGS23SupplySideRecommenderSystems.SourceNorm

- JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels
- JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous
- JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.paper_powered_unit_image
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

```
1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nonempty (Fin N)
5. assumption: Nonempty (Fin P)
6. assumption: Nontrivial (Fin P)
7. parameter: Fin N → JGS23SupplySideRecommenderSystems.Content D
8. parameter: JGS23SupplySideRecommenderSystems.SourceNorm D
9. parameter: ℝ
10. assumption: 1 ≤ β
11. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)
12. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)
13. assumption: JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous ν
14. assumption: JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν
15. assumption: Continuous ν.norm
16. conclusion: ((∃ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) (genre :
   JGS23SupplySideRecommenderSystems.Content D),
   JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
     (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ ∧
    JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ = {genre}) ↔
  JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition
   {y : Fin N → ℝ |
    JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
     (fun (p : JGS23SupplySideRecommenderSystems.Content D) =>
      JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧ ν.norm p ≤ 1)
     β y}) ∧
 (¬JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition
     {y : Fin N → ℝ |
      JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
       (fun (p : JGS23SupplySideRecommenderSystems.Content D) =>
        JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧ ν.norm p ≤ 1)
       β y} →
   ∀ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D)
    (genre : JGS23SupplySideRecommenderSystems.Content D),
    JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
      (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ →
     JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ ≠ {genre}) ∧
  ∀ {β' : ℝ},
   0 < β' →
    β' ≤ β →
     JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition
       {y : Fin N → ℝ |
        JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
         (fun (p : JGS23SupplySideRecommenderSystems.Content D) =>
          JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧ ν.norm p ≤ 1)
         β y} →
      JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition
       {y : Fin N → ℝ |
        JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
         (fun (p : JGS23SupplySideRecommenderSystems.Content D) =>
          JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧ ν.norm p ≤ 1)
         β' y}
```

Lean proof endpoint (built separately)

```
JGS23SupplySideRecommenderSystems.theoremSingleGenre
```

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: Anchor 415-423 gives the product-condition iff, multi-genre alternative, and exponent monotonicity. The approved target preserves all three under the recorded finite-dimensional regularity and zero-safe nonzero-support genre convention.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 442-449

Verbatim source input

```
\begin{restatable}[Informal]{lemma}{optimization}
\label{lemma:optimizationprogram}
There exists a symmetric equilibrium $\mu$ with $|\Genre(\mu)| = 1$ if and only if:
\begin{equation}
    \label{eq:optimizationprogrammain}
    \inf_{y \in \Set^{\beta}} \left(\sup_{y' \in \Set^{\beta}}   \sum_{i=1}^N  \frac{y'_i}{y_i} \right) = \sup_{y' \in \Set^{\beta}} \left(\inf_{y \in
  ↪  \Set^{\beta}} \sum_{i=1}^N  \frac{y'_i}{y_i}\right).
\end{equation}
\end{restatable}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

```
For nonnegative nonzero users and compact source-norm sublevels, the source product condition is equivalent to an attained positive feasible-ray
↪  source-Nash certificate.
```

Archival source anchor: cited publication, lines 442-449

Expanded PaperInterface specification

```
∀ {D N P : ℕ} [Nonempty (Fin N)] [Nonempty (Fin P)] [Nontrivial (Fin P)]
  {users : Fin N → JGS23SupplySideRecommenderSystems.Content D} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D)
  {β : ℝ},
  0 < β →
    (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
      (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)) →
        JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν →
          ((∃ (p : JGS23SupplySideRecommenderSystems.Content D),
              JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧
                ν.norm p ≤ 1 ∧
                  (∀ (i : Fin N), 0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p) ∧
                    JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
                      (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β)
                      (JGS23SupplySideRecommenderSystems.singleGenreContentLaw N P β (ν.normalizedContent p))) ↔
            JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition
              {y : Fin N → ℝ |
                JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
                  (fun (p : JGS23SupplySideRecommenderSystems.Content D) =>
                    JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧ ν.norm p ≤ 1)
                  β y})
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.SourceNorm.normalizedContent
- JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.paper_inferred_user_value
- JGS23SupplySideRecommenderSystems.paper_powered_unit_image
- JGS23SupplySideRecommenderSystems.singleGenreContentLaw
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nonempty (Fin N)
5. assumption: Nonempty (Fin P)
6. assumption: Nontrivial (Fin P)
7. parameter: Fin N → JGS23SupplySideRecommenderSystems.Content D
8. parameter: JGS23SupplySideRecommenderSystems.SourceNorm D
9. parameter: ℝ
10. assumption: 0 < β
11. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)
12. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)
13. assumption: JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν
14. conclusion: (∃ (p : JGS23SupplySideRecommenderSystems.Content D),
    JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧
      ν.norm p ≤ 1 ∧
        (∀ (i : Fin N), 0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p) ∧
          JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
            (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β)
            (JGS23SupplySideRecommenderSystems.singleGenreContentLaw N P β (ν.normalizedContent p))) ↔
  JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition
    {y : Fin N → ℝ |
      JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
        (fun (p : JGS23SupplySideRecommenderSystems.Content D) =>
          JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧ ν.norm p ≤ 1)
        β y}

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.lemmaOptimizationProgram

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: Anchor 442-449 has ratios with zero-denominator and attainment gaps. The approved target is the compact positive feasible-ray certificate equivalent to the source product condition, rather than the literal extended-real minimax display.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 7

Source locator: cited publication, lines 459-462

Verbatim source input

```
\begin{restatable}{corollary}{betaone}
\label{cor:beta1}
The threshold $\beta^*$ is always at least $1$. That is, if $\beta = 1$, there exists a single-genre equilibrium.
\end{restatable}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

```
For nonnegative nonzero users and a compact convex source norm, beta equal to one belongs to the source product-condition beta-star set and has an
↪    explicit positive-score single-ray symmetric mixed equilibrium.
```

Archival source anchor: cited publication, lines 459-462

Expanded PaperInterface specification

```
∀ {D N P : ℕ} [Nonempty (Fin N)] [Nonempty (Fin P)] [Nontrivial (Fin P)]
  {users : Fin N → JGS23SupplySideRecommenderSystems.Content D} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D),
  (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
    (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)) →
      JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν →
        JGS23SupplySideRecommenderSystems.SourceNormConvexSublevels ν →
          ∃ (p : JGS23SupplySideRecommenderSystems.Content D),
            JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧
              ν.norm p ≤ 1 ∧
                (∀ (i : Fin N), 0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p) ∧
                  JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
                      (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν 1)
                      (JGS23SupplySideRecommenderSystems.singleGenreContentLaw N P 1 (ν.normalizedContent p)) ∧
                    1 ≤ JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec users ν
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.SourceNorm.normalizedContent
- JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels
- JGS23SupplySideRecommenderSystems.SourceNormConvexSublevels
- JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.paper_inferred_user_value
- JGS23SupplySideRecommenderSystems.singleGenreContentLaw
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

```
1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nonempty (Fin N)
5. assumption: Nonempty (Fin P)
6. assumption: Nontrivial (Fin P)
7. parameter: Fin N → JGS23SupplySideRecommenderSystems.Content D
8. parameter: JGS23SupplySideRecommenderSystems.SourceNorm D
9. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)
10. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)
11. assumption: JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν
12. assumption: JGS23SupplySideRecommenderSystems.SourceNormConvexSublevels ν
13. conclusion: ∃ (p : JGS23SupplySideRecommenderSystems.Content D),
  JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧
    ν.norm p ≤ 1 ∧
      (∀ (i : Fin N), 0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p) ∧
        JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
            (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν 1)
            (JGS23SupplySideRecommenderSystems.singleGenreContentLaw N P 1 (ν.normalizedContent p)) ∧
          1 ≤ JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec users ν
```

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.corollaryBetaOne

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: The approved beta-one repair is stated for nonnegative nonzero users and a source norm with compact and convex sublevels. The expanded proposition uses exactly that domain and returns one nonnegative unit-ball direction with positive score for every user, the explicit beta-one power-law measure on its normalized ray, symmetric Nash optimality under the displayed uniform-tie payoff, and the beta-star lower bound 1. The ray-law and beta-star prerequisites preserve the approved product-condition and equilibrium meanings, so the Lean target matches the corrected target.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 470-473

Verbatim source input

```
\begin{restatable}{corollary}{betap}
\label{cor:betap}
Let the cost function be $c(\p) = \|\p\|_{q}^{\beta}$. For any set of user vectors, it holds that $\beta^* \ge q$. If the user vectors are equal to the standard
↪    basis vectors $\left\{e_1, \ldots, e_D \right\}$, then $\beta^*$ is equal to $q$.
\end{restatable}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

```
For an Lq source norm, nonnegative nonzero users, and q at least one, the source product-condition beta-star is at least q; a singleton
↪    nonzero-support-genre symmetric mixed equilibrium exists for every positive beta at most q. For at least two standard-basis users, beta-star equals q.
```

Archival source anchor: cited publication, lines 470-473

Expanded PaperInterface specification

```
(∀ {D N P : ℕ} [Nonempty (Fin N)] [Nonempty (Fin P)] [Nontrivial (Fin P)] {q β : ℝ}
    {users : Fin N → JGS23SupplySideRecommenderSystems.Content D} (hq : 1 ≤ q),
    0 < β →
      β ≤ q →
        (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
          (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)) →
            (∃ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) (genre :
                JGS23SupplySideRecommenderSystems.Content D),
                JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
                    (JGS23SupplySideRecommenderSystems.normPowerCostSpec
                      (JGS23SupplySideRecommenderSystems.SourceNorm.lp D
                        (JGS23SupplySideRecommenderSystems.corollaryBetaPSpec._proof_1 hq))
                      β)
                    μ ∧
                  JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec
                      (JGS23SupplySideRecommenderSystems.SourceNorm.lp D
                        (JGS23SupplySideRecommenderSystems.corollaryBetaPSpec._proof_1 hq))
                      μ =
                    {genre}) ∧
              ↑q ≤
                JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec users
                  (JGS23SupplySideRecommenderSystems.SourceNorm.lp D
                    (JGS23SupplySideRecommenderSystems.corollaryBetaPSpec._proof_1 hq))) ∧
  ∀ {D : ℕ} [Nontrivial (Fin D)] {q : ℝ} (hq : 1 ≤ q),
    JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec (JGS23SupplySideRecommenderSystems.standardBasisUsers D)
        (JGS23SupplySideRecommenderSystems.SourceNorm.lp D
          (JGS23SupplySideRecommenderSystems.corollaryBetaPSpec._proof_1 hq)) =
      ↑q
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SourceNorm.lp
- JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec
- JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.standardBasisUsers
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

```
1. conclusion: (∀ {D N P : ℕ} [Nonempty (Fin N)] [Nonempty (Fin P)] [Nontrivial (Fin P)] {q β : ℝ}
    {users : Fin N → JGS23SupplySideRecommenderSystems.Content D} (hq : 1 ≤ q),
    0 < β →
      β ≤ q →
        (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
          (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)) →
            (∃ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) (genre :
                JGS23SupplySideRecommenderSystems.Content D),
                JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
                    (JGS23SupplySideRecommenderSystems.normPowerCostSpec
                      (JGS23SupplySideRecommenderSystems.SourceNorm.lp D
```

```
            (JGS23SupplySideRecommenderSystems.corollaryBetaPSpec._proof_1 hq))
          β)
        μ ∧
      JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec
          (JGS23SupplySideRecommenderSystems.SourceNorm.lp D
            (JGS23SupplySideRecommenderSystems.corollaryBetaPSpec._proof_1 hq))
          μ =
        {genre}) ∧
    ↑q ≤
      JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec users
        (JGS23SupplySideRecommenderSystems.SourceNorm.lp D
          (JGS23SupplySideRecommenderSystems.corollaryBetaPSpec._proof_1 hq))) ∧
  ∀ {D : ℕ} [Nontrivial (Fin D)] {q : ℝ} (hq : 1 ≤ q),
    JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec (JGS23SupplySideRecommenderSystems.standardBasisUsers D)
        (JGS23SupplySideRecommenderSystems.SourceNorm.lp D
          (JGS23SupplySideRecommenderSystems.corollaryBetaPSpec._proof_1 hq)) =
      ↑q
```

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.corollaryBetaP

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: For the Lq repair, the Lean conjunction has the same two scopes as the approved target. With q at least one, positive beta at most q, and arbitrary nonnegative nonzero users, it produces a symmetric Nash measure with exactly one nonzero support direction and proves beta-star at least q. Separately, for the standard-basis family in nontrivial dimension, it proves beta-star equals q. The expanded Lq norm, powered unit image, nonzero-genre, and beta-star definitions introduce no different formula or domain restriction.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 9

Source locator: cited publication, lines 477-481

Verbatim source input

```
\begin{restatable}{corollary}{twousers}
\label{cor:2users}
Suppose that there are $N$ users split equally between two linearly independently vectors $u_1, u_2 \in \mathbb{R}_{\ge 0}^D$, and let $\theta^* :=
↪    \cos^{-1}\left( \frac{\langle u_1, u_2\rangle}{\|u_1\|_2 \|u_2\|} \right)$. Let the cost function be $c(\p) = \|\p\|_{2}^{\beta}$. Then it holds that:
\[\beta^*  = \frac{2}{1 - \cos(\theta^*)}. \]
\end{restatable}
```

Formalized review target The archival source above is not asserted equivalent to this formalized target.

```
For two equal nonzero user populations at angle theta with L2 power cost, the source product-condition beta-star is exactly the two-user phase threshold;
↪    the accompanying zero-safe single-genre symmetric-equilibrium condition holds exactly at exponents below that threshold.
```

Archival source anchor: cited publication, lines 548-554

Expanded PaperInterface specification

```
∀ {D K P : ℕ} [Nonempty (Fin P)] [Nontrivial (Fin P)] {users : Fin 2 → JGS23SupplySideRecommenderSystems.Content D}
  {β θ : ℝ},
  0 < K →
    (∀ (i : Fin 2), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
      JGS23SupplySideRecommenderSystems.NonzeroContent (users 0) →
        JGS23SupplySideRecommenderSystems.NonzeroContent (users 1) →
          JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users 0) (users 1) /
                (AppliedModelingLib.FiniteDimensionalNorms.l2 (users 0) *
                  AppliedModelingLib.FiniteDimensionalNorms.l2 (users 1)) =
              Real.cos θ →
            1 ≤ β →
              0 < θ →
                θ ≤ Real.pi / 2 →
                  ((∃ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) (genre :
                        JGS23SupplySideRecommenderSystems.Content D),
                        JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec
                            (JGS23SupplySideRecommenderSystems.twoPopulationUsers K users)
                            (JGS23SupplySideRecommenderSystems.normPowerCostSpec
                              (JGS23SupplySideRecommenderSystems.SourceNorm.l2 D) β)
                            μ ∧
                          JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec
                              (JGS23SupplySideRecommenderSystems.SourceNorm.l2 D) μ =
                            {genre}) ↔
                      β ≤ JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold θ) ∧
                    JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec
                        (JGS23SupplySideRecommenderSystems.twoPopulationUsers K users)
                        (JGS23SupplySideRecommenderSystems.SourceNorm.l2 D) =
                      ↑(JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold θ)
```

Retained governing declarations

- AppliedModelingLib.FiniteDimensionalNorms.l2
- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SourceNorm.l2
- JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec
- JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.paper_inferred_user_value
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec
- JGS23SupplySideRecommenderSystems.twoPopulationUsers
- JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold

Lean-elaborated claim roles

1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nonempty (Fin P)
5. assumption: Nontrivial (Fin P)
6. parameter: Fin 2 → JGS23SupplySideRecommenderSystems.Content D
7. parameter: ℝ
8. parameter: ℝ
9. assumption: 0 < K
10. assumption: ∀ (i : Fin 2), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)
11. assumption: JGS23SupplySideRecommenderSystems.NonzeroContent (users 0)
12. assumption: JGS23SupplySideRecommenderSystems.NonzeroContent (users 1)
13. assumption: JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users 0) (users 1) / (AppliedModelingLib.FiniteDimensionalNorms.l2 (users 0) * AppliedModelingLib.FiniteDimensionalNorms.l2 (users 1)) = Real.cos θ
14. assumption: 1 ≤ β
15. assumption: 0 < θ
16. assumption: θ ≤ Real.pi / 2
17. conclusion: ((∃ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) (genre : JGS23SupplySideRecommenderSystems.Content D), JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec (JGS23SupplySideRecommenderSystems.twoPopulationUsers K users) (JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 D) β) μ ∧ JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 D) μ = {genre}) ↔ β ≤ JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold θ) ∧ JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec (JGS23SupplySideRecommenderSystems.twoPopulationUsers K users) (JGS23SupplySideRecommenderSystems.SourceNorm.l2 D) = ↑(JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold θ)

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.corollaryTwoUsers

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: The source's equal split is represented as K copies of each of two nonzero nonnegative user types, with K positive. The Lean angle premise is their L2-normalized inner product, theta is restricted to the nonnegative independent-user range, and the supporting threshold expands to 2/(1-cos theta). In this domain the conclusion gives both the zero-safe singleton-nonzero-genre equilibrium equivalence beta <= threshold and exact product-condition beta-star equality, which is the full approved corrected target.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 10

Source locator: cited publication, lines 486-493

Verbatim source input

```
\begin{restatable}{corollary}{betageneral}
\label{cor:beta}
Let $\|\cdot\|_{*}$ denote the dual norm of $\|\cdot\|$, defined to be $\|\p\|_{*} = \max_{\|\p\| = 1, \p \in \bR_{\ge 0}^D} \langle q, \p \rangle$. Let $Z:=
↪    \|\sum_{n=1}^{N} \frac{u_n}{\|u_n\|_{*}} \|_{*}$. Then,
\begin{equation}
\label{eq:betacondition}
\beta^* \le \frac{\log (N)}{\log(N) - \log(Z)}.
\end{equation}
\end{restatable}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

```
For normalized nonnegative users with compact source-norm sublevels and score-one unit witnesses, the source product-condition beta-star is at most one
↪    when Z=1 and at most the displayed logarithmic threshold when 1<Z<N; above the finite threshold no compact regular source symmetric equilibrium
↪    has one nonzero support genre.
```

Archival source anchor: cited publication, lines 486-493

Expanded PaperInterface specification

```
(∀ {D N P : ℕ} [Nontrivial (Fin N)] [Nonempty (Fin P)] [Nontrivial (Fin P)] {β Z : ℝ}
    {users : Fin N → JGS23SupplySideRecommenderSystems.Content D} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D),
    (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
      (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)) →
        JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous ν →
          JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν →
            Continuous ν.norm →
              (∀ (p : JGS23SupplySideRecommenderSystems.Content D),
                  JGS23SupplySideRecommenderSystems.NonnegativeContent p →
                    ν.norm p ≤ 1 →
                      ∑ i : Fin N, JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p ≤ Z) →
                (∀ (i : Fin N),
                    ∃ (p : JGS23SupplySideRecommenderSystems.Content D),
                      JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧
                        ν.norm p = 1 ∧ JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p = 1) →
                  1 < Z →
                    Z < ↑N →
                      Real.log ↑N / (Real.log ↑N - Real.log Z) < β →
                        ¬∃ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) (genre :
                            JGS23SupplySideRecommenderSystems.Content D),
                            JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
                                (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ ∧
                              JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ = {genre}) ∧
  (∀ {D N : ℕ} [Nontrivial (Fin N)] {users : Fin N → JGS23SupplySideRecommenderSystems.Content D}
      (ν : JGS23SupplySideRecommenderSystems.SourceNorm D),
      (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
        JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν →
          (∀ (p : JGS23SupplySideRecommenderSystems.Content D),
              JGS23SupplySideRecommenderSystems.NonnegativeContent p →
                ν.norm p ≤ 1 →
                  ∑ i : Fin N, JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p ≤ 1) →
            (∀ (i : Fin N),
                ∃ (p : JGS23SupplySideRecommenderSystems.Content D),
                  JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧
                    ν.norm p = 1 ∧ JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p = 1) →
              JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec users ν ≤ 1) ∧
    ∀ {D N : ℕ} [Nontrivial (Fin N)] {Z : ℝ} {users : Fin N → JGS23SupplySideRecommenderSystems.Content D}
      (ν : JGS23SupplySideRecommenderSystems.SourceNorm D),
      (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
        JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν →
          (∀ (p : JGS23SupplySideRecommenderSystems.Content D),
              JGS23SupplySideRecommenderSystems.NonnegativeContent p →
                ν.norm p ≤ 1 →
                  ∑ i : Fin N, JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p ≤ Z) →
            (∀ (i : Fin N),
                ∃ (p : JGS23SupplySideRecommenderSystems.Content D),
                  JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧
                    ν.norm p = 1 ∧ JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p = 1) →
              1 < Z →
                Z < ↑N →
                  JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec users ν ≤
                    ↑(Real.log ↑N / (Real.log ↑N - Real.log Z))
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.NonzeroContent

- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels
- JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous
- JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec
- JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.paper_inferred_user_value
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

```
1. conclusion: (∀ {D N P : ℕ} [Nontrivial (Fin N)] [Nonempty (Fin P)] [Nontrivial (Fin P)] {β Z : ℝ}
    {users : Fin N → JGS23SupplySideRecommenderSystems.Content D} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D),
    (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
      (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)) →
        JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous ν →
          JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν →
            Continuous ν.norm →
              (∀ (p : JGS23SupplySideRecommenderSystems.Content D),
                  JGS23SupplySideRecommenderSystems.NonnegativeContent p →
                    ν.norm p ≤ 1 →
                      ∑ i : Fin N, JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p ≤ Z) →
                (∀ (i : Fin N),
                    ∃ (p : JGS23SupplySideRecommenderSystems.Content D),
                      JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧
                        ν.norm p = 1 ∧ JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p = 1) →
                  1 < Z →
                    Z < ↑N →
                      Real.log ↑N / (Real.log ↑N - Real.log Z) < β →
                        ¬∃ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) (genre :
                            JGS23SupplySideRecommenderSystems.Content D),
                            JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
                                (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ ∧
                              JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ = {genre}) ∧
  (∀ {D N : ℕ} [Nontrivial (Fin N)] {users : Fin N → JGS23SupplySideRecommenderSystems.Content D}
      (ν : JGS23SupplySideRecommenderSystems.SourceNorm D),
      (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
        JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν →
          (∀ (p : JGS23SupplySideRecommenderSystems.Content D),
              JGS23SupplySideRecommenderSystems.NonnegativeContent p →
                ν.norm p ≤ 1 →
                  ∑ i : Fin N, JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p ≤ 1) →
            (∀ (i : Fin N),
                ∃ (p : JGS23SupplySideRecommenderSystems.Content D),
                  JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧
                    ν.norm p = 1 ∧ JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p = 1) →
              JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec users ν ≤ 1) ∧
    ∀ {D N : ℕ} [Nontrivial (Fin N)] {Z : ℝ} {users : Fin N → JGS23SupplySideRecommenderSystems.Content D}
      (ν : JGS23SupplySideRecommenderSystems.SourceNorm D),
      (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
        JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν →
          (∀ (p : JGS23SupplySideRecommenderSystems.Content D),
              JGS23SupplySideRecommenderSystems.NonnegativeContent p →
                ν.norm p ≤ 1 →
                  ∑ i : Fin N, JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p ≤ Z) →
            (∀ (i : Fin N),
                ∃ (p : JGS23SupplySideRecommenderSystems.Content D),
                  JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧
                    ν.norm p = 1 ∧ JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p = 1) →
              1 < Z →
                Z < ↑N →
                  JGS23SupplySideRecommenderSystems.betaStarDefinitionSpec users ν ≤
                    ↑(Real.log ↑N / (Real.log ↑N - Real.log Z))
```

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.corollaryBeta

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: The corrected logarithmic target excludes the undefined Z = N quotient and retains the finite cases Z = 1 and 1 < Z < N. Lean exposes the source-norm compactness and perturbation/continuity conditions at the equilibrium-exclusion endpoint, represents Z by the displayed uniform upper bound on the sum of normalized user scores together with score-one unit witnesses, proves beta-star at most 1 in the Z = 1 case, and proves the stated log(N)/(log(N)-log(Z)) bound in the interior case. Its first conjunct also rules out a singleton nonzero-genre equilibrium strictly above that threshold, exactly as approved.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 11

Source locator: cited publication, lines 501-508

Verbatim source input

```
\begin{restatable}{corollary}{singlegenrestructure}
\label{cor:singlegenrestructure}
If there exists $\mu$ with $|\Genre(\mu)| = 1$, then the corresponding producer direction maximizes Nash social welfare of the users:
\begin{equation}
\label{eq:direction}
 \Genre(\mu) = \argmax_{\|\p\|=1 \mid \p \in \bR_{\ge 0}^D} \sum_{i=1}^N \log(\langle \p, u_i\rangle).
\end{equation}
\end{restatable}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

```
With a singleton nonzero-support genre, nonzero users, and finite-dimensional norm regularity, the genre's powered score row maximizes log Nash welfare
↪    on the positive nonnegative unit-sphere image.
```

Archival source anchor: cited publication, lines 501-508

Expanded PaperInterface specification

```
∀ {D N P : ℕ} [Nonempty (Fin N)] [Nonempty (Fin P)] [Nontrivial (Fin P)]
  {users : Fin N → JGS23SupplySideRecommenderSystems.Content D} {ν : JGS23SupplySideRecommenderSystems.SourceNorm D}
  {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D} {j : Fin P}
  {genre : JGS23SupplySideRecommenderSystems.Content D} {β : ℝ},
  JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
      (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ →
    JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ ⊆ {genre} →
      (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
        (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)) →
          JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous ν →
            0 < β →
              JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν →
                Continuous ν.norm →
                  AppliedModelingLib.Optimization.IsMaximizerOn
                    (fun (z : Fin N → ℝ) =>
                      z ∈
                          {y' : Fin N → ℝ |
                            JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
                              (fun (p : JGS23SupplySideRecommenderSystems.Content D) =>
                                JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧ ν.norm p = 1)
                              β y'} ∧
                        ∀ (i : Fin N), 0 < z i)
                    (fun (y : Fin N → ℝ) => ∑ i : Fin N, Real.log (y i)) fun (i : Fin N) =>
                    JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) genre ^ β
```

Retained governing declarations

- AppliedModelingLib.Optimization.IsMaximizerOn
- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels
- JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous
- JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.paper_inferred_user_value
- JGS23SupplySideRecommenderSystems.paper_powered_unit_image
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nonempty (Fin N)
5. assumption: Nonempty (Fin P)
6. assumption: Nontrivial (Fin P)
7. parameter: Fin N → JGS23SupplySideRecommenderSystems.Content D
8. parameter: JGS23SupplySideRecommenderSystems.SourceNorm D
9. parameter: JGS23SupplySideRecommenderSystems.MixedContentStrategy D
10. parameter: Fin P
11. parameter: JGS23SupplySideRecommenderSystems.Content D
12. parameter: ℝ
13. assumption: JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
  (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ
14. assumption: JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ ⊆ {genre}
15. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)
16. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)
17. assumption: JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous ν
18. assumption: 0 < β
19. assumption: JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν
20. assumption: Continuous ν.norm
21. conclusion: AppliedModelingLib.Optimization.IsMaximizerOn

```
  (fun (z : Fin N → ℝ) =>
    z ∈
        {y' : Fin N → ℝ |
          JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
            (fun (p : JGS23SupplySideRecommenderSystems.Content D) =>
              JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧ ν.norm p = 1)
            β y'} ∧
      ∀ (i : Fin N), 0 < z i)
  (fun (y : Fin N → ℝ) => ∑ i : Fin N, Real.log (y i)) fun (i : Fin N) =>
  JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) genre ^ β
```

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.corollarySingleGenreStructure

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: Anchor 501-508 identifies the singleton genre with the Nash-welfare maximizer; the approved target uses the zero-safe nonzero-support premise and positive powered-score log-welfare domain, with the required norm regularity explicit.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 12

Source locator: cited publication, lines 581-584

Verbatim source input

```
\begin{restatable}{proposition}{supportrestriction}
\label{prop:supportrestriction}
Suppose that there are $N$ users split equally between two linearly independently vectors $u_1, u_2 \in \mathbb{R}_{\ge 0}^2$, and let $\theta^* :=
↪    \cos^{-1}\left( \frac{\langle u_1, u_2\rangle}{\|u_1\|_2 \|u_2\|} \right)$ be the angle between the user vectors. Let the cost function be $c(\p) =
↪    \|p\|_2^{\beta}$, and let $P \ge 2$. Let $\mu$ be a symmetric Nash equilibrium such that the distributions $\langle u_1, \p\rangle$ and $\langle u_2,
↪    \p\rangle$ over $\mathbb{R}_{\ge 0}$ are absolutely continuous. As long as $\beta \neq 2$ or $\theta^* \neq \pi /2$, the support of $\mu$ does not
↪    contain an $\ell_2$-ball of radius $\epsilon$ for any $\epsilon > 0$.\footnote{The case of $\beta = 2$ and $\theta^* = \pi/2$ is degenerate and permits
↪    a range of possible equilibria.}
\end{restatable}
```

Formalized review target   The archival source above is not asserted equivalent to this formalized target.

```
After per-user positive score normalization, a nondegenerate two-user L2 source symmetric equilibrium with absolutely continuous score laws has no
↪    positive-radius open content ball in its support.
```

Archival source anchor: cited publication, lines 581-584

Expanded PaperInterface specification

```
∀ {P : ℕ} [Nonempty (Fin P)] {α β θ ε : ℝ} {u v p0 : JGS23SupplySideRecommenderSystems.Content 2}
  {μ : MeasureTheory.Measure (JGS23SupplySideRecommenderSystems.Content 2)} [MeasureTheory.IsProbabilityMeasure μ],
  1 < P →
    (∀ (i : Fin 2), JGS23SupplySideRecommenderSystems.NonnegativeContent (if i = 0 then u else v)) →
      0 < ε →
        (∀ (p : JGS23SupplySideRecommenderSystems.Content 2),
            (p 0 - p0 0) ^ 2 + (p 1 - p0 1) ^ 2 < ε ^ 2 → p ∈ μ.support) →
          JGS23SupplySideRecommenderSystems.score u u = 1 →
            JGS23SupplySideRecommenderSystems.score v v = 1 →
              JGS23SupplySideRecommenderSystems.score u v = Real.cos θ →
                0 < α →
                  0 < β →
                    0 < Real.sin θ →
                      0 ≤ θ →
                        θ ≤ Real.pi / 2 →
                          β ≠ 2 ∨ θ ≠ Real.pi / 2 →
                            JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash
                                (fun (i : Fin 2) => if i = 0 then u else v)
                                (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
                                  α *
                                    JGS23SupplySideRecommenderSystems.normPowerCostSpec
                                      (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2) β q)
                                μ →
                              (∀ (i : Fin 2),
                                  (MeasureTheory.Measure.map
                                        (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
                                          JGS23SupplySideRecommenderSystems.score (if i = 0 then u else v) q)
                                        μ).AbsolutelyContinuous
                                    MeasureTheory.volume) →
                                False
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.SourceNorm.l2
- JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.score

Lean-elaborated claim roles

```
1. parameter: ℕ
2. assumption: Nonempty (Fin P)
3. parameter: ℝ
4. parameter: ℝ
5. parameter: ℝ
6. parameter: ℝ
7. parameter: JGS23SupplySideRecommenderSystems.Content 2
8. parameter: JGS23SupplySideRecommenderSystems.Content 2
9. parameter: JGS23SupplySideRecommenderSystems.Content 2
10. parameter: MeasureTheory.Measure (JGS23SupplySideRecommenderSystems.Content 2)
11. assumption: MeasureTheory.IsProbabilityMeasure μ
12. assumption: 1 < P
13. assumption: ∀ (i : Fin 2), JGS23SupplySideRecommenderSystems.NonnegativeContent (if i = 0 then u else v)
14. assumption: 0 < ε
```

```
15. assumption: ∀ (p : JGS23SupplySideRecommenderSystems.Content 2), (p 0 - p0 0) ^ 2 + (p 1 - p0 1) ^ 2 < ε ^ 2 → p ∈ μ.support
16. assumption: JGS23SupplySideRecommenderSystems.score u u = 1
17. assumption: JGS23SupplySideRecommenderSystems.score v v = 1
18. assumption: JGS23SupplySideRecommenderSystems.score u v = Real.cos θ
19. assumption: 0 < α
20. assumption: 0 < β
21. assumption: 0 < Real.sin θ
22. assumption: 0 ≤ θ
23. assumption: θ ≤ Real.pi / 2
24. assumption: β ≠ 2 ∨ θ ≠ Real.pi / 2
25. assumption: JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash (fun (i : Fin 2) => if i = 0 then u else v)
  (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
    α * JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2) β q)
  μ
26. assumption: ∀ (i : Fin 2),
  (MeasureTheory.Measure.map
        (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
          JGS23SupplySideRecommenderSystems.score (if i = 0 then u else v) q)
        μ).AbsolutelyContinuous
    MeasureTheory.volume
27. conclusion: False
```

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.propositionSupportRestriction

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: Anchor 581-584 excludes positive support balls for arbitrary two-user magnitudes. The approved target is the same normalized ranking-game conclusion with unit-score, angle, probability, and absolute-continuity normalization data explicit.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 13

Source locator: cited publication, lines 590-597

Verbatim source input

```
\begin{restatable}{theorem}{phasetransitionformal}
\label{thm:phasetransitionformal}
Suppose that there are $N$ users split equally between two linearly independently vectors $u_1, u_2 \in \mathbb{R}_{\ge 0}^D$, and let $\theta^* :=
↪    \cos^{-1}\left( \frac{\langle u_1, u_2\rangle}{\|u_1\|_2 \|u_2\|} \right)$ be the angle between the user vectors. Let the cost function be $c(\p) =
↪    \|p\|_2^{\beta}$. Let $\mu$ be a a distribution on $\mathbb{R}^d$ such that the distributions $\langle u_1, \p\rangle$ and $\langle u_2, \p\rangle$
↪    over $\mathbb{R}_{\ge 0}$ over $\mathbb{R}_{\ge 0}$ for $p \sim \mu$ are absolutely continuous and twice continuously differentiable within their
↪    supports. There are two regimes based on $\beta$ and $\theta^*$:
\begin{enumerate}
  \item If $\beta < \beta^* = \frac{2}{1 - \cos(\theta^*)}$ and if $\mu$ is a symmetric mixed equilibrium, then $\mu$ satisfies $|\Genre(\mu)| = 1$.
  \item If $\beta > \beta^* = \frac{2}{1 - \cos(\theta^*)}$, if $|\Genre(\mu)| < \infty$, and if the conditional distribution of $\|p\|$ along each genre is
  ↪    continuously differentiable, then $\mu$ is not an equilibrium.
\end{enumerate}
\end{restatable}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

```
For arbitrary nonzero nonnegative equal-population two-user L2 markets with absolutely continuous score laws and twice differentiable score CDFs on
↪    support, beta below the phase threshold forces one nonzero support genre, while beta above it excludes every finite conditional norm-law genre
↪    representation.
```

Archival source anchor: cited publication, lines 586-602

Expanded PaperInterface specification

```
∀ {D K P : ℕ} [Nonempty (Fin P)] {β θ : ℝ} {u v : JGS23SupplySideRecommenderSystems.Content D}
  {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D} [MeasureTheory.IsProbabilityMeasure μ],
  0 < K →
    1 < P →
      JGS23SupplySideRecommenderSystems.NonzeroContent u →
        JGS23SupplySideRecommenderSystems.NonzeroContent v →
          JGS23SupplySideRecommenderSystems.NonnegativeContent u →
            JGS23SupplySideRecommenderSystems.NonnegativeContent v →
              JGS23SupplySideRecommenderSystems.score u v /
                    (AppliedModelingLib.FiniteDimensionalNorms.l2 u * AppliedModelingLib.FiniteDimensionalNorms.l2 v) =
                  Real.cos θ →
                1 ≤ β →
                  0 < θ →
                    θ ≤ Real.pi / 2 →
                      JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash
                          (JGS23SupplySideRecommenderSystems.twoPopulationUsers K fun (i : Fin 2) =>
                            if i = 0 then u else v)
                          (JGS23SupplySideRecommenderSystems.normPowerCostSpec
                            (JGS23SupplySideRecommenderSystems.SourceNorm.l2 D) β)
                          μ →
                        (∀ (i : Fin 2),
                            (MeasureTheory.Measure.map
                                  (fun (q : JGS23SupplySideRecommenderSystems.Content D) =>
                                    JGS23SupplySideRecommenderSystems.score (if i = 0 then u else v) q)
                                  μ).AbsolutelyContinuous
                              MeasureTheory.volume) →
                          (∀
                              x ∈
                                (MeasureTheory.Measure.map
                                    (fun (q : JGS23SupplySideRecommenderSystems.Content D) =>
                                      JGS23SupplySideRecommenderSystems.score u q)
                                    μ).support,
                              ContDiffAt ℝ 2
                                (AppliedModelingLib.Probability.lowerCDFMass
                                  (MeasureTheory.Measure.map
                                    (fun (q : JGS23SupplySideRecommenderSystems.Content D) =>
                                      JGS23SupplySideRecommenderSystems.score u q)
                                    μ))
                                x) →
                            (∀
                                x ∈
                                  (MeasureTheory.Measure.map
                                      (fun (q : JGS23SupplySideRecommenderSystems.Content D) =>
                                        JGS23SupplySideRecommenderSystems.score v q)
                                      μ).support,
                                ContDiffAt ℝ 2
                                  (AppliedModelingLib.Probability.lowerCDFMass
                                    (MeasureTheory.Measure.map
                                      (fun (q : JGS23SupplySideRecommenderSystems.Content D) =>
                                        JGS23SupplySideRecommenderSystems.score v q)
                                      μ))
                                  x) →
                              (β < JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold θ →
                                  JGS23SupplySideRecommenderSystems.SourceNonzeroSupportGenres
                                      (JGS23SupplySideRecommenderSystems.SourceNorm.l2 D) μ =
                                    {(JGS23SupplySideRecommenderSystems.SourceNorm.l2 D).normalizedContent
                                        (JGS23SupplySideRecommenderSystems.twoUserNormalizedContent u +
                                          JGS23SupplySideRecommenderSystems.twoUserNormalizedContent v)}) ∧
                                (JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold θ < β →
                                  ∀ {G : ℕ},
                                    Nonempty (Fin G) →
                                      ∀ (direction : Fin G → JGS23SupplySideRecommenderSystems.Content D)
```

```
(a :
  JGS23SupplySideRecommenderSystems.FiniteGenreConditionalNormLawAnyDim μ
    direction),
Function.Injective direction → False)
```

Retained governing declarations

- AppliedModelingLib.FiniteDimensionalNorms.l2
- AppliedModelingLib.Probability.lowerCDFMass
- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.FiniteGenreConditionalNormLawAnyDim
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SourceNonzeroSupportGenres
- JGS23SupplySideRecommenderSystems.SourceNorm.l2
- JGS23SupplySideRecommenderSystems.SourceNorm.normalizedContent
- JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.score
- JGS23SupplySideRecommenderSystems.twoPopulationUsers
- JGS23SupplySideRecommenderSystems.twoUserNormalizedContent
- JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold

Lean-elaborated claim roles

```
1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nonempty (Fin P)
5. parameter: ℝ
6. parameter: ℝ
7. parameter: JGS23SupplySideRecommenderSystems.Content D
8. parameter: JGS23SupplySideRecommenderSystems.Content D
9. parameter: JGS23SupplySideRecommenderSystems.MixedContentStrategy D
10. assumption: MeasureTheory.IsProbabilityMeasure μ
11. assumption: 0 < K
12. assumption: 1 < P
13. assumption: JGS23SupplySideRecommenderSystems.NonzeroContent u
14. assumption: JGS23SupplySideRecommenderSystems.NonzeroContent v
15. assumption: JGS23SupplySideRecommenderSystems.NonnegativeContent u
16. assumption: JGS23SupplySideRecommenderSystems.NonnegativeContent v
17. assumption: JGS23SupplySideRecommenderSystems.score u v /
    (AppliedModelingLib.FiniteDimensionalNorms.l2 u * AppliedModelingLib.FiniteDimensionalNorms.l2 v) =
  Real.cos θ
18. assumption: 1 ≤ β
19. assumption: 0 < θ
20. assumption: θ ≤ Real.pi / 2
21. assumption: JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash
  (JGS23SupplySideRecommenderSystems.twoPopulationUsers K fun (i : Fin 2) => if i = 0 then u else v)
  (JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 D) β) μ
22. assumption: ∀ (i : Fin 2),
  (MeasureTheory.Measure.map
      (fun (q : JGS23SupplySideRecommenderSystems.Content D) =>
        JGS23SupplySideRecommenderSystems.score (if i = 0 then u else v) q)
      μ).AbsolutelyContinuous
    MeasureTheory.volume
23. assumption: ∀
  x ∈
    (MeasureTheory.Measure.map
      (fun (q : JGS23SupplySideRecommenderSystems.Content D) => JGS23SupplySideRecommenderSystems.score u q)
      μ).support,
  ContDiffAt ℝ 2
    (AppliedModelingLib.Probability.lowerCDFMass
      (MeasureTheory.Measure.map
        (fun (q : JGS23SupplySideRecommenderSystems.Content D) => JGS23SupplySideRecommenderSystems.score u q) μ))
    x
24. assumption: ∀
  x ∈
    (MeasureTheory.Measure.map
      (fun (q : JGS23SupplySideRecommenderSystems.Content D) => JGS23SupplySideRecommenderSystems.score v q)
      μ).support,
  ContDiffAt ℝ 2
    (AppliedModelingLib.Probability.lowerCDFMass
      (MeasureTheory.Measure.map
        (fun (q : JGS23SupplySideRecommenderSystems.Content D) => JGS23SupplySideRecommenderSystems.score v q) μ))
```

x
25. conclusion: (β < JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold θ →
  JGS23SupplySideRecommenderSystems.SourceNonzeroSupportGenres (JGS23SupplySideRecommenderSystems.SourceNorm.l2 D) μ =
   {(JGS23SupplySideRecommenderSystems.SourceNorm.l2 D).normalizedContent
     (JGS23SupplySideRecommenderSystems.twoUserNormalizedContent u +
      JGS23SupplySideRecommenderSystems.twoUserNormalizedContent v)}) ∧
 (JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold θ < β →
  ∀ {G : ℕ},
   Nonempty (Fin G) →
    ∀ (direction : Fin G → JGS23SupplySideRecommenderSystems.Content D)
     (a : JGS23SupplySideRecommenderSystems.FiniteGenreConditionalNormLawAnyDim μ direction),
     Function.Injective direction → False)

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.theoremPhaseTransition

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: The Lean theorem retains arbitrary ambient dimension and represents the equal populations by two nonzero nonnegative types repeated K times. Its normalized-cosine premise and theta range give the source angle, while source Nash, absolute continuity, and support-local C2 score CDF hypotheses expose the approved regularity domain. Below 2/(1-cos theta) it identifies the nonzero support genres with the singleton normalized sum direction; above the threshold it contradicts every injectively enumerated finite positive-weight ray decomposition whose conditional radial laws have C1 CDFs. Those are exactly the two approved corrected phase-transition implications.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 14

Source locator: cited publication, lines 616-619

Verbatim source input

```
\begin{restatable}{proposition}{Ptwo}
\label{prop:P2}
Suppose that there are 2 users located at the standard basis vectors $e_1, e_2 \in \mathbb{R}^2$, and the cost function is $c(\p) = \|p\|_2^{\beta}$. For
↪    $P = 2$ and $\beta \ge \beta^* = 2$, there is an equilibrium $\mu$ supported on the quarter-circle of radius $(2 \beta^{-1})^{1/\beta}$, where the
↪    angle $\theta \in [0, \pi/2]$ has density $f(\theta) = 2 \cos(\theta) \sin(\theta)$.
\end{restatable}
```

Expanded PaperInterface specification

```
∀ {β : ℝ},
  2 ≤ β →
    JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec
        JGS23SupplySideRecommenderSystems.pTwoStandardBasisUsers
        (JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2) β)
        (JGS23SupplySideRecommenderSystems.pTwoSourceContentMeasure β) ∧
      MeasureTheory.Measure.map JGS23SupplySideRecommenderSystems.pTwoStandardBasisValueMap
          (JGS23SupplySideRecommenderSystems.pTwoSourceContentMeasure β) =
         JGS23SupplySideRecommenderSystems.pTwoSourceCircleMeasure β ∧
        (JGS23SupplySideRecommenderSystems.pTwoSourceCircleMeasure β).support =
            {z : ℝ × ℝ | 0 ≤ z.1 ∧ 0 ≤ z.2 ∧ z.1 ^ 2 + z.2 ^ 2 = (2 / β) ^ (2 / β)} ∧
          JGS23SupplySideRecommenderSystems.pTwoSourceCircleMeasure β =
              MeasureTheory.Measure.map
                (fun (θ : ℝ) => ((2 / β) ^ (1 / β) * Real.cos θ, (2 / β) ^ (1 / β) * Real.sin θ))
                (MeasureTheory.volume.withDensity
                  ((Set.Icc 0 (Real.pi / 2)).indicator fun (θ : ℝ) => ENNReal.ofReal (2 * Real.cos θ * Real.sin θ))) ∧
            ∀ (z : ℝ),
              (JGS23SupplySideRecommenderSystems.pTwoSourceCircleMeasure β) {point : ℝ × ℝ | point.1 ≤ z} =
                ENNReal.ofReal (if z < 0 then 0 else min ((2 / β) ^ (-(2 / β)) * z ^ 2) 1)
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.SourceNorm.l2
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.pTwoSourceCircleMeasure
- JGS23SupplySideRecommenderSystems.pTwoSourceContentMeasure
- JGS23SupplySideRecommenderSystems.pTwoStandardBasisUsers
- JGS23SupplySideRecommenderSystems.pTwoStandardBasisValueMap
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

```
1. parameter: ℝ
2. assumption: 2 ≤ β
3. conclusion: JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec JGS23SupplySideRecommenderSystems.pTwoStandardBasisUsers
    (JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2) β)
    (JGS23SupplySideRecommenderSystems.pTwoSourceContentMeasure β) ∧
  MeasureTheory.Measure.map JGS23SupplySideRecommenderSystems.pTwoStandardBasisValueMap
      (JGS23SupplySideRecommenderSystems.pTwoSourceContentMeasure β) =
     JGS23SupplySideRecommenderSystems.pTwoSourceCircleMeasure β ∧
    (JGS23SupplySideRecommenderSystems.pTwoSourceCircleMeasure β).support =
        {z : ℝ × ℝ | 0 ≤ z.1 ∧ 0 ≤ z.2 ∧ z.1 ^ 2 + z.2 ^ 2 = (2 / β) ^ (2 / β)} ∧
      JGS23SupplySideRecommenderSystems.pTwoSourceCircleMeasure β =
          MeasureTheory.Measure.map (fun (θ : ℝ) => ((2 / β) ^ (1 / β) * Real.cos θ, (2 / β) ^ (1 / β) * Real.sin θ))
            (MeasureTheory.volume.withDensity
              ((Set.Icc 0 (Real.pi / 2)).indicator fun (θ : ℝ) => ENNReal.ofReal (2 * Real.cos θ * Real.sin θ))) ∧
        ∀ (z : ℝ),
          (JGS23SupplySideRecommenderSystems.pTwoSourceCircleMeasure β) {point : ℝ × ℝ | point.1 ≤ z} =
            ENNReal.ofReal (if z < 0 then 0 else min ((2 / β) ^ (-(2 / β)) * z ^ 2) 1)
```

Lean proof endpoint (built separately)

```
JGS23SupplySideRecommenderSystems.propositionPTwo
```

Recorded source-to-Spec screening **Verdict:** matches

Reason: Anchor 616-619 gives the P=2, beta >= 2 standard-basis quarter-circle equilibrium with radius and angle density. The target retains that exact content law, support, density parameterization, and coordinate CDF.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 15

Source locator: cited publication, lines 635-642

Verbatim source input

```
We next vary the number of producers $P$ while fixing $\beta = 2$ (see Figure \ref{fig:equilibriaP}). \begin{restatable}{proposition}{finiteP}
\label{prop:finiteP}
Suppose that there are 2 users located at the standard basis vectors $e_1, e_2 \in \mathbb{R}^2$, with cost function $c(\p) = \|p\|_2^{\beta}$. For $\beta
↪    = 2$, there is a multi-genre equilibrium $\mu$ with support equal to
\begin{equation}
\big\{\big(x, (1 -  x^{\frac{2}{P-1}})^{\frac{P-1}{2}}\big) \mid x \in [0,1] \big\},
\end{equation}
and where the distribution of $x$ has cdf equal to $\min(1, x^{2/(P-1)})$.
\end{restatable}
```

Expanded PaperInterface specification

```
∀ {P : ℕ},
  2 ≤ P →
    JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec
        JGS23SupplySideRecommenderSystems.pTwoStandardBasisUsers
        (JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2) 2)
        (JGS23SupplySideRecommenderSystems.finitePSourceContentMeasure P) ∧
      MeasureTheory.Measure.map JGS23SupplySideRecommenderSystems.pTwoStandardBasisValueMap
            (JGS23SupplySideRecommenderSystems.finitePSourceContentMeasure P) =
          JGS23SupplySideRecommenderSystems.finitePSourceCurveMeasure P ∧
        (JGS23SupplySideRecommenderSystems.finitePSourceCurveMeasure P).support =
            {z : ℝ × ℝ | z.1 ∈ Set.Icc 0 1 ∧ z.2 = (1 - z.1 ^ (2 / (↑P - 1))) ^ ((↑P - 1) / 2)} ∧
          ∀ (z : ℝ),
            (JGS23SupplySideRecommenderSystems.finitePSourceCurveMeasure P) {point : ℝ × ℝ | point.1 ≤ z} =
              ENNReal.ofReal (if z < 0 then 0 else min 1 (z ^ (2 / (↑P - 1))))
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.SourceNorm.l2
- JGS23SupplySideRecommenderSystems.finitePSourceContentMeasure
- JGS23SupplySideRecommenderSystems.finitePSourceCurveMeasure
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.pTwoStandardBasisUsers
- JGS23SupplySideRecommenderSystems.pTwoStandardBasisValueMap
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

```
1. parameter: ℕ
2. assumption: 2 ≤ P
3. conclusion: JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec JGS23SupplySideRecommenderSystems.pTwoStandardBasisUsers
    (JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2) 2)
    (JGS23SupplySideRecommenderSystems.finitePSourceContentMeasure P) ∧
  MeasureTheory.Measure.map JGS23SupplySideRecommenderSystems.pTwoStandardBasisValueMap
        (JGS23SupplySideRecommenderSystems.finitePSourceContentMeasure P) =
      JGS23SupplySideRecommenderSystems.finitePSourceCurveMeasure P ∧
    (JGS23SupplySideRecommenderSystems.finitePSourceCurveMeasure P).support =
        {z : ℝ × ℝ | z.1 ∈ Set.Icc 0 1 ∧ z.2 = (1 - z.1 ^ (2 / (↑P - 1))) ^ ((↑P - 1) / 2)} ∧
      ∀ (z : ℝ),
        (JGS23SupplySideRecommenderSystems.finitePSourceCurveMeasure P) {point : ℝ × ℝ | point.1 ≤ z} =
          ENNReal.ofReal (if z < 0 then 0 else min 1 (z ^ (2 / (↑P - 1))))
```

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.propositionFiniteP

Recorded source-to-Spec screening **Verdict:** matches

Reason: Anchor 635-642 gives the beta=2 standard-basis finite-P equilibrium, its exact curve support, and x CDF. The target states the same coordinate/value-law representation for every P >= 2.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 16

Source locator: cited publication, lines 659-666

Verbatim source input

```
\begin{restatable}{theorem}{infinitegenre}[Informal version of Theorem \ref{thm:infinitegenreformal}]
\label{thm:infinitegenre}
Suppose that there are 2 users located at two linearly independently vectors $u_1, u_2 \in \mathbb{R}_{\ge 0}^D$, let $\theta^* := \cos^{-1}\Big(
↪    \frac{\langle u_1, u_2\rangle}{\|u_1\|_2 \|u_2\|} \Big)$ be the angle between them.  Suppose we have cost function $c(\p) = \|p\|_2^{\beta}$, $\beta >
↪    \beta^* = \frac{2}{1 - \cos(\theta^*)}$, and $P = \infty$ producers. Then, there exists an equilibrium with two genres:
\[\left\{[\cos(\theta^G + \theta_{\text{min}}), \sin(\theta^G + \theta_{\text{min}})], [\cos(\theta^* - \theta^G + \theta_{\text{min}}), \sin(\theta^* -
↪    \theta^G + \theta_{\text{min}})] \right\} \]
where $\theta^G := \argmax_{\theta \le \theta^*/2} \left(\cos^{\beta}(\theta) + \cos^{\beta}(\theta^* - \theta)\right)$ and $\theta_{\text{min}} :=
↪    \min\big(\cos^{-1}\big(\frac{\langle u_1, e_1\rangle}{\|u_1\|} \big) , \cos^{-1}\big(\frac{\langle u_2, e_1\rangle}{\|u_2\|} \big) \big)$.

For each genre, the conditional quality distribution (i.e.~the distribution of the maximum quality $\|p\|$ along a genre, conditional on all of the producers
↪    choosing that genre) has cdf given by a countably-infinite piecewise function, where each piece is either constant or grows proportionally to
↪    $\|p\|^{2\beta}$.
\end{restatable}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

```
For a strictly acute two-user geometry and beta strictly above the two-user phase threshold, a coherent normalized two-genre infinite-producer equilibrium
↪    exists with the explicit candidate conditional CDF and support-wise global optimality.
```

Archival source anchor: cited publication, lines 659-666

Expanded PaperInterface specification

```
∀ {β θ : ℝ},
  0 < θ →
    θ < Real.pi / 2 →
      JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold θ < β →
        ∃ (a : ℝ) (C1 : ℝ) (C2 : ℝ) (A : ℝ) (B : ℝ) (phi : ℝ),
          JGS23SupplySideRecommenderSystems.infiniteProducerCandidateSpec
              (fun (q : ℝ) =>
                if q ≤ 0 then 0
                else
                  if a ≤ q then 1
                  else
                    have k : ℕ := if h : ∃ (n : ℕ), a * C2 ^ n ≤ q then Nat.find h else 0;
                    if Even k then C2 ^ (↑k * β) else C1 ^ (-2) * C2 ^ (-2 * ↑(k / 2) * β) * q ^ (2 * β))
              β θ phi ∧
            phi ∈ Set.Icc 0 (θ / 2) ∧
              IsMaxOn (fun (x : ℝ) => Real.cos x ^ β + Real.cos (θ - x) ^ β) (Set.Icc 0 (θ / 2)) phi ∧
                0 < a ∧
                  0 < C2 ∧
                    C2 < 1 ∧ C1 = a ^ β ∧ C1 = 1 + C2 ^ β ∧ A = Real.cos phi ∧ B = Real.cos (θ - phi) ∧ C2 = B / A
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.infiniteProducerCandidateSpec
- JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold

Lean-elaborated claim roles

```
1. parameter: ℝ
2. parameter: ℝ
3. assumption: 0 < θ
4. assumption: θ < Real.pi / 2
5. assumption: JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold θ < β
6. conclusion: ∃ (a : ℝ) (C1 : ℝ) (C2 : ℝ) (A : ℝ) (B : ℝ) (phi : ℝ),
  JGS23SupplySideRecommenderSystems.infiniteProducerCandidateSpec
      (fun (q : ℝ) =>
        if q ≤ 0 then 0
        else
          if a ≤ q then 1
          else
            have k : ℕ := if h : ∃ (n : ℕ), a * C2 ^ n ≤ q then Nat.find h else 0;
            if Even k then C2 ^ (↑k * β) else C1 ^ (-2) * C2 ^ (-2 * ↑(k / 2) * β) * q ^ (2 * β))
      β θ phi ∧
    phi ∈ Set.Icc 0 (θ / 2) ∧
      IsMaxOn (fun (x : ℝ) => Real.cos x ^ β + Real.cos (θ - x) ^ β) (Set.Icc 0 (θ / 2)) phi ∧
        0 < a ∧ 0 < C2 ∧ C2 < 1 ∧ C1 = a ^ β ∧ C1 = 1 + C2 ^ β ∧ A = Real.cos phi ∧ B = Real.cos (θ - phi) ∧ C2 = B / A
```

Lean proof endpoint (built separately)

```
JGS23SupplySideRecommenderSystems.theoremInfiniteGenreInformal
```

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: The archival informal theorem specifies a two-genre infinite-producer equilibrium above 2/(1-cos theta) with an alternating constant/power conditional CDF. Its displayed approved correction calls for the coherent normalized two-genre construction; the Lean target imposes the acute threshold regime, that explicit CDF, probability conditional law, equal half weights, and support-wise global optimality.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 17

Source locator: cited publication, lines 679-693

Verbatim source input

```
\begin{restatable}{lemma}{necessarysuff}
\label{claim:necessarysuff}
Let $\UMatrix = [u_1; u_2; \ldots; u_N]$ be the $N \times D$ matrix of users vectors. Given a set $S \subseteq \mathbb{R}_{\ge 0}^N$ and distributions
↪    $H_1, \ldots, H_N$ over $\mathbb{R}_{\ge 0}$, suppose that the following conditions hold:
\begin{description}
    \item{(C1)} Every $z^* \in S$ is a maximizer of the equation:
    \begin{equation}
\label{eq:reparam}
 \max_{z \in \mathbb{R}_{\ge 0}^D} \sum_{i=1}^N H_i(z_i) - c_{\UMatrix}(z),
\end{equation}
where $c_{\UMatrix}(z) := \min \left\{c(\p) \mid \p \in \mathbb{R}_{\ge 0}^D, \UMatrix p=z  \right\}$.
    \item{(C2)} There exists a random variable $Z$ with support $S$, such that the marginal distribution $Z_i$ has cdf equal to $H_i(z)^{1/(P-1)}$.
    \item{(C3)} $Z$ is distributed as $\UMatrix Y$ with $Y \sim \mu$, for some distribution $\mu$ over $\mathbb{R}_{\ge 0}^D$.
\end{description}
Then, the distribution $\mu$ from (C3) is a symmetric mixed Nash equilibrium. Moreover, every symmetric mixed Nash equilibrium $\mu$ is associated with
↪    some $(H_1, \ldots, H_N, S)$ that satisfy (C1)-(C3).
\end{restatable}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

```
With measurable score maps and null mass at every score level, a source symmetric mixed Nash equilibrium is equivalent to nonnegative support and
↪    support-wise maximization of the explicit tie-aware expected-winner objective.
```

Archival source anchor: cited publication, lines 679-693

Expanded PaperInterface specification

```
∀ {D N P : ℕ} [Nonempty (Fin P)] {users : Fin N → JGS23SupplySideRecommenderSystems.Content D}
  {cost : JGS23SupplySideRecommenderSystems.Content D → ℝ}
  {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D} [MeasureTheory.IsProbabilityMeasure μ],
  (∀ (i : Fin N),
      Measurable fun (q : JGS23SupplySideRecommenderSystems.Content D) =>
        JGS23SupplySideRecommenderSystems.score (users i) q) →
    (∀ (i : Fin N) (z : ℝ),
        μ {q : JGS23SupplySideRecommenderSystems.Content D | JGS23SupplySideRecommenderSystems.score (users i) q = z} =
          0) →
      (JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users cost μ ↔
        MeasureTheory.Measure.support μ ⊆
            {p : JGS23SupplySideRecommenderSystems.Content D | JGS23SupplySideRecommenderSystems.NonnegativeContent p} ∧
          ∀ (j : Fin P),
            ∀ p ∈ MeasureTheory.Measure.support μ,
              ∀ (q : JGS23SupplySideRecommenderSystems.Content D),
                JGS23SupplySideRecommenderSystems.NonnegativeContent q →
                  ∑ i : Fin N,
                        MeasureTheory.Measure.real μ
                            {r : JGS23SupplySideRecommenderSystems.Content D |
                              JGS23SupplySideRecommenderSystems.score (users i) r <
                                JGS23SupplySideRecommenderSystems.score (users i) q} ^
                          (P - 1) -
                      cost q ≤
                    ∑ i : Fin N,
                        MeasureTheory.Measure.real μ
                            {r : JGS23SupplySideRecommenderSystems.Content D |
                              JGS23SupplySideRecommenderSystems.score (users i) r <
                                JGS23SupplySideRecommenderSystems.score (users i) p} ^
                          (P - 1) -
                      cost p)
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.score
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

```
1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nonempty (Fin P)
5. parameter: Fin N → JGS23SupplySideRecommenderSystems.Content D
6. parameter: JGS23SupplySideRecommenderSystems.Content D → ℝ
7. parameter: JGS23SupplySideRecommenderSystems.MixedContentStrategy D
8. assumption: MeasureTheory.IsProbabilityMeasure μ
```

```
9. assumption: ∀ (i : Fin N),
  Measurable fun (q : JGS23SupplySideRecommenderSystems.Content D) =>
    JGS23SupplySideRecommenderSystems.score (users i) q
10. assumption: ∀ (i : Fin N) (z : ℝ),
  μ {q : JGS23SupplySideRecommenderSystems.Content D | JGS23SupplySideRecommenderSystems.score (users i) q = z} = 0
11. conclusion: JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users cost μ ↔
  MeasureTheory.Measure.support μ ⊆
      {p : JGS23SupplySideRecommenderSystems.Content D | JGS23SupplySideRecommenderSystems.NonnegativeContent p} ∧
    ∀ (j : Fin P),
      ∀ p ∈ MeasureTheory.Measure.support μ,
        ∀ (q : JGS23SupplySideRecommenderSystems.Content D),
          JGS23SupplySideRecommenderSystems.NonnegativeContent q →
            ∑ i : Fin N,
                  MeasureTheory.Measure.real μ
                      {r : JGS23SupplySideRecommenderSystems.Content D |
                        JGS23SupplySideRecommenderSystems.score (users i) r <
                          JGS23SupplySideRecommenderSystems.score (users i) q} ^
                    (P - 1) -
                cost q ≤
              ∑ i : Fin N,
                  MeasureTheory.Measure.real μ
                      {r : JGS23SupplySideRecommenderSystems.Content D |
                        JGS23SupplySideRecommenderSystems.score (users i) r <
                          JGS23SupplySideRecommenderSystems.score (users i) p} ^
                    (P - 1) -
                cost p
```

Lean proof endpoint (built separately)

```
JGS23SupplySideRecommenderSystems.claimNecessarySufficient
```

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: Anchor 679-693 uses a CDF payoff shortcut; the approved correction explicitly adds score-level nullness, and the target gives the corresponding strict-score, uniform-tie-aware support-action iff.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 18

Source locator: cited publication, lines 727-730

Verbatim source input

```
\begin{corollary}
\label{cor:2usersprofit}
Suppose that there are 2 users located at the standard basis vectors $e_1, e_2 \in \mathbb{R}^2$, and the cost function is $c(\p) = \|p\|_2^{\beta}$. For
↪    $P = 2$ and $\beta \ge \beta^* = 2$, there is an equilibrium $\mu$ where $\ProfitEq(\mu) = 1 - \frac{2}{\beta}$.
\end{corollary}
```

Expanded PaperInterface specification

```
∀ {β : ℝ},
  2 ≤ β →
    JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec
        JGS23SupplySideRecommenderSystems.pTwoStandardBasisUsers
        (JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2) β)
        (JGS23SupplySideRecommenderSystems.pTwoSourceContentMeasure β) ∧
      ∫ (p : JGS23SupplySideRecommenderSystems.Content 2),
          JGS23SupplySideRecommenderSystems.SourceMixedPurePayoff
            (JGS23SupplySideRecommenderSystems.ExpectedUsersWonAgainstSymmetricMixed
              JGS23SupplySideRecommenderSystems.pTwoStandardBasisUsers 0)
            (JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2) β)
            p
            (JGS23SupplySideRecommenderSystems.pTwoSourceContentMeasure
              β) ∂JGS23SupplySideRecommenderSystems.pTwoSourceContentMeasure β =
        1 - 2 / β
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.ExpectedUsersWonAgainstSymmetricMixed
- JGS23SupplySideRecommenderSystems.SourceMixedPurePayoff
- JGS23SupplySideRecommenderSystems.SourceNorm.l2
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.pTwoSourceContentMeasure
- JGS23SupplySideRecommenderSystems.pTwoStandardBasisUsers
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

```
1. parameter: ℝ
2. assumption: 2 ≤ β
3. conclusion: JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec JGS23SupplySideRecommenderSystems.pTwoStandardBasisUsers
    (JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2) β)
    (JGS23SupplySideRecommenderSystems.pTwoSourceContentMeasure β) ∧
  ∫ (p : JGS23SupplySideRecommenderSystems.Content 2),
      JGS23SupplySideRecommenderSystems.SourceMixedPurePayoff
        (JGS23SupplySideRecommenderSystems.ExpectedUsersWonAgainstSymmetricMixed
          JGS23SupplySideRecommenderSystems.pTwoStandardBasisUsers 0)
        (JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2) β) p
        (JGS23SupplySideRecommenderSystems.pTwoSourceContentMeasure
          β) ∂JGS23SupplySideRecommenderSystems.pTwoSourceContentMeasure β =
    1 - 2 / β
```

Lean proof endpoint (built separately)

```
JGS23SupplySideRecommenderSystems.corollaryTwoUsersProfit
```

Recorded source-to-Spec screening **Verdict:** matches

Reason: Anchor 727-730 states the P=2 standard-basis L2 equilibrium profit 1 - 2/beta for beta at least 2; the target states that same law and its expected source payoff.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 19

Source locator: cited publication, lines 734-742

Verbatim source input

```
\begin{restatable}{proposition}{utility}
\label{thm:utility}
Suppose that
\begin{equation}
\label{eq:positiveprofit}
\max_{\|\p\| \le 1} \min_{1 \le i \le N} \Big\langle \p, \frac{u_i}{||u_i||}\Big\rangle < N^{-P/\beta}.
\end{equation}
Then for any symmetric equilibrium $\mu$, the profit $\ProfitEq(\mu)$ is strictly positive.
\end{restatable}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

For the normalized Euclidean source game, the displayed strict geometric Q bound implies strictly positive common expected equilibrium profit.

Archival source anchor: cited publication, lines 734-742

Expanded PaperInterface specification

```
∀ {D N P : ℕ} [Nonempty (Fin N)] [Nontrivial (Fin P)] {users : Fin N → JGS23SupplySideRecommenderSystems.Content D}
  {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D} {β profit Q : ℝ},
  JGS23SupplySideRecommenderSystems.equilibriumProfitModelSpec users (JGS23SupplySideRecommenderSystems.SourceNorm.l2 D)
      (JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 D) β) μ
      profit Q →
    (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
      0 < β → Q < (1 / ↑N) ^ (↑P / β) → 0 < profit
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.SourceNorm.l2
- JGS23SupplySideRecommenderSystems.equilibriumProfitModelSpec
- JGS23SupplySideRecommenderSystems.normPowerCostSpec

Lean-elaborated claim roles

```
1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nonempty (Fin N)
5. assumption: Nontrivial (Fin P)
6. parameter: Fin N → JGS23SupplySideRecommenderSystems.Content D
7. parameter: JGS23SupplySideRecommenderSystems.MixedContentStrategy D
8. parameter: ℝ
9. parameter: ℝ
10. parameter: ℝ
11. assumption: JGS23SupplySideRecommenderSystems.equilibriumProfitModelSpec users (JGS23SupplySideRecommenderSystems.SourceNorm.l2 D)
  (JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 D) β) μ profit Q
12. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)
13. assumption: 0 < β
14. assumption: Q < (1 / ↑N) ^ (↑P / β)
15. conclusion: 0 < profit
```

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.propositionUtility

Recorded source-to-Spec screening

Verdict: matches_approved_corrected_target

Reason: The displayed equilibrium-profit predicate unfolds Q as the supremum over nonnegative L2-unit-ball actions of their minimum score against L2-normalized nonzero users, binds one profit value to every producer's source payoff integral, and includes the full symmetric Nash condition. With nonnegative users and positive beta, the Lean endpoint turns Q < (1/N)^(P/beta), the same as the source's N^(-P/beta) threshold, into strictly positive profit. This is the normalized Euclidean profit statement authorized by the approved bridge repair.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 20

Source locator: cited publication, lines 749-752

Verbatim source input

```
\begin{restatable}{proposition}{zeroutilitysinglegenre}
\label{prop:zeroutilitysinglegenre}
If $\mu$ is a single-genre equilibrium, then the profit $\ProfitEq(\mu)$ is equal to $0$.
\end{restatable}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

```
Under the explicit source-Nash, atomlessness, singleton-nonzero-genre, positive-score, measurable proper-norm, and compactness hypotheses, the
↪    common expected equilibrium profit is zero.
```

Archival source anchor: cited publication, lines 748-990

Expanded PaperInterface specification

```
∀ {D N P : ℕ} [Nonempty (Fin N)] [Nontrivial (Fin P)] {users : Fin N → JGS23SupplySideRecommenderSystems.Content D}
  {ν : JGS23SupplySideRecommenderSystems.SourceNorm D} {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D}
  {j : Fin P} {genre : JGS23SupplySideRecommenderSystems.Content D} {β profit : ℝ},
  JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
      (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ →
    ∫ (q : JGS23SupplySideRecommenderSystems.Content D),
          JGS23SupplySideRecommenderSystems.SourceMixedPurePayoff
            (JGS23SupplySideRecommenderSystems.ExpectedUsersWonAgainstSymmetricMixed users j)
            (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) q μ ∂μ =
        profit →
      MeasureTheory.NoAtoms μ →
        JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ ⊆ {genre} →
          (∀ (i : Fin N), 0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) genre) →
            Measurable ν.norm →
              (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
                0 < β → JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν → Continuous ν.norm → profit = 0
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.ExpectedUsersWonAgainstSymmetricMixed
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.SourceMixedPurePayoff
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels
- JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.paper_inferred_user_value
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

```
1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nonempty (Fin N)
5. assumption: Nontrivial (Fin P)
6. parameter: Fin N → JGS23SupplySideRecommenderSystems.Content D
7. parameter: JGS23SupplySideRecommenderSystems.SourceNorm D
8. parameter: JGS23SupplySideRecommenderSystems.MixedContentStrategy D
9. parameter: Fin P
10. parameter: JGS23SupplySideRecommenderSystems.Content D
11. parameter: ℝ
12. parameter: ℝ
13. assumption: JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
  (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ
14. assumption: ∫ (q : JGS23SupplySideRecommenderSystems.Content D),
    JGS23SupplySideRecommenderSystems.SourceMixedPurePayoff
      (JGS23SupplySideRecommenderSystems.ExpectedUsersWonAgainstSymmetricMixed users j)
      (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) q μ ∂μ =
  profit
15. assumption: MeasureTheory.NoAtoms μ
16. assumption: JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ ⊆ {genre}
17. assumption: ∀ (i : Fin N), 0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) genre
```

18. assumption: Measurable ν.norm
19. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)
20. assumption: 0 < β
21. assumption: JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν
22. assumption: Continuous ν.norm
23. conclusion: profit = 0

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.propositionZeroUtilitySingleGenre

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: The corrected zero-profit target makes all source-proof regularity visible. Lean assumes the uniform-tie symmetric Nash condition, identifies profit with the focal producer's mixed payoff integral, requires an atomless law, at most one normalized nonzero support direction with strictly positive scores for every user, a measurable continuous source norm with compact sublevels, nonnegative users, and positive beta, and concludes that profit is zero. The expanded support-genre and payoff definitions implement the approved zero-safe singleton-genre reading without changing the conclusion.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 21

Source locator: cited publication, lines 931-937

Verbatim source input

```
\begin{restatable}{lemma}{cdf}
\label{lemma:cdf}
Suppose that $\mu$ is a symmetric equilibrium such that $\Genre(\mu)$ contains a single vector. Let $F$ be the cdf of the distribution over $\|\p\|$ where
↪   $\p \sim \mu$. Then, it holds that:
\begin{equation}
F(r) = \min\left(1, \left(\frac{r^{\beta}}{N}\right)^{1/(P-1)}\right).
\end{equation}
\end{restatable}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

```
For a measurable source norm and a unit genre, the norm pushforward of the single-genre ray law has the corrected clipped radial CDF at every
↪   nonnegative radius.
```

Archival source anchor: cited publication, lines 931-937

Expanded PaperInterface specification

```
∀ {D N P : ℕ} [Nonempty (Fin N)] [Nontrivial (Fin P)] {β z : ℝ} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D)
  {genre : JGS23SupplySideRecommenderSystems.Content D},
  Measurable ν.norm →
    ν.norm genre = 1 →
      0 < β →
        0 ≤ z →
          (MeasureTheory.Measure.map ν.norm (JGS23SupplySideRecommenderSystems.singleGenreContentLaw N P β genre))
              (Set.Iic z) =
            ENNReal.ofReal (min 1 ((z ^ β / ↑N) ^ (↑P - 1)⁻¹))
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.singleGenreContentLaw

Lean-elaborated claim roles

```
1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nonempty (Fin N)
5. assumption: Nontrivial (Fin P)
6. parameter: ℝ
7. parameter: ℝ
8. parameter: JGS23SupplySideRecommenderSystems.SourceNorm D
9. parameter: JGS23SupplySideRecommenderSystems.Content D
10. assumption: Measurable ν.norm
11. assumption: ν.norm genre = 1
12. assumption: 0 < β
13. assumption: 0 ≤ z
14. conclusion: (MeasureTheory.Measure.map ν.norm (JGS23SupplySideRecommenderSystems.singleGenreContentLaw N P β genre)) (Set.Iic z) =
  ENNReal.ofReal (min 1 ((z ^ β / ↑N) ^ (↑P - 1)⁻¹))
```

Lean proof endpoint (built separately)

```
JGS23SupplySideRecommenderSystems.lemmaCdf
```

Recorded source-to-Spec screening

Verdict: matches_approved_corrected_target

Reason: Anchor 931-937 gives the radial CDF formula; the approved target transparently states its corrected clipped form for the source-norm pushforward of the single-ray law.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 22

Source locator: cited publication, lines 958-963

Verbatim source input

```
\begin{lemma}
\label{lemma:nonzero}
Suppose that $\mu$ is a symmetric equilibrium such that $\Genre(\mu)$ contains a single vector $\p^*$. Then $p^* \in \text{span}(u_1, \ldots, u_N)$
↪    (which also means that $\langle \p^*, u_i \rangle > 0$ for all $i$.)
\end{lemma}
\begin{proof}
Assume for sake of contradiction that $\langle p^*, u_i \rangle = 0$ for some $i$. Suppose that $p' \in \text{supp}(\mu)$, and consider the vector $\p' +
↪    \epsilon \frac{u_i}{||u_i||}$. We see that $\p' + \epsilon \frac{u_i}{||u_i||}$ wins user $u_i$ with probability 1 whereas $\p'$ wins user $u_i$ with
↪    probability $1/P$. The probability that $\p + \epsilon u_i$ wins any other user is also at least the probability that $\p'$ wins $u_i$. By leveraging this
↪    discontinuity, we see there exists $\epsilon$ such that $\Profit(\p' + \epsilon \frac{u_i}{||u_i||}; [\mu, \ldots, \mu]) > \Profit(\p'; [\mu, \ldots, \mu]) + (1 -
↪    \frac{1}{P})$ which is a contradiction.
```

Formalized review target   The archival source above is not asserted equivalent to this formalized target.

```
In a source symmetric equilibrium with one nonzero support genre, every nonzero nonnegative user's score against that genre is strictly positive when
↪    perturbation costs are continuous.
```

Archival source anchor: cited publication, lines 958-963

Expanded PaperInterface specification

```
∀ {D N P : ℕ} [Nonempty (Fin N)] [Nontrivial (Fin P)] {users : Fin N → JGS23SupplySideRecommenderSystems.Content D}
  {ν : JGS23SupplySideRecommenderSystems.SourceNorm D} {cost : JGS23SupplySideRecommenderSystems.Content D → ℝ}
  {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D} {genre : JGS23SupplySideRecommenderSystems.Content D},
  JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users cost μ →
    JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ ⊆ {genre} →
      (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
        (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)) →
          JGS23SupplySideRecommenderSystems.PerturbationCostContinuous cost →
            ∀ (i : Fin N), 0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) genre
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.PerturbationCostContinuous
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec
- JGS23SupplySideRecommenderSystems.paper_inferred_user_value
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

```
1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nonempty (Fin N)
5. assumption: Nontrivial (Fin P)
6. parameter: Fin N → JGS23SupplySideRecommenderSystems.Content D
7. parameter: JGS23SupplySideRecommenderSystems.SourceNorm D
8. parameter: JGS23SupplySideRecommenderSystems.Content D → ℝ
9. parameter: JGS23SupplySideRecommenderSystems.MixedContentStrategy D
10. parameter: JGS23SupplySideRecommenderSystems.Content D
11. assumption: JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users cost μ
12. assumption: JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ ⊆ {genre}
13. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)
14. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)
15. assumption: JGS23SupplySideRecommenderSystems.PerturbationCostContinuous cost
16. parameter: Fin N
17. conclusion: 0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) genre
```

Lean proof endpoint (built separately)

```
JGS23SupplySideRecommenderSystems.lemmaNonzero
```

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: Anchor 958-963 incorrectly treats span membership as positive score. The approved target states the economically used positive-score conclusion under visible source-Nash, nonzero-user, and perturbation-continuity conditions.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 23

Source locator: cited publication, lines 1002-1009

Verbatim source input

```
\begin{lemma}[Formalization of Lemma \ref{lemma:optimizationprogram}]
\label{lemma:optimizationprogramrestated}
There exists a symmetric equilibrium $\mu$ with $|\Genre(\mu)| = 1$ if and only if:
\begin{equation}
    \label{eq:optimizationprogramrestated}
\inf_{p^* \in \Ballm} \sup_{y' \in \Set^{\beta}} \sum_{i=1}^N  \frac{y'_i}{(\langle \p^*, u_i \rangle)^{\beta}} = \sup_{y' \in \Set^{\beta}}\inf_{p^* \in
↪    \Ballm}  \sum_{i=1}^N  \frac{y'_i}{(\langle \p^*, u_i \rangle)^{\beta}}.
\end{equation}
\end{lemma}
```

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For nonnegative nonzero users and compact source-norm sublevels, the appendix Ball_m formulation is equivalent to the same attained positive
↪   feasible-ray source-Nash certificate and source product condition.

Archival source anchor: cited publication, lines 1002-1009

Expanded PaperInterface specification

```
∀ {D N P : ℕ} [Nonempty (Fin N)] [Nonempty (Fin P)] [Nontrivial (Fin P)]
  {users : Fin N → JGS23SupplySideRecommenderSystems.Content D} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D)
  {β : ℝ},
  0 < β →
    (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
      (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)) →
        JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν →
          ((∃ (p : JGS23SupplySideRecommenderSystems.Content D),
              JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧
                ν.norm p ≤ 1 ∧
                  (∀ (i : Fin N), 0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p) ∧
                    JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
                      (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β)
                      (JGS23SupplySideRecommenderSystems.singleGenreContentLaw N P β (ν.normalizedContent p))) ↔
            JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition
              {y : Fin N → ℝ |
                JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
                  (fun (p : JGS23SupplySideRecommenderSystems.Content D) =>
                    JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧ ν.norm p ≤ 1)
                  β y})
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.SourceNorm.normalizedContent
- JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.paper_inferred_user_value
- JGS23SupplySideRecommenderSystems.paper_powered_unit_image
- JGS23SupplySideRecommenderSystems.singleGenreContentLaw
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nonempty (Fin N)
5. assumption: Nonempty (Fin P)
6. assumption: Nontrivial (Fin P)
7. parameter: Fin N → JGS23SupplySideRecommenderSystems.Content D
8. parameter: JGS23SupplySideRecommenderSystems.SourceNorm D
9. parameter: ℝ
10. assumption: 0 < β
11. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)
12. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)
13. assumption: JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν
14. conclusion: (∃ (p : JGS23SupplySideRecommenderSystems.Content D),
    JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧
      ν.norm p ≤ 1 ∧
        (∀ (i : Fin N), 0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p) ∧
          JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
            (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β)
            (JGS23SupplySideRecommenderSystems.singleGenreContentLaw N P β (ν.normalizedContent p))) ↔
  JGS23SupplySideRecommenderSystems.SingleGenreProductSupCondition
    {y : Fin N → ℝ |
      JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
        (fun (p : JGS23SupplySideRecommenderSystems.Content D) =>
          JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧ ν.norm p ≤ 1)
        β y}

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.lemmaOptimizationProgramRestated

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: Anchor 1002-1009 is the Ball_m minimax restatement. The approved target retains its positive feasible-ray domain and replaces the unsupported ratio minimax equality by the attained source-Nash certificate/product-condition iff.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 24

Source locator: cited publication, lines 1012-1019

Verbatim source input

```
\begin{lemma}
\label{lemma:conditiongen}
There exists a symmetric equilibrium $\mu$ with $|\Genre(\mu)| = 1$ if and only if:
\begin{equation}
\label{eq:conditioninter}
\inf_{p^* \in \Ballm} \sup_{y' \in \Set^{\beta}} \sum_{i=1}^N  \frac{y'_i}{(\langle \p^*, u_i \rangle)^{\beta}} \le N.
\end{equation}
\end{lemma}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

```
For nonnegative nonzero users and the explicit perturbation-continuity, compact-sublevel, and norm-continuity package, a source symmetric equilibrium
↪    with one nonzero support genre exists if and only if there is a positive-score nonnegative source-unit-ball point satisfying the displayed ratio bound.
```

Archival source anchor: cited publication, lines 1012-1019

Expanded PaperInterface specification

```
∀ {D N P : ℕ} [Nonempty (Fin N)] [Nonempty (Fin P)] [Nontrivial (Fin P)] {β : ℝ}
  {users : Fin N → JGS23SupplySideRecommenderSystems.Content D} (ν : JGS23SupplySideRecommenderSystems.SourceNorm D),
  0 < β →
    (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
      (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)) →
        JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous ν →
          JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν →
            Continuous ν.norm →
              ((∃ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) (genre :
                  JGS23SupplySideRecommenderSystems.Content D),
                  JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
                      (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ ∧
                    JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ = {genre}) ↔
                ∃ (p : JGS23SupplySideRecommenderSystems.Content D),
                  JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧
                    ν.norm p ≤ 1 ∧
                      (∀ (i : Fin N), 0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p) ∧
                        ∀ (y' : Fin N → ℝ),
                          JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
                              (fun (q : JGS23SupplySideRecommenderSystems.Content D) =>
                                JGS23SupplySideRecommenderSystems.NonnegativeContent q ∧ ν.norm q ≤ 1)
                              β y' →
                            ∑ i : Fin N,
                                y' i / JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p ^ β ≤
                              ↑N)
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels
- JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous
- JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.paper_inferred_user_value
- JGS23SupplySideRecommenderSystems.paper_powered_unit_image
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nonempty (Fin N)
5. assumption: Nonempty (Fin P)
6. assumption: Nontrivial (Fin P)
7. parameter: ℝ
8. parameter: Fin N → JGS23SupplySideRecommenderSystems.Content D
9. parameter: JGS23SupplySideRecommenderSystems.SourceNorm D
10. assumption: 0 < β
11. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)
12. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)
13. assumption: JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous ν
14. assumption: JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν
15. assumption: Continuous ν.norm
16. conclusion: (∃ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D) (genre : JGS23SupplySideRecommenderSystems.Content D),
  JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
    (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ ∧
   JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ = {genre}) ↔
 ∃ (p : JGS23SupplySideRecommenderSystems.Content D),
  JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧
   ν.norm p ≤ 1 ∧
    (∀ (i : Fin N), 0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p) ∧
     ∀ (y' : Fin N → ℝ),
      JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
        (fun (q : JGS23SupplySideRecommenderSystems.Content D) =>
         JGS23SupplySideRecommenderSystems.NonnegativeContent q ∧ ν.norm q ≤ 1)
        β y' →
       ∑ i : Fin N, y' i / JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) p ^ β ≤ ↑N

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.lemmaConditionGen

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: Anchor 1012-1019 uses an unattained infimum over the positive ball. The approved target replaces it by the equivalent positive feasible witness and explicit compactness, continuity, and perturbation hypotheses.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 25

Source locator: cited publication, lines 1023-1030

Verbatim source input

```
\begin{lemma}
\label{lemma:optsingledirection}
There is a symmetric equilibrium $\mu$ with $\Genre(\mu) = \left\{p^*\right\}$ if and only if:
\begin{equation}
\label{eq:optsingle}
  \sup_{y' \in \Set^{\beta}} \sum_{i=1}^N  \frac{y'_i}{(\langle \p^*, u_i \rangle)^{\beta}} \le N.
\end{equation}
\end{lemma}
```

Formalized review target   The archival source above is not asserted equivalent to this formalized target.

```
For a nonnegative unit genre with strictly positive user scores, nonnegative nonzero users, and the explicit perturbation-continuity, compact-sublevel, and
↪   norm-continuity package, a source symmetric equilibrium with exactly that one nonzero support genre exists if and only if every powered nonnegative
↪   unit-ball score row satisfies the displayed ratio bound.
```

Archival source anchor: cited publication, lines 1023-1030

Expanded PaperInterface specification

```
∀ {D N P : ℕ} [Nonempty (Fin N)] [Nonempty (Fin P)] [Nontrivial (Fin P)] {β : ℝ}
  {users : Fin N → JGS23SupplySideRecommenderSystems.Content D} {genre : JGS23SupplySideRecommenderSystems.Content D}
  (ν : JGS23SupplySideRecommenderSystems.SourceNorm D),
  0 < β →
    ν.norm genre = 1 →
      JGS23SupplySideRecommenderSystems.NonnegativeContent genre →
        (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)) →
          (∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)) →
            (∀ (i : Fin N), 0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) genre) →
              JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous ν →
                JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν →
                  Continuous ν.norm →
                    ((∃ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D),
                        JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
                            (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ ∧
                          JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ = {genre}) ↔
                      ∀ (y' : Fin N → ℝ),
                        JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
                            (fun (p : JGS23SupplySideRecommenderSystems.Content D) =>
                              JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧ ν.norm p ≤ 1)
                            β y' →
                          ∑ i : Fin N,
                              y' i / JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) genre ^ β ≤
                            ↑N)
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels
- JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous
- JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.paper_inferred_user_value
- JGS23SupplySideRecommenderSystems.paper_powered_unit_image
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. assumption: Nonempty (Fin N)
5. assumption: Nonempty (Fin P)
6. assumption: Nontrivial (Fin P)
7. parameter: ℝ
8. parameter: Fin N → JGS23SupplySideRecommenderSystems.Content D
9. parameter: JGS23SupplySideRecommenderSystems.Content D
10. parameter: JGS23SupplySideRecommenderSystems.SourceNorm D
11. assumption: 0 < β
12. assumption: ν.norm genre = 1
13. assumption: JGS23SupplySideRecommenderSystems.NonnegativeContent genre
14. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonnegativeContent (users i)
15. assumption: ∀ (i : Fin N), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)
16. assumption: ∀ (i : Fin N), 0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) genre
17. assumption: JGS23SupplySideRecommenderSystems.SourceNormPerturbationContinuous ν
18. assumption: JGS23SupplySideRecommenderSystems.SourceNormCompactSublevels ν
19. assumption: Continuous ν.norm
20. conclusion: (∃ (μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D),
    JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec users
        (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β) μ ∧
      JGS23SupplySideRecommenderSystems.nonzeroSupportGenresSpec ν μ = {genre}) ↔
  ∀ (y' : Fin N → ℝ),
    JGS23SupplySideRecommenderSystems.paper_powered_unit_image users
        (fun (p : JGS23SupplySideRecommenderSystems.Content D) =>
          JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧ ν.norm p ≤ 1)
        β y' →
      ∑ i : Fin N, y' i / JGS23SupplySideRecommenderSystems.paper_inferred_user_value (users i) genre ^ β ≤ ↑N

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.lemmaOptSingleDirection

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: Anchor 1023-1030 gives the fixed-direction ratio criterion. The approved target keeps that criterion and adds the positive unit-direction, zero-safe singleton, compactness, continuity, and perturbation premises used by the ray-law identification.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 26

Source locator: cited publication, lines 1083-1093

Verbatim source input

```
\begin{lemma}
\label{lemma:supinf}
For any set $\mathcal{R} \subseteq \bR_{> 0}^N$, it holds that:
\[\sup_{y' \in \mathcal{R}} \inf_{y \in \mathcal{R}}  \sum_{i=1}^N  \frac{y'_i}{y_i} = N. \]
\end{lemma}
\begin{proof}
By taking $y' = y$, we see that:
\[\sup_{y' \in \mathcal{R}} \inf_{y \in \mathcal{R}}  \sum_{i=1}^N  \frac{y'_i}{y_i} \le N. \]
To show equality, notice by AM-GM that:
\[\sum_{i=1}^N  \frac{y'_i}{y_i} \ge N \left(\prod_{i=1}^n \frac{y'_i}{y_i}\right)^{1/N} = N \left(\frac{\prod_{i=1}^n y'_i}{\prod_{i=1}^N
↪    y_i}\right)^{1/N}. \]
We can take $y' = \argmax_{y'' \in \mathcal{R}} \prod_{i=1}^n y''_i$, and obtain a lower bound of $N$ as desired. (If the $\argmax$ does not exist, then
↪    note that if we take $y'$ where $\prod_{i=1}^n y'_i$ is sufficiently close to the optimum $\sup_{y'' \in \mathcal{R}} \prod_{i=1}^n y''_i$, we have
↪    that $\inf_{y \in \mathcal{R}}  \left(\frac{\prod_{i=1}^n y'_i}{\prod_{i=1}^N y_i}\right)^{1/N}$ is sufficiently close to $1$ as desired.)
```

Formalized review target   The archival source above is not asserted equivalent to this formalized target.

```
For a positive feasible row set with an attained coordinate-product maximum, that maximizer attains the source ratio objective's minimum with value equal
↪    to the number of coordinates.
```

Archival source anchor: cited publication, lines 1083-1093

Expanded PaperInterface specification

```
∀ {N : ℕ} [Nonempty (Fin N)] {R : Set (Fin N → ℝ)} {yStar : Fin N → ℝ},
  JGS23SupplySideRecommenderSystems.IsCoordinateProductMaximizer R yStar →
    (∀ z ∈ R, ∀ (i : Fin N), 0 < z i) →
      AppliedModelingLib.Optimization.IsMinimizerOn (fun (y : Fin N → ℝ) => y ∈ R)
          (fun (y : Fin N → ℝ) => JGS23SupplySideRecommenderSystems.paper_ratio_objective y yStar) yStar ∧
        JGS23SupplySideRecommenderSystems.paper_ratio_objective yStar yStar = ↑N
```

Retained governing declarations

- AppliedModelingLib.Optimization.IsMinimizerOn
- JGS23SupplySideRecommenderSystems.IsCoordinateProductMaximizer
- JGS23SupplySideRecommenderSystems.paper_ratio_objective

Lean-elaborated claim roles

```
1. parameter: ℕ
2. assumption: Nonempty (Fin N)
3. parameter: Set (Fin N → ℝ)
4. parameter: Fin N → ℝ
5. assumption: JGS23SupplySideRecommenderSystems.IsCoordinateProductMaximizer R yStar
6. assumption: ∀ z ∈ R, ∀ (i : Fin N), 0 < z i
7. conclusion: AppliedModelingLib.Optimization.IsMinimizerOn (fun (y : Fin N → ℝ) => y ∈ R)
    (fun (y : Fin N → ℝ) => JGS23SupplySideRecommenderSystems.paper_ratio_objective y yStar) yStar ∧
  JGS23SupplySideRecommenderSystems.paper_ratio_objective yStar yStar = ↑N
```

Lean proof endpoint (built separately)

```
JGS23SupplySideRecommenderSystems.lemmaSupInf
```

Recorded source-to-Spec screening   **Verdict:** matches_approved_corrected_target

Reason: Anchor 1083-1093 asserts a false arbitrary-set sup-inf equality. The approved target is the attained positive coordinate-product-maximizer form, with minimum value N, exactly the valid AM-GM conclusion.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 27

Source locator: cited publication, lines 1279-1282

Verbatim source input

```
\begin{claim}
\label{claim:equivalence}
A distribution $\mu$ is an equilibria for a marketplace with 2 populations of users of size $N/2$ located at vectors $u_1$ and $u_2$ and with producer cost
↪    function $c(\p) = \|p\|_2^{\beta}$ if and only if $\mu$ is an equilibria for a marketplace with 2 users located at vectors $\frac{u_1}{\|u_1\|}$ and
↪    $\frac{u_2}{\|u_2\|}$ and with producer cost function $c(\p) = \frac{2}{N} \|p\|_2^{\beta}$.
\end{claim}
```

Expanded PaperInterface specification

```
∀ {D K P : ℕ} (users : Fin 2 → JGS23SupplySideRecommenderSystems.Content D)
  {cost : JGS23SupplySideRecommenderSystems.Content D → ℝ}
  {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy D},
  0 < K →
    (∀ (i : Fin 2), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)) →
      (JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec
          (JGS23SupplySideRecommenderSystems.twoPopulationUsers K users) cost μ ↔
        JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec
          (JGS23SupplySideRecommenderSystems.l2NormalizedUsers users)
          (fun (q : JGS23SupplySideRecommenderSystems.Content D) => 2 / ↑(2 * K) * cost q) μ)
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonzeroContent
- JGS23SupplySideRecommenderSystems.l2NormalizedUsers
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec
- JGS23SupplySideRecommenderSystems.twoPopulationUsers

Lean-elaborated claim roles

```
1. parameter: ℕ
2. parameter: ℕ
3. parameter: ℕ
4. parameter: Fin 2 → JGS23SupplySideRecommenderSystems.Content D
5. parameter: JGS23SupplySideRecommenderSystems.Content D → ℝ
6. parameter: JGS23SupplySideRecommenderSystems.MixedContentStrategy D
7. assumption: 0 < K
8. assumption: ∀ (i : Fin 2), JGS23SupplySideRecommenderSystems.NonzeroContent (users i)
9. conclusion: JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec
    (JGS23SupplySideRecommenderSystems.twoPopulationUsers K users) cost μ ↔
  JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec
    (JGS23SupplySideRecommenderSystems.l2NormalizedUsers users)
    (fun (q : JGS23SupplySideRecommenderSystems.Content D) => 2 / ↑(2 * K) * cost q) μ
```

Lean proof endpoint (built separately)

```
JGS23SupplySideRecommenderSystems.claimEquivalence
```

Recorded source-to-Spec screening **Verdict:** matches

Reason: Anchor 1279-1282 gives the equal-two-population iff; with N = 2K, the target's 2/(2K) cost factor is exactly 2/N, and its general cost form specializes to the source L2-power cost.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 28

Source locator: cited publication, lines 1311-1315

Verbatim source input

```
\begin{lemma}
\label{lemma:inducedcost}
Let there be 2 users located at $u_1, u_2 \in \mathbb{R}_{\ge 0}^D$ such that  $\|u_1\| = \|u_2\| = 1$, and let $\theta^* := \cos^{-1}\left( \langle u_1,
↪    u_2\rangle \right) > 0$ be the angle between the user vectors. Let the cost function be $c(\p) = \alpha \|p\|_2^{\beta}$ for $\alpha > 0$. For any $z \in
↪    \left\{\UMatrix p \mid p \in \mathbb{R}_{\ge 0}^D \right\}$, the induced cost function is given by:
\[c_{\UMatrix}(z) = \alpha \sin^{-\beta}(\theta^*) \left(z_1^2 + z_2^2 - 2 z_1 z_2 \cos(\theta^*) \right)^{\frac{\beta}{2}}.\]
\end{lemma}
```

Formalized review target   The archival source above is not asserted equivalent to this formalized target.

```
For a feasible score pair in the canonical two-user nonnegative cone, the minimum squared L2 cost among content vectors realizing that pair equals the
↪    stated induced-cost quadratic.
```

Archival source anchor: cited publication, lines 1311-1315

Expanded PaperInterface specification

```
∀ {θ z1 z2 : ℝ},
  0 < Real.sin θ →
    JGS23SupplySideRecommenderSystems.NonnegativeContent
        (JGS23SupplySideRecommenderSystems.canonicalTwoUserContentOfValues θ z1 z2) →
      sInf
          {t : ℝ |
            ∃ (p : JGS23SupplySideRecommenderSystems.Content 2),
              JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧
                JGS23SupplySideRecommenderSystems.score JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst p = z1 ∧
                  JGS23SupplySideRecommenderSystems.score (JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
                        p =
                      z2 ∧
                    t = JGS23SupplySideRecommenderSystems.score p p} =
        (z1 ^ 2 + z2 ^ 2 - 2 * z1 * z2 * Real.cos θ) / Real.sin θ ^ 2
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.canonicalTwoUserContentOfValues
- JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
- JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond
- JGS23SupplySideRecommenderSystems.score

Lean-elaborated claim roles

```
1. parameter: ℝ
2. parameter: ℝ
3. parameter: ℝ
4. assumption: 0 < Real.sin θ
5. assumption: JGS23SupplySideRecommenderSystems.NonnegativeContent
  (JGS23SupplySideRecommenderSystems.canonicalTwoUserContentOfValues θ z1 z2)
6. conclusion: sInf
    {t : ℝ |
      ∃ (p : JGS23SupplySideRecommenderSystems.Content 2),
        JGS23SupplySideRecommenderSystems.NonnegativeContent p ∧
          JGS23SupplySideRecommenderSystems.score JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst p = z1 ∧
            JGS23SupplySideRecommenderSystems.score (JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ) p =
                z2 ∧
              t = JGS23SupplySideRecommenderSystems.score p p} =
  (z1 ^ 2 + z2 ^ 2 - 2 * z1 * z2 * Real.cos θ) / Real.sin θ ^ 2
```

Lean proof endpoint (built separately)

```
JGS23SupplySideRecommenderSystems.lemmaInducedCost
```

Recorded source-to-Spec screening   **Verdict:** matches_approved_corrected_target

Reason: Anchor 1311-1315 overstates an arbitrary-dimensional score-fibre identity. The approved target is the exact nonnegative-fibre minimum in the canonical two-user cone, with the displayed Gram quadratic retained.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 29

Source locator: cited publication, lines 1318-1343

Verbatim source input

```
\begin{lemma}
\label{lemma:FOC}
Let there be 2 users located at $u_1, u_2 \in \mathbb{R}_{\ge 0}^D$ such that  $\|u_1\| = \|u_2\| = 1$, and let $\theta^* := \cos^{-1}\left( \langle u_1,
↪    u_2\rangle \right) > 0$ be the angle between the user vectors. Let the cost function be $c(\p) = \alpha \|p\|_2^{\beta}$ for $\alpha > 0$. For any $z \in
↪    \left\{\UMatrix p \mid p \in \mathbb{R}_{\ge 0}^D \right\}$, the first-order condition of equation \eqref{eq:reparam} can be written as:
\[\begin{bmatrix}
h_1(z_1) \\
h_2(z_2)
\end{bmatrix}
= \nabla_{z}(c_{\UMatrix}(z)).\]
More specifically, it holds that:
\[\begin{bmatrix}
h_1(z_1) \\
h_2(z_2)
\end{bmatrix} = \beta \alpha \sin^{-\beta}(\theta^*) \left(z_1^2 + z_2^2 - 2 z_1 z_2 \cos(\theta^*) \right)^{\frac{\beta}{2} - 1} \begin{bmatrix}
z_1 - z_2 \cos(\theta^*) \\
z_2 - z_1 \cos(\theta^*)
\end{bmatrix},
\]
and if we represent $z = \UMatrix [r\cos(\theta), r \sin(\theta)]$, then it also holds that:
\[\begin{bmatrix}
h_1(z_1) \\
h_2(z_2)
\end{bmatrix}  = \beta \alpha r^{\beta - 1} \begin{bmatrix}
\frac{\sin(\theta^* - \theta)}{\sin(\theta^*)} \\
\frac{\sin(\theta)}{\sin(\theta^*)}
\end{bmatrix}.   \]
\end{lemma}
```

Formalized review target   The archival source above is not asserted equivalent to this formalized target.

At a canonical two-user score pair where the power-cost derivative is defined, both first partial derivatives equal the displayed induced-cost formulas; at
↪   every nonzero strictly interior C1 support value of a source symmetric Nash law with absolutely continuous score laws and support-local C2 raw score
↪   CDFs, those formulas equal the two iid maximum-score CDF derivatives.

Archival source anchor: cited publication, lines 1318-1333

Expanded PaperInterface specification

```
∀ {α β θ z1 z2 : ℝ},
  JGS23SupplySideRecommenderSystems.twoUserInducedCostNumerator θ z1 z2 ≠ 0 ∨ 1 ≤ β / 2 →
    HasDerivAt (fun (x : ℝ) => JGS23SupplySideRecommenderSystems.paper_two_user_induced_cost α β θ x z2)
        (β * α * Real.sin θ ^ (-β) *
            JGS23SupplySideRecommenderSystems.twoUserInducedCostNumerator θ z1 z2 ^ (β / 2 - 1) *
          (z1 - z2 * Real.cos θ))
        z1 ∧
      HasDerivAt (fun (y : ℝ) => JGS23SupplySideRecommenderSystems.paper_two_user_induced_cost α β θ z1 y)
          (β * α * Real.sin θ ^ (-β) *
              JGS23SupplySideRecommenderSystems.twoUserInducedCostNumerator θ z1 z2 ^ (β / 2 - 1) *
            (z2 - z1 * Real.cos θ))
          z2 ∧
        ∀ {P : ℕ} [Nonempty (Fin P)] {α β θ : ℝ} {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy 2}
          [MeasureTheory.IsProbabilityMeasure μ],
          0 < Real.sin θ →
            JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash
                (fun (i : Fin 2) =>
                  if i = 0 then JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
                  else JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
                (fun (p : JGS23SupplySideRecommenderSystems.Content 2) =>
                  α *
                    JGS23SupplySideRecommenderSystems.normRpowCost (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2) β
                      p)
                μ →
              (∀ (i : Fin 2),
                  (MeasureTheory.Measure.map
                        (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
                          JGS23SupplySideRecommenderSystems.score
                            (if i = 0 then JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
                            else JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
                            q)
                        μ).AbsolutelyContinuous
                    MeasureTheory.volume) →
                (∀ (i : Fin 2),
                    ∀
                      x ∈
                        (MeasureTheory.Measure.map
                            (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
                              JGS23SupplySideRecommenderSystems.score
                                (if i = 0 then JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
                                else JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
                                q)
                            μ).support,
                      ContDiffAt ℝ 2
                        (AppliedModelingLib.Probability.lowerCDFMass
                          (MeasureTheory.Measure.map
                            (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
```

```
            JGS23SupplySideRecommenderSystems.score
              (if i = 0 then JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
              else JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
              q)
          μ))
        x) →
  (∀
      z ∈
        (MeasureTheory.Measure.map (JGS23SupplySideRecommenderSystems.canonicalTwoUserValueMap θ)
            μ).support,
      z ≠ (0, 0) → 0 < z.1 ∧ z.1 * Real.cos θ < z.2) →
    ∀
      z ∈
        (MeasureTheory.Measure.map (JGS23SupplySideRecommenderSystems.canonicalTwoUserValueMap θ)
            μ).support,
      z ≠ (0, 0) →
        β * α * Real.sin θ ^ (-β) *
              JGS23SupplySideRecommenderSystems.twoUserInducedCostNumerator θ z.1 z.2 ^ (β / 2 - 1) *
            (z.1 - z.2 * Real.cos θ) =
          deriv
            (fun (x : ℝ) =>
              AppliedModelingLib.Probability.iidMaximumCdf
                (MeasureTheory.Measure.map
                  (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
                    JGS23SupplySideRecommenderSystems.score
                      JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst q)
                  μ)
                x)
            z.1 ∧
          β * α * Real.sin θ ^ (-β) *
              JGS23SupplySideRecommenderSystems.twoUserInducedCostNumerator θ z.1 z.2 ^ (β / 2 - 1) *
            (z.2 - z.1 * Real.cos θ) =
          deriv
            (fun (x : ℝ) =>
              AppliedModelingLib.Probability.iidMaximumCdf
                (MeasureTheory.Measure.map
                  (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
                    JGS23SupplySideRecommenderSystems.score
                      (JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ) q)
                  μ)
                x)
            z.2
```

Retained governing declarations

- AppliedModelingLib.Probability.iidMaximumCdf
- AppliedModelingLib.Probability.lowerCDFMass
- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.SourceNorm.l2
- JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash
- JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
- JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond
- JGS23SupplySideRecommenderSystems.canonicalTwoUserValueMap
- JGS23SupplySideRecommenderSystems.normRpowCost
- JGS23SupplySideRecommenderSystems.paper_two_user_induced_cost
- JGS23SupplySideRecommenderSystems.score
- JGS23SupplySideRecommenderSystems.twoUserInducedCostNumerator

Lean-elaborated claim roles

```
1. parameter: ℝ
2. parameter: ℝ
3. parameter: ℝ
4. parameter: ℝ
5. parameter: ℝ
6. assumption: JGS23SupplySideRecommenderSystems.twoUserInducedCostNumerator θ z1 z2 ≠ 0 ∨ 1 ≤ β / 2
7. conclusion: HasDerivAt (fun (x : ℝ) => JGS23SupplySideRecommenderSystems.paper_two_user_induced_cost α β θ x z2)
    (β * α * Real.sin θ ^ (-β) * JGS23SupplySideRecommenderSystems.twoUserInducedCostNumerator θ z1 z2 ^ (β / 2 - 1) *
      (z1 - z2 * Real.cos θ))
    z1 ∧
  HasDerivAt (fun (y : ℝ) => JGS23SupplySideRecommenderSystems.paper_two_user_induced_cost α β θ z1 y)
      (β * α * Real.sin θ ^ (-β) * JGS23SupplySideRecommenderSystems.twoUserInducedCostNumerator θ z1 z2 ^ (β / 2 - 1) *
        (z2 - z1 * Real.cos θ))
      z2 ∧
    ∀ {P : ℕ} [Nonempty (Fin P)] {α β θ : ℝ} {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy 2}
      [MeasureTheory.IsProbabilityMeasure μ],
      0 < Real.sin θ →
        JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash
          (fun (i : Fin 2) =>
```

```
    if i = 0 then JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
    else JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
   (fun (p : JGS23SupplySideRecommenderSystems.Content 2) =>
    α *
     JGS23SupplySideRecommenderSystems.normRpowCost (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2) β p)
   μ →
  (∀ (i : Fin 2),
    (MeasureTheory.Measure.map
       (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
        JGS23SupplySideRecommenderSystems.score
         (if i = 0 then JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
         else JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
         q)
       μ).AbsolutelyContinuous
     MeasureTheory.volume) →
   (∀ (i : Fin 2),
     ∀
      x ∈
       (MeasureTheory.Measure.map
         (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
          JGS23SupplySideRecommenderSystems.score
           (if i = 0 then JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
           else JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
           q)
         μ).support,
      ContDiffAt ℝ 2
       (AppliedModelingLib.Probability.lowerCDFMass
        (MeasureTheory.Measure.map
         (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
          JGS23SupplySideRecommenderSystems.score
           (if i = 0 then JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
           else JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
           q)
         μ))
       x) →
    (∀
      z ∈
       (MeasureTheory.Measure.map (JGS23SupplySideRecommenderSystems.canonicalTwoUserValueMap θ)
         μ).support,
      z ≠ (0, 0) → 0 < z.1 ∧ z.1 * Real.cos θ < z.2) →
     ∀
      z ∈
       (MeasureTheory.Measure.map (JGS23SupplySideRecommenderSystems.canonicalTwoUserValueMap θ)
         μ).support,
      z ≠ (0, 0) →
       β * α * Real.sin θ ^ (-β) *
           JGS23SupplySideRecommenderSystems.twoUserInducedCostNumerator θ z.1 z.2 ^ (β / 2 - 1) *
          (z.1 - z.2 * Real.cos θ) =
         deriv
          (fun (x : ℝ) =>
           AppliedModelingLib.Probability.iidMaximumCdf
            (MeasureTheory.Measure.map
             (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
              JGS23SupplySideRecommenderSystems.score
               JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst q)
             μ)
            x)
          z.1 ∧
        β * α * Real.sin θ ^ (-β) *
           JGS23SupplySideRecommenderSystems.twoUserInducedCostNumerator θ z.1 z.2 ^ (β / 2 - 1) *
          (z.2 - z.1 * Real.cos θ) =
         deriv
          (fun (x : ℝ) =>
           AppliedModelingLib.Probability.iidMaximumCdf
            (MeasureTheory.Measure.map
             (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
              JGS23SupplySideRecommenderSystems.score
               (JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ) q)
             μ)
            x)
          z.2
```

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.lemmaFoc

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: The first Lean conjunct states both partial derivatives of the canonical induced cost with the source formula and imposes precisely the approved rpow-base alternative. The second uses canonical two-user scores, a source symmetric Nash measure, absolutely continuous score laws, C2 lower CDFs on support, and the explicit strict interior score-cone inequalities; at every nonzero realized score it equates the two cost partials to derivatives of the P-1-draw iid maximum CDFs. Thus the formulas, derivative domains, Nash/C1 route, and regularity premises match the approved FOC correction.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 30

Source locator: cited publication, lines 1346-1350

Verbatim source input

```
\begin{lemma}
\label{lemma:secondderiv}
Let there be 2 users located at $u_1, u_2 \in \mathbb{R}_{\ge 0}^D$ such that  $\|u_1\| = \|u_2\| = 1$, and let $\theta^* := \cos^{-1}\left( \langle u_1,
↪    u_2\rangle \right) > 0$ be the angle between the user vectors. Let the cost function be $c(\p) = \alpha \|p\|_2^{\beta}$ for $\alpha > 0$. If $z$ is of
↪    the form $[r \cos(\theta), r \cos(\theta^* - \theta)]$ for $\theta \in [0, \theta^*]$, then the sign of $\frac{\partial^2  c_{\UMatrix}(z)}{\partial z_1
↪    \partial z_2}$ is equal to the sign of:
\[\frac{\beta - 2}{\beta} \cos(\theta^* - 2 \theta) - \cos(\theta^*).\]
\end{lemma}
```

Formalized review target The archival source above is not asserted equivalent to this formalized target.

At a positive-radius nonsingular canonical score point, the induced-cost mixed partial equals the displayed scalar multiple of the second-derivative bracket.

Archival source anchor: cited publication, lines 1336-1347

Expanded PaperInterface specification

```
∀ {α β θ φ r : ℝ},
  β ≠ 0 →
    0 < r ^ 2 * Real.sin θ ^ 2 →
      JGS23SupplySideRecommenderSystems.paper_two_user_induced_cost_cross_partial α β θ (r * Real.cos φ)
          (r * Real.cos (θ - φ)) =
        β * α * Real.sin θ ^ (-β) *
          ((r ^ 2 * Real.sin θ ^ 2) ^ (β / 2 - 1) *
            (β / 2 * JGS23SupplySideRecommenderSystems.paper_two_user_second_deriv_sign_bracket β θ φ))
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.paper_two_user_induced_cost_cross_partial
- JGS23SupplySideRecommenderSystems.paper_two_user_second_deriv_sign_bracket

Lean-elaborated claim roles

```
1. parameter: ℝ
2. parameter: ℝ
3. parameter: ℝ
4. parameter: ℝ
5. parameter: ℝ
6. assumption: β ≠ 0
7. assumption: 0 < r ^ 2 * Real.sin θ ^ 2
8. conclusion: JGS23SupplySideRecommenderSystems.paper_two_user_induced_cost_cross_partial α β θ (r * Real.cos φ)
    (r * Real.cos (θ - φ)) =
  β * α * Real.sin θ ^ (-β) *
    ((r ^ 2 * Real.sin θ ^ 2) ^ (β / 2 - 1) *
      (β / 2 * JGS23SupplySideRecommenderSystems.paper_two_user_second_deriv_sign_bracket β θ φ))
```

Lean proof endpoint (built separately)

```
JGS23SupplySideRecommenderSystems.lemmaSecondDerivative
```

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: Anchor 1346-1350 gives a sign comparison. The approved target gives the exact mixed-partial scalar-factor identity at a positive-radius nonsingular canonical point, which entails the intended sign information.
Reviewer check: ☐ Matches formalized target
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 31

Source locator: cited publication, lines 1351-1355

Verbatim source input

```
\begin{lemma}
\label{lemma:regionscolor}
Let there be 2 users located at $u_1, u_2 \in \mathbb{R}_{\ge 0}^D$ such that  $\|u_1\| = \|u_2\| = 1$, and let $\theta^* := \cos^{-1}\left( \langle u_1,
↪    u_2\rangle \right) > 0$ be the angle between the user vectors. Let the cost function be $c(\p) = \alpha \|p\|_2^{\beta}$ for $\alpha > 0$. Suppose that
↪    condition (C1) is satisfied for $(H_1, H_2, S)$. If $S$ contains a curve of the form $\left\{(z_1, g(z_1)) \mid x \in I \right\}$ for any open interval $I$ and
↪    any differentiable function $g$, then for any $z_1 \in I$, it holds that:
\[g'(z_1) \cdot \left(\frac{\beta - 2}{\beta} \cos(\theta^* - 2 \theta) - \cos(\theta^*) \right) \le 0.\]
\end{lemma}
```

Formalized review target The archival source above is not asserted equivalent to this formalized target.

```
For two nonnegative unit users in the canonical two-dimensional score realization, a differentiable C1 support curve in the canonical nonnegative score
↪    cone has slope times the displayed induced-cost bracket at most zero.
```

Archival source anchor: cited publication, lines 1351-1355

Expanded PaperInterface specification

```
∀ {u v : JGS23SupplySideRecommenderSystems.Content 2} {H1 H2 g : ℝ → ℝ} {S : Set (ℝ × ℝ)} {α β θ φ r a b x slope : ℝ},
  JGS23SupplySideRecommenderSystems.NonnegativeContent u →
    JGS23SupplySideRecommenderSystems.NonnegativeContent v →
      JGS23SupplySideRecommenderSystems.score u u = 1 →
        JGS23SupplySideRecommenderSystems.score v v = 1 →
          JGS23SupplySideRecommenderSystems.score u v = Real.cos θ →
            0 < α →
              0 < β →
                0 < Real.sin θ →
                  Monotone H1 →
                    Monotone H2 →
                      (∀ z ∈ S, 0 ≤ z.1 ∧ 0 ≤ z.2) →
                        x ∈ Set.Ioo a b →
                          x = r * Real.cos φ →
                            g x = r * Real.cos (θ - φ) →
                              HasDerivAt g slope x →
                                (∀ w ∈ Set.Ioo a b, (w, g w) ∈ S) →
                                  (∀ z ∈ S,
                                      IsMaxOn
                                        (fun (w : ℝ × ℝ) =>
                                          H1 w.1 + H2 w.2 -
                                            JGS23SupplySideRecommenderSystems.twoUserInducedCost α β θ w.1 w.2)
                                        (JGS23SupplySideRecommenderSystems.twoUserScoreFeasibleSet u v) z) →
                                    slope *
                                        JGS23SupplySideRecommenderSystems.paper_two_user_second_deriv_sign_bracket β θ
                                          φ ≤
                                      0
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.paper_two_user_second_deriv_sign_bracket
- JGS23SupplySideRecommenderSystems.score
- JGS23SupplySideRecommenderSystems.twoUserInducedCost
- JGS23SupplySideRecommenderSystems.twoUserScoreFeasibleSet

Lean-elaborated claim roles

```
1. parameter: JGS23SupplySideRecommenderSystems.Content 2
2. parameter: JGS23SupplySideRecommenderSystems.Content 2
3. parameter: ℝ → ℝ
4. parameter: ℝ → ℝ
5. parameter: ℝ → ℝ
6. parameter: Set (ℝ × ℝ)
7. parameter: ℝ
8. parameter: ℝ
9. parameter: ℝ
10. parameter: ℝ
11. parameter: ℝ
12. parameter: ℝ
13. parameter: ℝ
14. parameter: ℝ
15. parameter: ℝ
16. assumption: JGS23SupplySideRecommenderSystems.NonnegativeContent u
17. assumption: JGS23SupplySideRecommenderSystems.NonnegativeContent v
18. assumption: JGS23SupplySideRecommenderSystems.score u u = 1
```

19. assumption: JGS23SupplySideRecommenderSystems.score v v = 1
20. assumption: JGS23SupplySideRecommenderSystems.score u v = Real.cos θ
21. assumption: 0 < α
22. assumption: 0 < β
23. assumption: 0 < Real.sin θ
24. assumption: Monotone H1
25. assumption: Monotone H2
26. assumption: ∀ z ∈ S, 0 ≤ z.1 ∧ 0 ≤ z.2
27. assumption: x ∈ Set.Ioo a b
28. assumption: x = r * Real.cos φ
29. assumption: g x = r * Real.cos (θ - φ)
30. assumption: HasDerivAt g slope x
31. assumption: ∀ w ∈ Set.Ioo a b, (w, g w) ∈ S
32. assumption: ∀ z ∈ S,
  IsMaxOn (fun (w : ℝ × ℝ) => H1 w.1 + H2 w.2 - JGS23SupplySideRecommenderSystems.twoUserInducedCost α β θ w.1 w.2)
    (JGS23SupplySideRecommenderSystems.twoUserScoreFeasibleSet u v) z
33. conclusion: slope * JGS23SupplySideRecommenderSystems.paper_two_user_second_deriv_sign_bracket β θ φ ≤ 0

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.lemmaRegionsColor

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: The successor source bundle expressly records the arbitrary-dimensional archival statement as non-equivalent to the approved canonical two-dimensional repair under defect JGS23-INDUCED-COST-GENERAL-DIMENSION-01. The expanded Lean target is exactly on Content 2: u and v are nonnegative unit users with inner product cos(theta), the feasible score region is the image of nonnegative Content 2 under their two-score map, every support point satisfies the displayed C1 maximization condition, and a differentiable curve point with polar parameter phi concludes slope * (((beta-2)/beta) cos(theta-2 phi) - cos(theta)) <= 0. This matches the approved corrected target rather than claiming the false arbitrary-D induced-cost equality.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 32

Source locator: cited publication, lines 1458-1461

Verbatim source input

```
\begin{restatable}{proposition}{uniqueness}
\label{prop:uniqueness}
Consider the setup in Theorem \ref{thm:phasetransitionformal}. If $\beta < \beta^* = \frac{2}{1 - \cos(\theta^*)}$ and $\mu$ is a symmetric mixed
↪    equilibrium, then $\mu$ satisfies $|\Genre(\mu)| = 1$.
\end{restatable}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

```
In the canonical two-user source-Nash model below the phase threshold, absolutely continuous twice-differentiable score laws force the stated unique
↪    nonzero support genre.
```

Archival source anchor: cited publication, lines 1458-1461

Expanded PaperInterface specification

```
∀ {P : ℕ} [Nonempty (Fin P)] {β θ : ℝ} {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy 2}
  [MeasureTheory.IsProbabilityMeasure μ],
  1 < P →
    1 ≤ β →
      0 < θ →
        θ ≤ Real.pi / 2 →
          JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash
              (fun (i : Fin 2) =>
                if i = 0 then JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
                else JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
              (JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2)
                β)
              μ →
            (∀ (i : Fin 2),
                (MeasureTheory.Measure.map
                      (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
                        JGS23SupplySideRecommenderSystems.score
                          (if i = 0 then JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
                          else JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
                          q)
                      μ).AbsolutelyContinuous
                  MeasureTheory.volume) →
              (∀
                  x ∈
                    (MeasureTheory.Measure.map
                        (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
                          JGS23SupplySideRecommenderSystems.score
                            JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst q)
                        μ).support,
                  ContDiffAt ℝ 2
                    (AppliedModelingLib.Probability.lowerCDFMass
                      (MeasureTheory.Measure.map
                        (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
                          JGS23SupplySideRecommenderSystems.score
                            JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst q)
                        μ))
                    x) →
                (∀
                    x ∈
                      (MeasureTheory.Measure.map
                          (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
                            JGS23SupplySideRecommenderSystems.score
                              (JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ) q)
                          μ).support,
                    ContDiffAt ℝ 2
                      (AppliedModelingLib.Probability.lowerCDFMass
                        (MeasureTheory.Measure.map
                          (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
                            JGS23SupplySideRecommenderSystems.score
                              (JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ) q)
                          μ))
                      x) →
                  β < JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold θ →
                    JGS23SupplySideRecommenderSystems.SourceNonzeroSupportGenres
                        (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2) μ =
                      {(JGS23SupplySideRecommenderSystems.SourceNorm.l2 2).normalizedContent
                          (JGS23SupplySideRecommenderSystems.canonicalTwoUserContentOfValues θ 1 1)}
```

Retained governing declarations

- AppliedModelingLib.Probability.lowerCDFMass
- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.SourceNonzeroSupportGenres

- JGS23SupplySideRecommenderSystems.SourceNorm.l2
- JGS23SupplySideRecommenderSystems.SourceNorm.normalizedContent
- JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash
- JGS23SupplySideRecommenderSystems.canonicalTwoUserContentOfValues
- JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
- JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.score
- JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold

Lean-elaborated claim roles

```
1. parameter: ℕ
2. assumption: Nonempty (Fin P)
3. parameter: ℝ
4. parameter: ℝ
5. parameter: JGS23SupplySideRecommenderSystems.MixedContentStrategy 2
6. assumption: MeasureTheory.IsProbabilityMeasure μ
7. assumption: 1 < P
8. assumption: 1 ≤ β
9. assumption: 0 < θ
10. assumption: θ ≤ Real.pi / 2
11. assumption: JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash
  (fun (i : Fin 2) =>
    if i = 0 then JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
    else JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
  (JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2) β) μ
12. assumption: ∀ (i : Fin 2),
  (MeasureTheory.Measure.map
        (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
          JGS23SupplySideRecommenderSystems.score
            (if i = 0 then JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
            else JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
            q)
        μ).AbsolutelyContinuous
    MeasureTheory.volume
13. assumption: ∀
  x ∈
    (MeasureTheory.Measure.map
        (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
          JGS23SupplySideRecommenderSystems.score JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst q)
        μ).support,
  ContDiffAt ℝ 2
    (AppliedModelingLib.Probability.lowerCDFMass
      (MeasureTheory.Measure.map
        (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
          JGS23SupplySideRecommenderSystems.score JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst q)
        μ))
    x
14. assumption: ∀
  x ∈
    (MeasureTheory.Measure.map
        (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
          JGS23SupplySideRecommenderSystems.score (JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ) q)
        μ).support,
  ContDiffAt ℝ 2
    (AppliedModelingLib.Probability.lowerCDFMass
      (MeasureTheory.Measure.map
        (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
          JGS23SupplySideRecommenderSystems.score (JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ) q)
        μ))
    x
15. assumption: β < JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold θ
16. conclusion: JGS23SupplySideRecommenderSystems.SourceNonzeroSupportGenres (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2) μ =
  {(JGS23SupplySideRecommenderSystems.SourceNorm.l2 2).normalizedContent
      (JGS23SupplySideRecommenderSystems.canonicalTwoUserContentOfValues θ 1 1)}
```

Lean proof endpoint (built separately)

```
JGS23SupplySideRecommenderSystems.propositionUniqueness
```

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: Anchor 1458-1461 is the below-threshold branch. The approved target retains its unique nonzero support genre in the canonical two-user source-Nash model and explicitly supplies absolute-continuity and twice-differentiable score-law hypotheses.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 33

Source locator: cited publication, lines 1462-1465

Verbatim source input

```
\begin{restatable}{proposition}{finitegenre}
\label{prop:finitegenre}
Consider the setup in Theorem \ref{thm:phasetransitionformal}. If $\beta > \beta^* = \frac{2}{1 - \cos(\theta^*)}$, if $|\Genre(\mu)| < \infty$, and if the
↪   conditional distribution of $\|p\|$ along each genre is continuous differentiable, then $\mu$ is not an equilibrium.
\end{restatable}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

```
In the canonical two-user source-Nash model above the phase threshold, no finite conditional radial-law genre representation with the stated score-law
↪   regularity can occur.
```

Archival source anchor: cited publication, lines 1462-1465

Expanded PaperInterface specification

```
∀ {P : ℕ} [Nonempty (Fin P)] {β θ : ℝ} {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy 2}
  [MeasureTheory.IsProbabilityMeasure μ],
  1 < P →
    1 ≤ β →
      0 < θ →
        θ ≤ Real.pi / 2 →
          JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash
              (fun (i : Fin 2) =>
                if i = 0 then JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
                else JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
              (JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2)
                β)
              μ →
            (∀ (i : Fin 2),
                (MeasureTheory.Measure.map
                    (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
                      JGS23SupplySideRecommenderSystems.score
                        (if i = 0 then JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
                        else JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
                        q)
                    μ).AbsolutelyContinuous
                  MeasureTheory.volume) →
              (∀
                  x ∈
                    (MeasureTheory.Measure.map
                        (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
                          JGS23SupplySideRecommenderSystems.score
                            JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst q)
                        μ).support,
                  ContDiffAt ℝ 2
                    (AppliedModelingLib.Probability.lowerCDFMass
                      (MeasureTheory.Measure.map
                        (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
                          JGS23SupplySideRecommenderSystems.score
                            JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst q)
                        μ))
                    x) →
                (∀
                    x ∈
                      (MeasureTheory.Measure.map
                          (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
                            JGS23SupplySideRecommenderSystems.score
                              (JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ) q)
                          μ).support,
                    ContDiffAt ℝ 2
                      (AppliedModelingLib.Probability.lowerCDFMass
                        (MeasureTheory.Measure.map
                          (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
                            JGS23SupplySideRecommenderSystems.score
                              (JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ) q)
                          μ))
                      x) →
                  JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold θ < β →
                    ∀ {G : ℕ},
                      Nonempty (Fin G) →
                        ∀ (angle : Fin G → ℝ)
                          (law : JGS23SupplySideRecommenderSystems.finiteGenreConditionalNormLawSpec μ angle),
                          Function.Injective angle → (∀ (i : Fin G), angle i ∈ Set.Icc 0 (Real.pi / 2)) → False
```

Retained governing declarations

- AppliedModelingLib.Probability.lowerCDFMass
- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.SourceNorm.l2

- JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash
- JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
- JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond
- JGS23SupplySideRecommenderSystems.finiteGenreConditionalNormLawSpec
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.score
- JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold

Lean-elaborated claim roles

```
1. parameter: ℕ
2. assumption: Nonempty (Fin P)
3. parameter: ℝ
4. parameter: ℝ
5. parameter: JGS23SupplySideRecommenderSystems.MixedContentStrategy 2
6. assumption: MeasureTheory.IsProbabilityMeasure μ
7. assumption: 1 < P
8. assumption: 1 ≤ β
9. assumption: 0 < θ
10. assumption: θ ≤ Real.pi / 2
11. assumption: JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash
  (fun (i : Fin 2) =>
    if i = 0 then JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
    else JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
  (JGS23SupplySideRecommenderSystems.normPowerCostSpec (JGS23SupplySideRecommenderSystems.SourceNorm.l2 2) β) μ
12. assumption: ∀ (i : Fin 2),
  (MeasureTheory.Measure.map
        (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
          JGS23SupplySideRecommenderSystems.score
            (if i = 0 then JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst
            else JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ)
            q)
        μ).AbsolutelyContinuous
    MeasureTheory.volume
13. assumption: ∀
  x ∈
    (MeasureTheory.Measure.map
        (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
          JGS23SupplySideRecommenderSystems.score JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst q)
        μ).support,
  ContDiffAt ℝ 2
    (AppliedModelingLib.Probability.lowerCDFMass
      (MeasureTheory.Measure.map
        (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
          JGS23SupplySideRecommenderSystems.score JGS23SupplySideRecommenderSystems.canonicalTwoUserFirst q)
        μ))
    x
14. assumption: ∀
  x ∈
    (MeasureTheory.Measure.map
        (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
          JGS23SupplySideRecommenderSystems.score (JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ) q)
        μ).support,
  ContDiffAt ℝ 2
    (AppliedModelingLib.Probability.lowerCDFMass
      (MeasureTheory.Measure.map
        (fun (q : JGS23SupplySideRecommenderSystems.Content 2) =>
          JGS23SupplySideRecommenderSystems.score (JGS23SupplySideRecommenderSystems.canonicalTwoUserSecond θ) q)
        μ))
    x
15. assumption: JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold θ < β
16. parameter: ℕ
17. assumption: Nonempty (Fin G)
18. parameter: Fin G → ℝ
19. parameter: JGS23SupplySideRecommenderSystems.finiteGenreConditionalNormLawSpec μ angle
20. assumption: Function.Injective angle
21. assumption: ∀ (i : Fin G), angle i ∈ Set.Icc 0 (Real.pi / 2)
22. conclusion: False
```

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.propositionFiniteGenre

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: Anchor 1462-1465 is the above-threshold finite-genre exclusion. The approved target keeps that conclusion in the canonical two-user model and explicitly requires the finite conditional radial-law and score-law regularity used by the proof.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 34

Source locator: cited publication, lines 1759-1771

Verbatim source input

```
\begin{restatable}{theorem}{infinitegenreformal}[Formal version of Theorem \ref{thm:infinitegenre}]
\label{thm:infinitegenreformal}
Suppose that there are 2 users located at two linearly independently vectors $u_1, u_2 \in \mathbb{R}_{\ge 0}^D$, let $\theta^* := \cos^{-1}\Big(
↪   \frac{\langle u_1, u_2\rangle}{\|u_1\|_2 \|u_2\|} \Big) < 0$ be the angle between them.  Suppose we have cost function $c(\p) = \|p\|_2^{\beta}$,
↪   $\beta > \beta^* = \frac{2}{1 - \cos(\theta^*)}$, and $P = \infty$ producers. Then, the genres $d_1, d_2$, conditional quality distributions $\FMax_1
↪   = \FMax$ and $\FMax_2 = \FMax$, and weights $\alpha_1 = \alpha_2 = 2$ form an equilibrium (as per  Definition \ref{def:infinitegenre}), where
\[\left\{d_1, d_2 \right\} := \left\{[\cos(\theta^G + \theta_{\text{min}}), \sin(\theta^G + \theta_{\text{min}})], [\cos(\theta^* - \theta^G +
↪   \theta_{\text{min}}), \sin(\theta^* - \theta^G + \theta_{\text{min}})] \right\} \]
such that $\theta^G := \argmax_{\theta \le \theta^*/2} \left(\cos^{\beta}(\theta) + \cos^{\beta}(\theta^* - \theta)\right)$ and $\theta_{\text{min}} :=
↪   \min\left(\cos^{-1}\left(\frac{\langle u_1, e_1\rangle}{\|u_1\|} \right) , \cos^{-1}\left(\frac{\langle u_2, e_1\rangle}{\|u_2\|} \right) \right)$, and where
\[\FMax(q) :=
\begin{cases}
C_2^{(2n+2) \beta} & \text{ if } q \in C_1^{1/\beta} C_2^{2n+1}  \left[C_2, 1 \right] \text{ for } n \ge 0 \\
 C_1^{-2}C_2^{-2 n\beta} q^{2 \beta}  & \text{ if } q \in C_1^{1/\beta} C_2^{2n}  \left[C_2, 1 \right] \text{ for } n \ge 0 \\
1 & \text{ if } q \ge C_1^{1/\beta},
\end{cases},\]
such that the constants are defined by $C_1  := \frac{\sin(\theta^*) \cos(\theta^G)}{\sin(\theta^* - \theta_G)}$ and $C_2 := \frac{\cos(\theta^* -
↪   \theta^G)}{\cos(\theta^G)}$.
\end{restatable}
```

Formalized review target   The archival source above is not asserted equivalent to this formalized target.

```
For 0 < theta < pi/2 and beta strictly above the two-user phase threshold, there are two normalized genres with equal probability weights, a corrected
↪   conditional-quality CDF, and support-wise global best-response optimality in the coherent finite-genre infinite-producer equilibrium predicate.
```

Archival source anchor: cited publication, lines 1759-1771

Expanded PaperInterface specification

```
∀ {β θ : ℝ},
  0 < θ →
    θ < Real.pi / 2 →
      JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold θ < β →
        ∃ (a : ℝ) (C1 : ℝ) (C2 : ℝ) (A : ℝ) (B : ℝ) (phi : ℝ),
          JGS23SupplySideRecommenderSystems.infiniteProducerCandidateSpec
              (fun (q : ℝ) =>
                if q ≤ 0 then 0
                else
                  if a ≤ q then 1
                  else
                    have k : ℕ := if h : ∃ (n : ℕ), a * C2 ^ n ≤ q then Nat.find h else 0;
                    if Even k then C2 ^ (↑k * β) else C1 ^ (-2) * C2 ^ (-2 * ↑(k / 2) * β) * q ^ (2 * β))
              β θ phi ∧
            phi ∈ Set.Icc 0 (θ / 2) ∧
              IsMaxOn (fun (x : ℝ) => Real.cos x ^ β + Real.cos (θ - x) ^ β) (Set.Icc 0 (θ / 2)) phi ∧
                0 < a ∧
                  0 < C2 ∧
                    C2 < 1 ∧ C1 = a ^ β ∧ C1 = 1 + C2 ^ β ∧ A = Real.cos phi ∧ B = Real.cos (θ - phi) ∧ C2 = B / A
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.infiniteProducerCandidateSpec
- JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold

Lean-elaborated claim roles

```
1. parameter: ℝ
2. parameter: ℝ
3. assumption: 0 < θ
4. assumption: θ < Real.pi / 2
5. assumption: JGS23SupplySideRecommenderSystems.twoUserPhaseThreshold θ < β
6. conclusion: ∃ (a : ℝ) (C1 : ℝ) (C2 : ℝ) (A : ℝ) (B : ℝ) (phi : ℝ),
  JGS23SupplySideRecommenderSystems.infiniteProducerCandidateSpec
      (fun (q : ℝ) =>
        if q ≤ 0 then 0
        else
          if a ≤ q then 1
          else
            have k : ℕ := if h : ∃ (n : ℕ), a * C2 ^ n ≤ q then Nat.find h else 0;
            if Even k then C2 ^ (↑k * β) else C1 ^ (-2) * C2 ^ (-2 * ↑(k / 2) * β) * q ^ (2 * β))
      β θ phi ∧
    phi ∈ Set.Icc 0 (θ / 2) ∧
      IsMaxOn (fun (x : ℝ) => Real.cos x ^ β + Real.cos (θ - x) ^ β) (Set.Icc 0 (θ / 2)) phi ∧
        0 < a ∧ 0 < C2 ∧ C2 < 1 ∧ C1 = a ^ β ∧ C1 = 1 + C2 ^ β ∧ A = Real.cos phi ∧ B = Real.cos (θ - phi) ∧ C2 = B / A
```

Lean proof endpoint (built separately)

```
JGS23SupplySideRecommenderSystems.theoremInfiniteGenre
```

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: The archival formal theorem has the impossible arccosine condition theta < 0 and weights alpha_1 = alpha_2 = 2. The displayed approved correction replaces these with the acute regime and normalized equal weights; the Lean target exactly uses 0 < theta < pi/2, half weights, the candidate CDF, normalized genres, and support-wise global best responses.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 35

Source locator: cited publication, lines 1906-1911

Verbatim source input

```
\begin{lemma}
\label{lemma:standardbasismax}
For any $\beta \ge 2$, the expression
\[\max_{z_1, z_2 \ge 0} \left(\left(\min\left(\left(\frac{2}{\beta} \right)^{-2/\beta} z_1^2, 1\right) + \min\left(\left(\frac{2}{\beta} \right)^{-2/\beta}
↪    z_2^2, 1\right)\right) - (z_1^2 + z_2^2)^{\beta / 2}\right)\]
is maximized for any $(z_1, z_2)$ such that $z_1^2 + z_2^2 = \left(\frac{2}{\beta}\right)^{2/\beta}$.
\end{lemma}
```

Expanded PaperInterface specification

```
∀ {β z1 z2 w1 w2 : ℝ},
  2 ≤ β →
    0 ≤ z1 →
      0 ≤ z2 →
        0 ≤ w1 →
          0 ≤ w2 →
            z1 ^ 2 + z2 ^ 2 = (2 / β) ^ (2 / β) →
              min ((2 / β) ^ (-(2 / β)) * w1 ^ 2) 1 + min ((2 / β) ^ (-(2 / β)) * w2 ^ 2) 1 -
                  (w1 ^ 2 + w2 ^ 2) ^ (β / 2) ≤
                min ((2 / β) ^ (-(2 / β)) * z1 ^ 2) 1 + min ((2 / β) ^ (-(2 / β)) * z2 ^ 2) 1 -
                  (z1 ^ 2 + z2 ^ 2) ^ (β / 2)
```

Lean-elaborated claim roles

```
1. parameter: ℝ
2. parameter: ℝ
3. parameter: ℝ
4. parameter: ℝ
5. parameter: ℝ
6. assumption: 2 ≤ β
7. assumption: 0 ≤ z1
8. assumption: 0 ≤ z2
9. assumption: 0 ≤ w1
10. assumption: 0 ≤ w2
11. assumption: z1 ^ 2 + z2 ^ 2 = (2 / β) ^ (2 / β)
12. conclusion: min ((2 / β) ^ (-(2 / β)) * w1 ^ 2) 1 + min ((2 / β) ^ (-(2 / β)) * w2 ^ 2) 1 - (w1 ^ 2 + w2 ^ 2) ^ (β / 2) ≤
  min ((2 / β) ^ (-(2 / β)) * z1 ^ 2) 1 + min ((2 / β) ^ (-(2 / β)) * z2 ^ 2) 1 - (z1 ^ 2 + z2 ^ 2) ^ (β / 2)
```

Lean proof endpoint (built separately)

```
JGS23SupplySideRecommenderSystems.lemmaStandardBasisMax
```

Recorded source-to-Spec screening **Verdict:** matches

Reason: Anchor 1906-1911 states that every nonnegative point on the displayed radius maximizes the two-coordinate objective for beta >= 2; the target is that same global comparison, written pointwise against arbitrary nonnegative coordinates.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 36

Source locator: cited publication, lines 345-349

Verbatim source input

```
\begin{example}[1-dimensional setup]
\label{example:1d}
Let $D = 1$, and suppose that there is a single user $\u_1 =1$. Suppose the cost function is $c(\p) = |\p|^{\beta}$. The unique symmetric mixed
↪    equilibrium $\mu$ is supported on the full interval $[0, 1]$ and has cumulative distribution function $F(\p) = \left(\p /N\right)^{\beta / (P-1)}$. We
↪    defer the derivation to Appendix \ref{sec:derivationexample1d}.
\end{example}
```

Formalized review target The archival source above is not asserted equivalent to this formalized target.

```
For the literal one-user, one-dimensional Euclidean source model, every symmetric mixed equilibrium equals the displayed single-ray law; its power-law
↪    CDF is therefore unique. The generic unit-genre construction remains available for the homogeneous one-dimensional specialization.
```

Archival source anchor: cited publication, lines 346-349

Expanded PaperInterface specification

```
∀ {P : ℕ} [Nonempty (Fin P)] [Nontrivial (Fin P)] {β : ℝ} {u genre : JGS23SupplySideRecommenderSystems.Content 1}
  (ν : JGS23SupplySideRecommenderSystems.SourceNorm 1),
  0 < β →
    JGS23SupplySideRecommenderSystems.NonnegativeContent u →
      JGS23SupplySideRecommenderSystems.NonnegativeContent genre →
        ν.norm genre = 1 →
          0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value u genre →
            (∀ (d : JGS23SupplySideRecommenderSystems.Content 1),
                JGS23SupplySideRecommenderSystems.NonnegativeContent d →
                  ν.norm d = 1 →
                    JGS23SupplySideRecommenderSystems.paper_inferred_user_value u d ≤
                      JGS23SupplySideRecommenderSystems.paper_inferred_user_value u genre) →
              Measurable ν.norm →
                JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec (fun (x : Fin 1) => u)
                    (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β)
                    (JGS23SupplySideRecommenderSystems.singleGenreContentLaw 1 P β genre) ∧
                  (JGS23SupplySideRecommenderSystems.singleGenreContentLaw 1 P β genre).support ⊆
                      {p : JGS23SupplySideRecommenderSystems.Content 1 |
                        ∃ q ∈ Set.Icc 0 1, p = JGS23SupplySideRecommenderSystems.scaleContent q genre} ∧
                    (∀ (z : ℝ),
                        0 ≤ z →
                          (MeasureTheory.Measure.map ν.norm
                                (JGS23SupplySideRecommenderSystems.singleGenreContentLaw 1 P β genre))
                              (Set.Iic z) =
                            ENNReal.ofReal (min 1 ((z ^ β / 1) ^ (↑P - 1)⁻¹))) ∧
                      ∀ {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy 1},
                        JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash
                            (fun (x : Fin 1) => JGS23SupplySideRecommenderSystems.oneDimensionalUnitContent)
                            (JGS23SupplySideRecommenderSystems.normRpowCost
                              (JGS23SupplySideRecommenderSystems.SourceNorm.l2 1) β)
                            μ →
                          μ =
                            JGS23SupplySideRecommenderSystems.singleGenreContentLaw 1 P β
                              JGS23SupplySideRecommenderSystems.oneDimensionalUnitContent
```

Retained governing declarations

- JGS23SupplySideRecommenderSystems.Content
- JGS23SupplySideRecommenderSystems.MixedContentStrategy
- JGS23SupplySideRecommenderSystems.NonnegativeContent
- JGS23SupplySideRecommenderSystems.SourceNorm
- JGS23SupplySideRecommenderSystems.SourceNorm.l2
- JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash
- JGS23SupplySideRecommenderSystems.normPowerCostSpec
- JGS23SupplySideRecommenderSystems.normRpowCost
- JGS23SupplySideRecommenderSystems.oneDimensionalUnitContent
- JGS23SupplySideRecommenderSystems.paper_inferred_user_value
- JGS23SupplySideRecommenderSystems.scaleContent
- JGS23SupplySideRecommenderSystems.singleGenreContentLaw
- JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec

Lean-elaborated claim roles

```
1. parameter: ℕ
2. assumption: Nonempty (Fin P)
3. assumption: Nontrivial (Fin P)
4. parameter: ℝ
5. parameter: JGS23SupplySideRecommenderSystems.Content 1
6. parameter: JGS23SupplySideRecommenderSystems.Content 1
7. parameter: JGS23SupplySideRecommenderSystems.SourceNorm 1
8. assumption: 0 < β
9. assumption: JGS23SupplySideRecommenderSystems.NonnegativeContent u
10. assumption: JGS23SupplySideRecommenderSystems.NonnegativeContent genre
11. assumption: ν.norm genre = 1
12. assumption: 0 < JGS23SupplySideRecommenderSystems.paper_inferred_user_value u genre
13. assumption: ∀ (d : JGS23SupplySideRecommenderSystems.Content 1),
  JGS23SupplySideRecommenderSystems.NonnegativeContent d →
    ν.norm d = 1 →
      JGS23SupplySideRecommenderSystems.paper_inferred_user_value u d ≤
        JGS23SupplySideRecommenderSystems.paper_inferred_user_value u genre
14. assumption: Measurable ν.norm
15. conclusion: JGS23SupplySideRecommenderSystems.symmetricMixedNashModelSpec (fun (x : Fin 1) => u)
    (JGS23SupplySideRecommenderSystems.normPowerCostSpec ν β)
    (JGS23SupplySideRecommenderSystems.singleGenreContentLaw 1 P β genre) ∧
  (JGS23SupplySideRecommenderSystems.singleGenreContentLaw 1 P β genre).support ⊆
      {p : JGS23SupplySideRecommenderSystems.Content 1 |
        ∃ q ∈ Set.Icc 0 1, p = JGS23SupplySideRecommenderSystems.scaleContent q genre} ∧
    (∀ (z : ℝ),
        0 ≤ z →
          (MeasureTheory.Measure.map ν.norm (JGS23SupplySideRecommenderSystems.singleGenreContentLaw 1 P β genre))
              (Set.Iic z) =
            ENNReal.ofReal (min 1 ((z ^ β / 1) ^ (↑P - 1)⁻¹))) ∧
      ∀ {μ : JGS23SupplySideRecommenderSystems.MixedContentStrategy 1},
        JGS23SupplySideRecommenderSystems.SourceSymmetricMixedNash
            (fun (x : Fin 1) => JGS23SupplySideRecommenderSystems.oneDimensionalUnitContent)
            (JGS23SupplySideRecommenderSystems.normRpowCost (JGS23SupplySideRecommenderSystems.SourceNorm.l2 1) β) μ →
          μ =
            JGS23SupplySideRecommenderSystems.singleGenreContentLaw 1 P β
              JGS23SupplySideRecommenderSystems.oneDimensionalUnitContent
```

Lean proof endpoint (built separately)

JGS23SupplySideRecommenderSystems.exampleOneDimensionalSetup

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: The approved one-dimensional correction asks for the explicit homogeneous ray construction and uniqueness only in the literal one-user Euclidean model. Lean constructs the mapped uniform-power law under positive beta, a nonnegative unit genre, positive/maximal user score, and norm measurability; it states Nash optimality, unit-ray support, and the clipped radial CDF corresponding to exponent beta/(P-1). Its final clause says every source symmetric Nash measure for user 1 and L2 power cost is exactly that measure. These conclusions and the displayed sampler agree with the repaired CDF and uniqueness target.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation